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SIMPLIFIED MODELING OF THE EDGE PLASMA PHYSICS AND NEUTRAL GAS EXHAUST IN THE THERMONUCLEAR REACTOR WENDELSTEIN 7-X

by

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SIMPLIFIED MODELING OF THE EDGE PLASMA PHYSICS AND NEUTRAL GAS EXHAUST IN THE THERMONUCLEAR FUSION REACTOR WENDELSTEIN 7-X

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Abstract

Magnetically confined thermonuclear fusion reactors produce energy by confining an ionized gas and extracting the heat generated by the fusion reaction. The inevitable sputtering from plasma-solid interactions (PSI) and the byproducts of the reaction introduce impurities that dilute the core plasma by cooling it and reducing fusion reaction probabilities. For this reason, the effective removal of exhaust plasma and impurities is essential for the reactor's efficiency and stable operation. Thus, models that allow for a computationally inexpensive study of the edge plasma transport and pumping system physics are crucial for future design improvements.

In the present work, an effort is made to provide some simplified modeling in order to correlate the upstream plasma parameters with target and subdivertor parameters. The plasma flow along the Scrape-Off Layer (SOL) is analyzed with a reduced-order model for ionized gases, the so-called Two-Point Model (TPM), which provides qualitative information about temperature, pressure, and velocity scaling. To produce this simplified 0D model, important physical phenomena, such as convection, impurity radiation, charge exchange, and ionization and momentum losses, are neglected and therefore, several correction parameters to account for these missing phenomena, are introduced.

Furthermore, to analyze the neutral gas flow in the subdivertor region, a particle balance model in the subdivertor is presented, yielding a closed-form expression for the subdivertor pressure, given the neutral gas flow that passes through the pumping gaps. The results are compared with literature models and experimental data, showing qualitative agreement.

Additionally, the open question of relating the incoming plasma flux at the targets to neutral gas flux in the divertor region has been considered. A simplified model that connects the plasma flux to the recycling neutral gas flux under attached conditions is proposed. This model establishes a complete semi-analytical framework that connects the upstream plasma properties directly to the divertor and subdivertor pressures. The pressure predictions for the divertor and subdivertor regions are in good agreement with the limited experimental data available, though further validation of the model is certainly necessary.

In general, the correction parameters used in these simplified models can be estimated for a specific magnetic configuration and plasma conditions using experimental data. Consequently, a Bayesian inference maximum a posteriori (MAP) optimization problem is formulated to solve for the desired free parameters.

Key words: Fusion engineering, magnetically confined plasma, edge plasma physics, two-point model, neutral gas exhaust system, neutral gas pressure, Bayesian inference, MAP problem

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Περίληψη

Οι αντιδραστήρες θερμοπυρηνικής σύντηξης μαγνητικού περιορισμού παράγουν ενέργεια περιορίζοντας ένα ιονισμένο αέριο και εξάγοντας την παραγόμενη ενέργεια από την αντίδραση σύντηξης. Τα εκτοξευόμενα σωματίδια από την αλληλεπίδραση πλάσματος-στερεού (PSI) και τα υποπροϊόντα της αντίδρασης εισάγουν προσμίξεις (impurities), οι οποίες αραιώνουν το κεντρικό πλάσμα ψύχοντάς το και μειώνοντας την πιθανότητα αντίδρασης σύντηξης. Για τον λόγο αυτόν, η αποτελεσματική απομάκρυνση του πλάσματος εξαγωγής (exhaust plasma) και των προσμίξεων είναι απαραίτητη για την αποδοτική και σταθερή λειτουργία του αντιδραστήρα. Συνεπώς, τα μοντέλα που επιτρέπουν την υπολογιστικά οικονομική μελέτη της ροής του πλάσματος στα τοιχώματα (edge plasma transport) και της φυσικής του συστήματος άντλησης, είναι κρίσιμα για μελλοντικές βελτιώσεις στον σχεδιασμό.

Στην παρούσα εργασία, γίνεται προσπάθεια παροχής απλοποιημένων μοντέλων που συνδέουν τις ιδιότητες του ανάντη πλάσματος (upstream plasma) με τις παραμέτρους του κατάντη πλάσματος (target plasma) και του υπο-εκτροπέα (subdivertor). Η ροή του πλάσματος κατά μήκος της στοιβάδας παράλληλα με το τοίχωμα του αντιδραστήρα, γνωστή ως Scrape-Off Layer, αναλύεται με ένα μειωμένης τάξης μοντέλο για ιονισμένα αέρια, το επονομαζόμενο μοντέλο δυο σημείων (Two-Point Model), το οποίο παρέχει ποιοτική πληροφόρηση σχετικά με τις μεταβολές της θερμοκρασίας, της πίεσης και της ταχύτητας. Για την παραγωγή αυτού του απλοποιημένου 0Δ (μηδενικής διάστασης) μοντέλου, σημαντικά φυσικά φαινόμενα, όπως η συναγωγή (convection), η ακτινοβολία των προσμίξεων (impurity radiation), η ανταλλαγή φορτίων (charge exchange), καθώς και ο ιονισμός και οι απώλειες ορμής, αγνοούνται και, ως εκ τούτου, εισάγονται κάποιες διορθωτικές παράμετροι για να ληφθούν υπόψη προσεγγιστικά.

Επιπλέον, για την ανάλυση της ροής των ουδέτερων αερίων στην περιοχή του υπο-εκτροπέα, παρουσιάζεται ένα μοντέλο ισοζυγίου σωματιδίων (particle balance), το οποίο οδηγεί σε έκφραση κλειστής μορφής (closed-form expression) για την πίεση του υπο-εκτροπέα, δεδομένης της ροής του αερίου που διέρχεται από τα κενά του συστήματος άντλησης (pumping gaps). Τα αποτελέσματα συγκρίνονται με αντίστοιχα υπολογιστικά και πειραματικά δεδομένα, επιδεικνύοντας ποιοτική συμφωνία.

Επιπρόσθετα, εξετάζεται το ανοιχτό ερώτημα σχετικά με τη συσχέτιση της εισερχόμενης ροής πλασμάτος προς τα τοιχώματα με τη ροή του ουδέτερου αερίου στην περιοχή του εκτροπέα (divertor). Προτείνεται ένα απλοποιημένο μοντέλο που συνδέει τη ροή του πλάσματος με τη ροή ανακύκλωσης του ουδέτερου αερίου (recycling neutral gas flux) υπό συνθήκες προσκόλλησης (attached conditions). Το μοντέλο αυτό, καθιερώνει μια πλήρη ημι-αναλυτική σειρά εκφράσεων, η οποία μπορεί να συνδέσει άμεσα τις ιδιότητες του ανάντη πλάσματος με τις πιέσεις στον υπο-εκτροπέα. Οι προβλέψεις του μοντέλου για τις πιέσεις του εκτροπέα και υπο-εκτροπέα βρίσκονται σε καλή συμφωνία με τα λίγα διαθέσιμα πειραματικά δεδομένα και βεβαίως η επιπλέον πιστοποίηση του μοντέλου θεωρείται αναγκαία.

Γενικά, οι διορθωτικές παράμετροι που εφαρμόζονται σε αυτά τα απλοποιημένα μοντέλα μπορούν να εκτιμηθούν για μια συγκεκριμένη μαγνητική κατάσταση (magnetic configuration) και συνθήκες πλάσματος, χρησιμοποιώντας πειραματικά δεδομένα. Στο πλαίσιο αυτό, αναπτύσσεται ένα πρόβλημα Μπεϋζιανής συμπερασματολογίας (Bayesian inference) για τη μεγιστοποίηση της εκ των υστέρων πιθανότητας (maximum a posteriori - MAP) με σκοπό τον προσδιορισμό των επιθυμητών ελεύθερων παραμέτρων.

Λέξεις κλειδιά: Αντιδραστήρες σύντηξης, μαγνητικά περιορισμένο πλάσμα, φυσική ροής πλάσματος στα τοιχώματα, μοντέλο δύο σημείων, σύστημα απαγωγής ουδέτερων αερίων, πίεση ουδέτερων αερίων Μπεϋζιανή συμπερασματολογία, μεγιστοποίηση της εκ των υστέρων πιθανότητας

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1 Introduction to magnetically confined fusion

"Anyone who looks for a source of power in the transformation of atoms is talking moonshine."
— Ernest Rutherford

1.1 Magnetically confined plasma mechanisms

In 1920 Francis Aston used his newly invented mass spectrograph to measure the masses of atoms and their isotopes, finding a groundbreaking result: the total mass of four protons (^1_1H) was larger than that of a Helium (^4_2He) nucleus. Therefore, if four protons are combined into a single Helium nucleus, the resulting Helium atom will have a mass about 0.7% smaller than the original protons. Using Einstein's principle that mass is related to energy, through the famous expression $E = mc^2$, it can be concluded that this "lost" mass is released as energy. This mechanism powers the stars.

In the early 1950s the scientific community started searching for methods to replicate stellar processes on Earth by confining plasma within a solid vessel and extracting the resulting energy as heat. This task is extremely challenging, as it requires putting the sun inside a solid box. Magnetically confined fusion (MCF) and inertial confinement fusion (ICF) are, currently, the most promising technologies being researched for future energy generation. In this thesis, our analysis will focus exclusively on MCF devices.

The first major plasma confinement design came from the renowned plasma physicist Lyman Spitzer, who proposed the stellarator and initiated Project Matterhorn at the Princeton Plasma Physics Laboratory (PPPL) (Figure 1.1). The project was named after the Matterhorn mountain because Spitzer compared the project's difficulty to climbing it. The design was extremely complex since it relied on externally twisted magnetic coils to create the necessary conditions required for the plasma confinement. For such a design to work, a precise optimization of the 3D magnetic coils was mandatory. The equations were computationally intractable at the time, so the idea was abandoned.

In 1951, Andrei Sakharov and Igor Tamm proposed the tokamak design (**To**roidal'naya **Ka**mera s **Ma**gnitnymi **K**atushkami). This concept was much simpler than the stellarator due to its toroidal axisymmetry. To eliminate the complex, twisted magnetic coils and finally achieve axisymmetry, an electric current is driven directly through the plasma to generate the confining poloidal magnetic field. The inherently 2D nature of the design allowed for an analytical treatment of the equations, making tokamaks the dominant design for decades.

However, while the presence of the electric current provides tokamaks with their biggest advantage (analytical tractability and design simplicity), it also introduces a fundamental disadvantage that remains to this day. The internal current makes the confined plasma significantly vulnerable to disruptions, impeding steady-state operation.

Figure 1.1: The first stellarator device at PPPL (Model A), with Lyman Spitzer, a plasma physics and fusion pioneer.

In recent decades, modern supercomputers and advancements in plasma theory have finally enabled the robust numerical treatment of 3D stellarator magnetic-coil optimization, which is required for stable operations, possible. This ability has allowed the fusion community to design and build experimental stellarator devices that aim for a steady-state operation without the plasma disruption risk caused by the strong current [1].

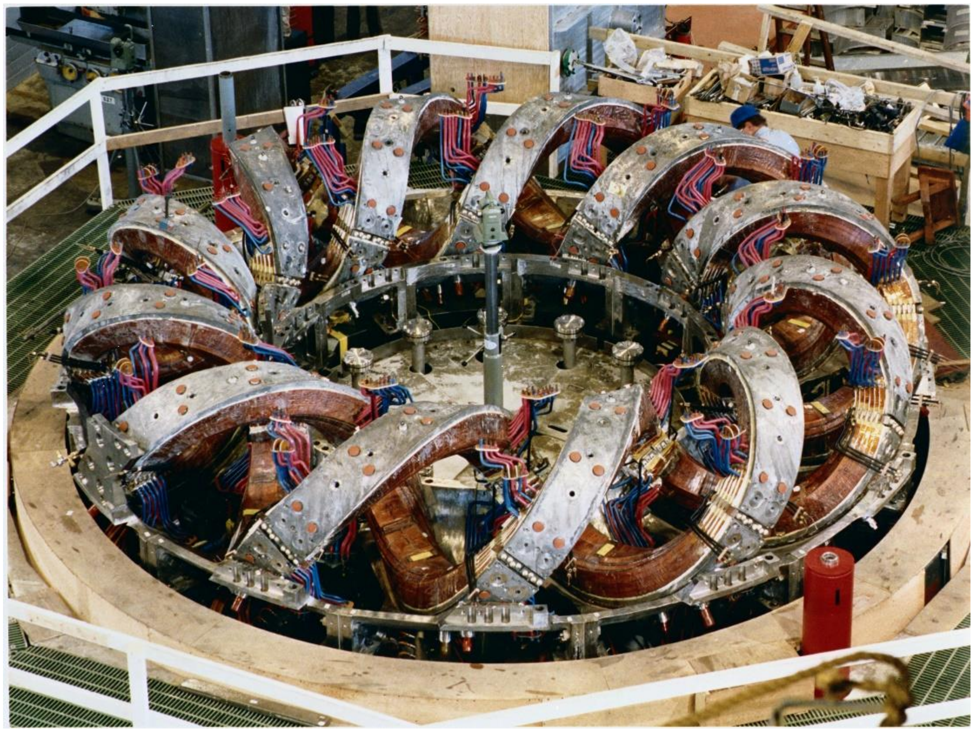

Figure 1.2: Advanced Toroidal Facility (ATF) stellarator at the Oak Ridge National Laboratory (Tennessee, USA).

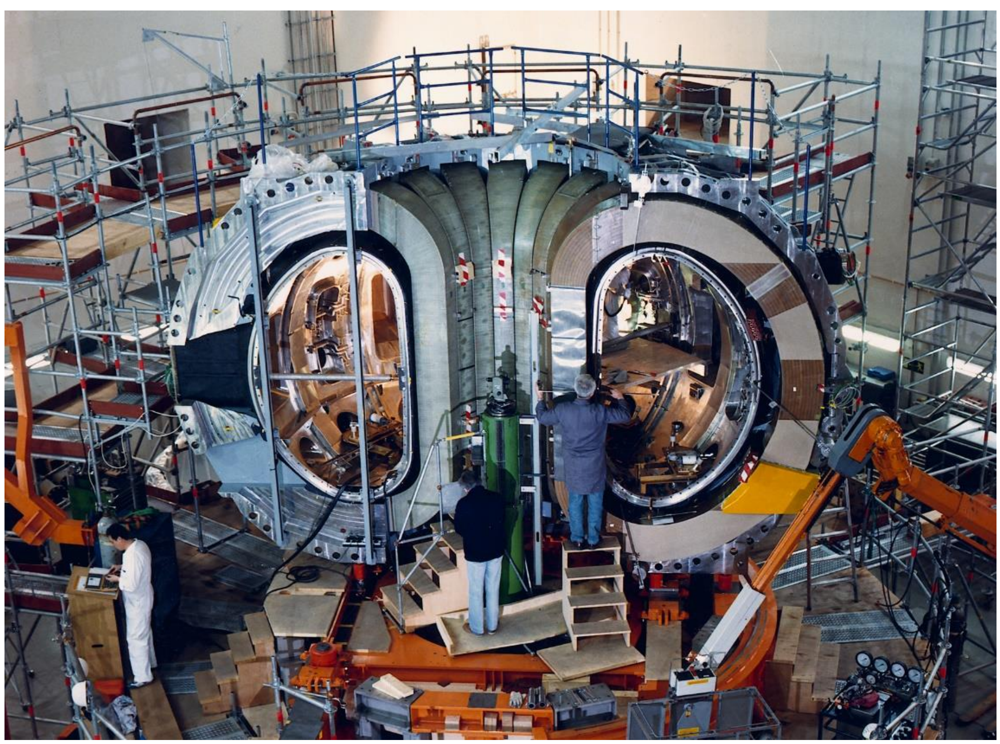

Figure 1.3: ASDEX Upgrade tokamak device at the Max-Planck Institute IPP.

1.2 The Wendelstein 7-X (W7-X) stellarator reactor

Today, the Max-Planck Institute for Plasma Physics (IPP) in Greifswald, Germany, operates the world's largest and most advanced stellarator: the Wendelstein 7-X (W7-X). The device is named after the Wendelstein mountain in the Bavarian Alps in Germany, honoring Lyman Spitzer's original Project Matterhorn. The goal of W7-X, as an experimental reactor, is to prove that under careful optimization stellarators can operate in steady-state and produce a large triple product (Lawson's criterion).

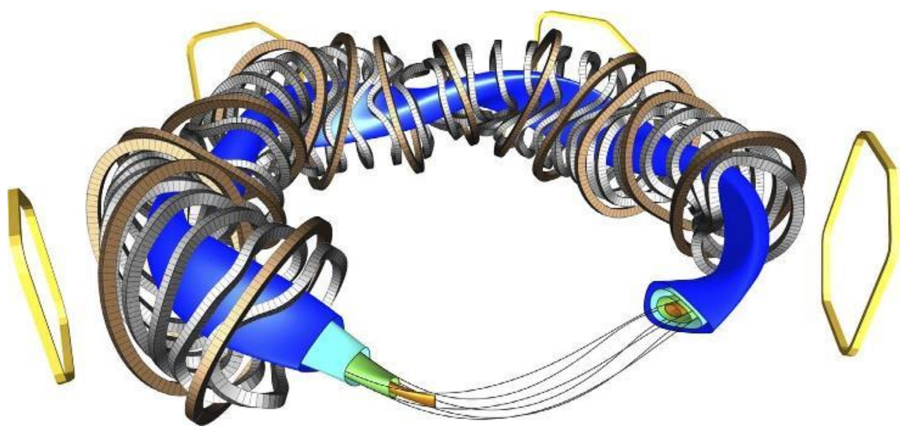

Figure 1.4: CAD of the W7-X reactor. The twisted external magnetic coils are in grey, the reactor vessel in blue and the nested magnetic surfaces in different colors [1].

However, maintaining a hot plasma for extended periods creates severe challenges regarding heat and particle exhaust. Even with highly optimized magnetic configurations, a significant fraction of the plasma will eventually cross the Last Closed Flux Surface (LCFS), enters the Scrape-Off Layer (SOL) and reaches the vessel walls. The current inability to effectively exhaust heat and particles from the reactor causes serious operational limitations.

A primary concern associated with this exhaust is the *impurity production* [2]. This is usually caused by the sputtering of solid plasma-facing components (PFC) due to plasma-surface interactions (PSI), or from helium ash accumulation (only in D-T reactions) as a byproduct of the fusion reaction. If these impurity atoms are not removed properly, they can radiate a significant amount of the core plasma energy, cooling it down and terminate the confinement, or even diluting the purity of the plasma, degrading its performance [3].

1.3 Magnetic fields

As previously mentioned, one significant difference between tokamaks and stellarators is the method for generating the poloidal magnetic field. Tokamaks use a toroidal electric current that passes through the plasma, while stellarators use sophisticated magnetic-coil configurations to create it.

Notwithstanding these differences, both devices are characterized by a total magnetic field $\mathbf{B}_{tot}$, that is the sum of a toroidal magnetic field $\mathbf{B}_{\phi}$ and a poloidal one $\mathbf{B}_{\theta}$, given by

$$\mathbf{B}_{tot} = \mathbf{B}_{\phi} + \mathbf{B}_{\theta}. \tag{1.1}$$

The resulting magnetic field lines form nested flux surfaces, as shown in Figure 1.5. The last closed flux surface (LCFS), is the last flux surface that does not come into contact with any solid structure.

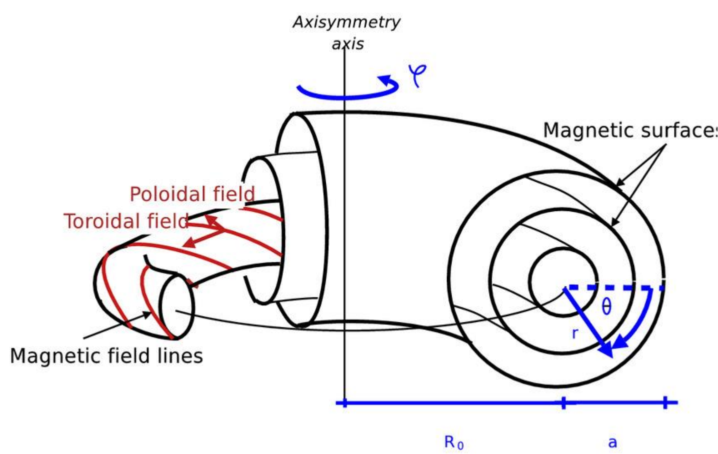

Figure 1.5: Magnetic flux surfaces formed by the poloidal and toroidal fields with a *denoting the minor radius and R (or R_0) the major radius.*

The magnetic surfaces are characterized by their shape, enclosed volume, and by the rotational transform, $\bar{\imath}$, which represents the poloidal turns that a magnetic field line does per toroidal turn on a toroidal flux surface [4], given by

$$\bar{\imath} \equiv \frac{\|\mathbf{B}_{\theta}\|}{\|\mathbf{B}_{\phi}\|} \frac{R}{\mathrm{a}}. \tag{1.2}$$

The theta pitch angle Θ measures the ratio of poloidal displacement to toroidal displacement along the magnetic field lines $\mathbf{B}_{tot}$. It provides a measure of the poloidal twist of the field lines and is calculated using the expression

$$\tan(\Theta) = \frac{\|\mathbf{B}_{\theta}\|}{\|\mathbf{B}_{\phi}\|}. \tag{1.3}$$

Assuming that for typical reactors $\|\mathbf{B}_{\phi}\| \gg \|\mathbf{B}_{\theta}\|$, the expression simplifies to [3]

$$\Theta \approx \frac{\|\mathbf{B}_{\theta}\|}{\|\mathbf{B}_{\phi}\|} \approx \frac{\|\mathbf{B}_{\theta}\|}{\|\mathbf{B}_{tot}\|}. \tag{1.4}$$

1.4 The island divertor

The necessity of heat and particle exhaust is a critical challenge in both tokamaks and stellarators. The two primary methods for managing plasma exhaust are the implementation of a limiter or a divertor (Figure 1.6). Both concepts rely on the idea of removing the particles that escape the LCFS before they reach the main vessel walls.

Limiters, are solid plasma-facing components (PFC) placed directly into the plasma, thus dictating the position of the LCFS. Any plasma particle that escapes, strikes the limiter and then removed. They are rarely used in modern devices since they concentrate plasma-surface interactions (PSI) into a very small area, causing extreme, localized heat fluxes that exceed material limits.

Divertors, on the other hand, use a modified magnetic-field topology that diverts the plasma particles entering the SOL onto the target plates. The magnetic fields interaction establishes a null point (the X-point), which creates a separate region (the private plasma region) that allows the plasma to rapidly cool before striking the walls, thereby increasing the local pressure and allowing for better pumping. If the particles cool enough that they undergo volumetric recombination before reaching the target, the plasma enters a state known as *divertor detachment*. Ultimately, by physically separating the PSI from the core, the divertor design improves both particle and heat exhaust [4].

As a general rule, an effective divertor plasma must achieve three primary objectives: *high radiation capabilities* to reduce and redistribute extreme localized heat fluxes, *high neutral gas* *pressure* in order to maximize pumping efficiency, and *high impurity retention* to prevent contamination from returning to the core [5].

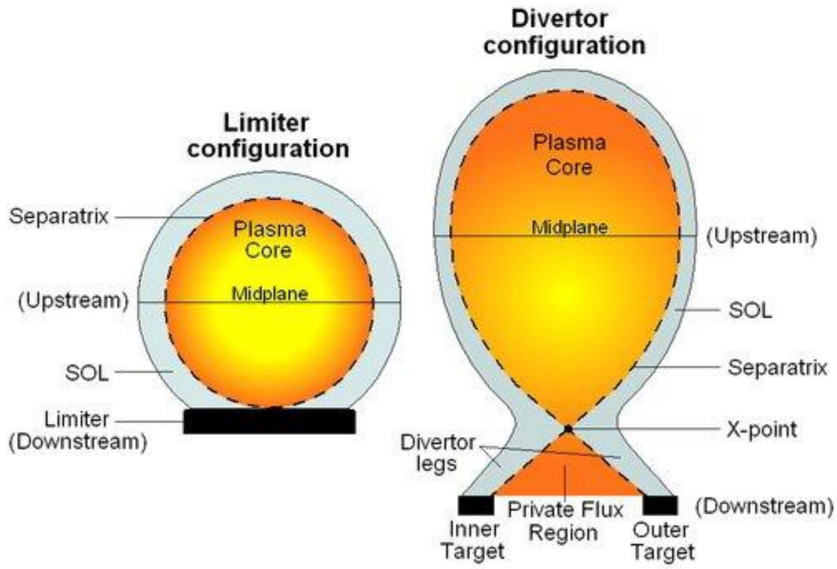

Figure 1.6: Schematic comparison of limiter and divertor configurations in a tokamak device. Limiters define the LCFS, while divertors form a private plasma region which separates the PSI from the core.

In contrast to tokamaks that require external magnetic coils to generate an X-point, stellarators use a different approach known as the *island divertor*. By specifically tailoring the magnetic configuration to naturally form a chain of magnetic island outside the LCFS (Figure 1.7), every particle that escapes this flux surface is guided through the SOL to these island structures. The particles flow along these open field lines until eventually hitting the target plates. This effectively cools the plasma, mitigates the intensity of the PSI, and most importantly, separates the PSI from the core.

Summarizing, the W7-X reactor uses the island divertor for achieving heat and particle exhaust. This is realized by optimizing the magnetic configuration to generate the edge magnetic islands at the intersection with the target elements [4]. Figure 1.8 illustrates this configuration.

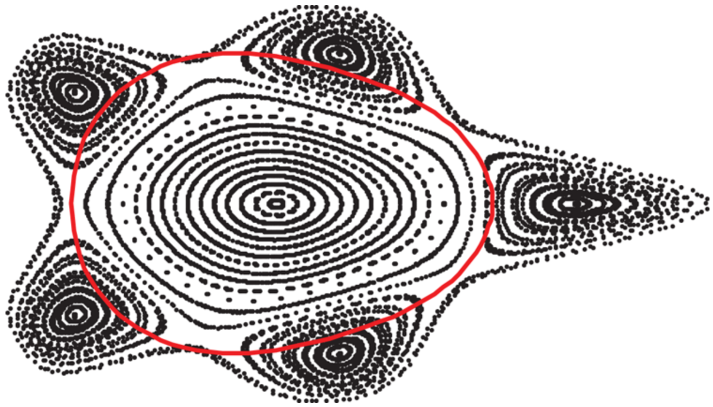

Figure 1.7: A simplified Poincare plot of an island divertor stellarator toroidal cross-section featuring an island divertor. The red circle represents the Last Closed Flux Surface (LCFS), surrounded by a series of five magnetic islands [6].

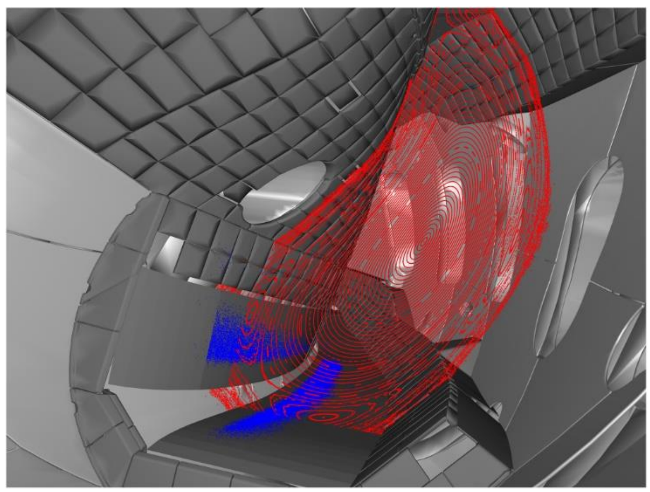

Figure 1.8: Poincare plot of the W7-X reactor. The five magnetic islands are visible peripherally in the vessel walls, the blue areas represent the strike lines of particles in the two target plates [7].

1.5 The divertor module and subdivertor of W7-X

The Wendelstein 7-X stellarator utilizes ten divertor modules [4]. Each module, as shown in Figure 1.9, is equipped with vertical and horizontal targets that form two gaps: the large and the small pumping gap. Particles entering through these gaps are guided into the subdivertor region, where they enter a specific port that pumps them out. The large pumping gap pivots them either to the AEI port or the AEH port, while the small pumping pivots them to the AEP one. Finally, the divertor module is divided into the high iota, the middle, and the low iota section.

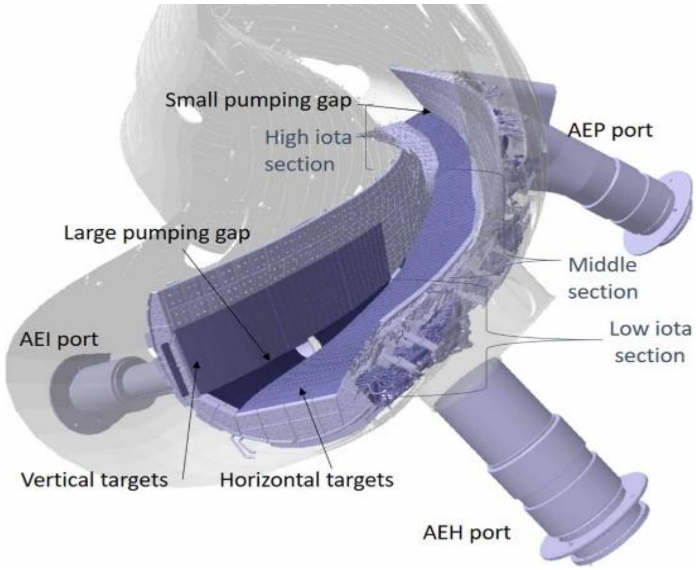

Figure 1.9: The divertor module of W7-X reactor [7].

Changing the shape and location of the magnetic islands alters the magnetic configuration, and moves the strike lines to different areas of the target plates (Figure 1.10). The magnetic configuration is expressed by the rotational transform Eq.(1.2). While infinite configurations may exist, primary interest lies in the standard configuration ($\bar{\iota} = 5/5$) which concentrates the strike lines in the middle section, the high iota configuration ($\bar{\iota} = 5/4$) that guides the particles into the high iota section, while the low iota ($\bar{\iota} = 5/6$) and high mirror configurations pivots the strike lines into the low iota one [7], as shown in Figure 1.10. The significance of the magnetic configuration is that it determines the location of the strike lines and thus the heat flux, and the particle flux entering the pumping gaps. As detailed in Chapter 3, the particle flux that surpasses the pumping gaps (Γ_{in}) is the most crucial parameter for estimating the neutral gas pressure, i.e., the pressure in the subdivertor vessel.

An optimized magnetic configuration, will help particles enter the gaps more effectively, increasing Γ_{in} and the neutral gas pressure. A high neutral gas pressure, coupled with high available effective pumping speed can lead to a powerful particle exhaust.

In [8], a simplified 3D CAD model of the AEH and AEP divertor sections is analysed. The most useful geometric parameters for the present thesis, are summarized in Table 1.1.

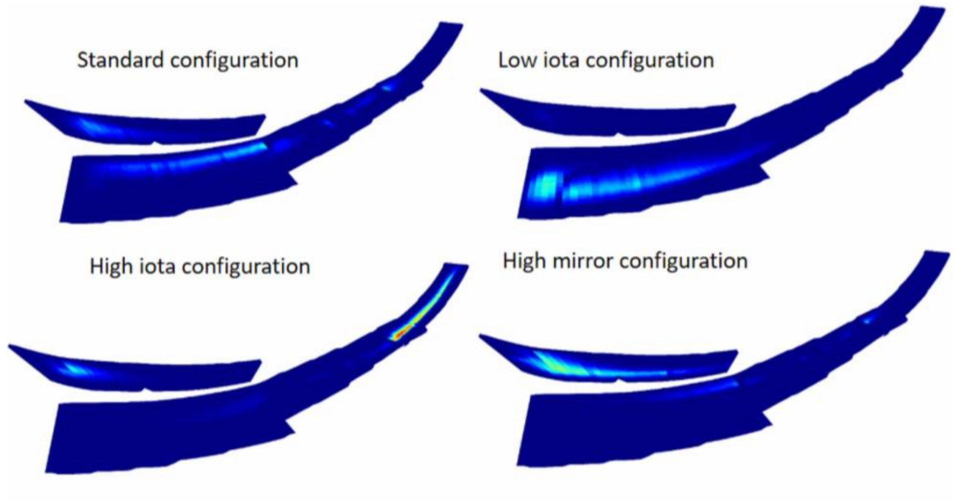

Figure 1.10: Magnetic configurations and their thermal loads intensity, which is proportional to the strike lines intensity [7].

Table 1.1: W7-X simplified divertor geometric areas.

Areas in $[m^2]$	AEH section	AEP section
Total leakage/openings (A_{leak})	7.702×10^{-2}	5.520×10^{-2}
Pumping gap (A_{pg})	1.53×10^{-1}	4.50×10^{-2}
Pumping surface	1.24×10^{-1}	1.24×10^{-1}

1.6 Scope and contents of thesis

The present thesis aims to provide some simplified models that connect the core (upstream) plasma properties with those in the divertor targets and subdivertor regions. Such simplified models may allow a qualitative parametric investigation of the SOL and divertor physics, as well as of the neutral gas exhaust, while bypassing the necessity for running high-fidelity kinetic or fluid codes, such as EIRENE [9] or DIVGAS [10], [11]. To achieve these objectives, the thesis is structured as follows:

In Chapter 2, a semi-analytical formulation of the SOL transport, the so-called the two-point model, for tokamak and stellarator reactors is derived, enabling the correlation of upstream and target parameters. The differences and similarities between tokamak and stellarator edge physics, as well as the different operation regimes, are studied.

In Chapter 3, simplified models that are solved for the subdivertor pressure, given the particle flux entering the pumping gaps, are obtained. Furthermore, a model that, under attached conditions, can correlate the upstream properties directly to the subdivertor pressure is proposed.

In Chapter 4, a simple Bayesian inference to compute the Maximum A Posteriori (MAP) of the desired free parameters based on experimental data, is derived.

Finally, in Chapter 5 the main concluding remarks are stated and some tentative future work is outlined.

2 Scrape-Off Layer (SOL) transport

> *"In the early decades of fusion energy research, the SOL was little considered, apparently in the hope that the edge would just sort itself out with little need for intervention or understanding. This hope was misplaced and by the 1980s it had been recognized that certain edge problems were sufficiently serious as to jeopardize the achievement of controlled fusion energy using magnetically confined plasmas."*
> — P.C. Stangeby, The Plasma Boundary of Magnetic Fusion Devices

The main transport equations that govern the plasma flow along the Scrape-Off Layer (SOL) are derived. The goal is to simplify these balance equations, and reach to algebraic expressions for the plasma parameters, such as the temperature, the plasma density and the pressure.

Usually, the poloidal direction is constricted between two points. The first point is in the upstream where plasma enters the SOL from the core and the second one at the targets that plasma (or neutrals) hit the solid surface.

2.1 Mathematical formulation of the 1D SOL transport

The governing equation for the Scrape-Off Layer (SOL) transport is the 3D time-dependent Boltzmann equation, where the external forces that accelerate the particles are the Lorentz forces. For electron particles (e is the negative electron charge), the equation is

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla_{\mathbf{r}} f + \frac{e}{m}(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \nabla_{\mathbf{v}} f = \left.\frac{\partial f}{\partial t}\right|_{coll} + S(\mathbf{r}, \mathbf{v}, t), \tag{2.1}$$

where $f(\mathbf{r}, \mathbf{v}, t)$ is the one-particle distribution function (see Appendix A). Steady-state (SS) conditions and one-dimensional spatial flow are assumed. Thus, the electric and magnetic fields are $\mathbf{E} = E\hat{\mathbf{i}}$, $\mathbf{B} = B\hat{\mathbf{i}}$, respectively, while the velocity vector is $\mathbf{v} = v_x \hat{\mathbf{i}}$. Under these assumptions, gyromotion can be neglected:

$$\mathbf{v} \times \mathbf{B} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ v_x & 0 & 0 \\ B & 0 & 0 \end{vmatrix} = 0 \tag{2.2}$$

Therefore, the Boltzmann equation takes the following form [3]:

$$v_x \frac{\partial f}{\partial x} + \frac{eE}{m}\frac{\partial f}{\partial v_x} = \left.\frac{\partial f}{\partial t}\right|_{coll} + S(x,\mathbf{v}) \tag{2.3}$$

This equation is also known as the *Fokker-Planck (collisional) kinetic equation.*

Considering that in plasmas the main source of particles is the ionization of neutrals, it is assumed that $S(x,\mathbf{v})$ comes mainly from the neutral ionization. Furthermore, it is assumed that the neutrals have a drifting Maxwellian distribution and the source term is approximate as

$$S(x,\mathbf{v}) = S_p(x)\left(\frac{\beta_n}{\pi}\right)^{3/2} e^{-\beta_n(v_x-v_n)^2} e^{-\beta_n v_y^2} e^{-\beta_n v_z^2}\ , \tag{2.4}$$

where v_n is the drift velocity and T_v the corresponding temperature, while $S_p(x)$ is the spatial variation of the particle source and $\beta_n \equiv m_n / 2k_B T_n$. It is also assumed that the plasma is a continuum media that flows as a two-fluid: electrons and ions. The two-fluid governing equations are derived by taking the proper moments of Eq. (2.3). Then, a series of assumptions and simplifications is introduced to obtain the so-called system of <u>*Semi-1D Fluid Equations for Plasma Flow*</u>, given by

$$\boxed{\begin{aligned} &\frac{d}{ds_{\|}}\left[nv\right] = S_p(s_{\|}) \\ &\frac{d}{ds_{\|}}[m_i n v^2 + 2k_B n T] = -m_i v\overline{\sigma v_{in}} n_n n \\ &\frac{\partial q_{\|}}{\partial s_{\|}} + \frac{\partial q_{\perp}}{\partial s_{\perp}} = Q_R + Q_E \end{aligned}} \tag{2.5}$$

The unknowns of the system (2.5) are the bulk velocity (v), the plasma density (n), and the temperature (T) in the parallel direction. The mathematical manipulation is long, tedious and well-known and is provided in [3] (see also Appendix C).

2.1.1 Importance comparison: conduction vs convection

Furthermore, additional robust approximations can be made by comparing the typical orders of magnitude of the energy conservation equation terms. The left-hand side (LHS) of the energy equation (2.5) is written in an explicit form

$$\frac{\partial}{\partial s_{\|}}\left[\frac{1}{2}m_i n v^3 + 5nk_B T v - k_{0,e}T^{5/2}\frac{dT}{ds_{\|}}\right] + \frac{\partial}{\partial s_{\perp}}\left[-\frac{5}{2}k_B T D_{\perp}\frac{dn}{ds_{\perp}} - x_{\perp} n \frac{d(k_B T)}{ds_{\perp}}\right], \qquad (2.6)$$

as shown in Eq. (C.50).

It is noted that only two terms of the parallel heat convection are multiplied by the velocity. This indicates that they are both essentially negligible everywhere except at the targets and therefore their typical values at the target are compared by taking their ratio:

$$\frac{1/2\, m_i n v^3}{5nk_B T v} \approx \frac{1}{10}\frac{m_i v_{se}{}^2}{k_B T_{se}} = \frac{1}{10}\frac{m_i\, 2k_B T_{se}/m_i}{k_B T_{se}} = \frac{1}{5}.$$

Here, the temperature T_{se} both in the nominator and denominator is in $[K]$ and is cancelled to find $q_{\|conv} \simeq 1/2 m_i n v^3$.

Next, the dominant parallel convection and conduction terms are compared by taking their ratio:

$$\frac{1/2 m_i n v^3}{k_{0,e}T^{5/2}dT/dx} \sim \frac{1/2 m_i n_{se}\left(\sqrt{2eT_{se}/m_i}\right)^3}{k_{0,e}T_{se}{}^{7/2}/L_c} = \frac{\sqrt{2}e^{3/2}}{\sqrt{m_i}k_{0,e}}\frac{n_{se}L_c}{T_{se}{}^2[eV]} \sim 1.11\times 10^{-18}\frac{n_{se}L_c}{T_{se}{}^2[eV]}.$$

Typical values for particle density and temperature at the targets are $n \sim 10^{20}[m^{-3}]$, and $T \sim 5[eV]$. Thus, the fraction becomes $4.44 L_c$. The connection length L_c differs between tokamaks ($\sim 1m$) and stellarators ($\sim 400m$). In both cases, it is observed that even though in the majority of the SOL the convection mechanism is practically zero, at the sheath edge it dominates over conduction. Keeping that in mind, it is assumed that heat is transferred mainly via conduction throughout the SOL and a proper boundary condition is applied at the sheath-edge (see Section 2.1.2). The parallel conductive heat flux is given by

$$q_{\|} \approx -k_{0,e}T^{5/2}\frac{dT}{dx}. \qquad (2.7)$$

Then, the perpendicular convection and conduction terms are compared by taking their ratio:

$$\frac{5/2\, k_B T_{[K]} D_{\perp}}{x_{\perp} n}\frac{dn/ds_{\perp}}{d(k_B T_{[K]})/ds_{\perp}} \sim \frac{5/2\, k_B T D_{\perp}}{x_{\perp} n}\frac{n/\lambda_{SOL}}{k_B T/\lambda_{SOL}} = \frac{5}{2}. \qquad (2.8)$$

Here, λ_{SOL} is the average SOL cross-sectional length, and as a first approximation, $x_{\perp} \approx D_{\perp}$. This shows that perpendicular convection is of the same order of magnitude as conduction. Retaining the parallel convection term would couple the momentum and energy conservation equations due to the particle density gradient term, restricting an analytical treatment. Therefore, this term is dropped, keeping in mind that if the perpendicular heat conduction term is proven to be of significant importance, a convection loss parameter must be added.

At this point it is worth recalling the definition of the magnetic field line pitch (or theta pitch angle) in Eq. (1.4). It is readily shown that

$$\Theta = \frac{ds_{\perp}}{ds_{\parallel}},$$

where $\Theta \sim 10^{-3}$ for stellarators and $\Theta \sim 10^{-1}$ for tokamaks. Based on all the above, the RHS term of (2.6) takes the form

$$\frac{\partial}{\partial s_{\parallel}}\left[-k_{0,e}T^{5/2}\frac{dT}{ds_{\parallel}}\right]+\frac{\partial}{\partial s_{\parallel}}\left[-\frac{1}{\Theta^2}x_{\perp}n\frac{d(k_BT)}{ds_{\parallel}}\right].$$

Finally, the parallel and perpendicular heat fluxes are compared by taking their ratio:

$$\frac{q_{\parallel}}{q_{\perp}} \approx \Theta^2\frac{k_{0,e}T^{5/2}}{x_{\perp}ne}. \tag{2.9}$$

In order to compare the two quantities, the same temperature unit is used and thus, the perpendicular temperature term is expressed in eV, by using eT instead of the typical k_BT. The critical particle density and temperature (n,T) values at which the two heat fluxes are equal ($q_{\parallel} = q_{\perp}$), are plotted from (2.9), in Figure 2.1, based on the deduced expression

$$T^{5/2} = \frac{x_{\perp}e}{k_0\Theta^2}n \tag{2.10}$$

For $x_{\perp} \simeq 3[m^2/s]$ and Spitzer- Härm heat conductivities, given by $k_{0,e} \approx 2000\left[Wm^{-1}eV^{-7/2}\right]$ and $k_{0,i} \approx 60\left[Wm^{-1}eV^{-7/2}\right]$, the critical particle density and temperature plots are shown in Figure 2.1.

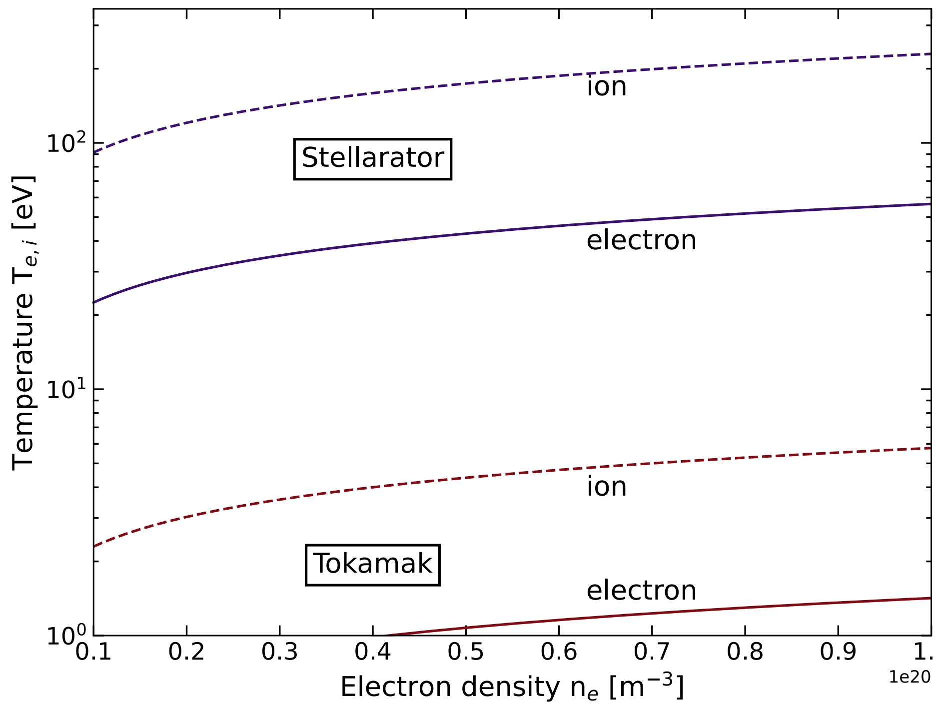

Figure 2.1: Critical particle density and temperature relation so that parallel conduction to be equal to perpendicular conduction, for tokamaks and stellarators.

Typical fusion reactors operate in extremely large plasma temperatures, with the lowest values usually obtained at the targets, having an order of about $2-10[eV]$. This leads to the conclusion that tokamaks usually operate well above the ion-line, validating the assumption of neglecting the perpendicular conduction. On the other hand, stellarators operate in approximately the same temperature regions, which makes the perpendicular conduction often stronger that the parallel one [12].

In conclusion, it is a valid simplification to neglect perpendicular diffusion, as well as convection, in tokamaks, but not in stellarators. This makes the stellarator analysis harder and yields, as it will be shown, to non-analytically solvable equations. In this case, since the perpendicular convection term is important and its effect must be considered, a corrective convection parameter is added to be able to simplify the system by removing this term (see Section 2.2.3). The main reason for this significant difference between tokamaks and stellarators is the divertor geometry of these two devices, since stellarators, typically using an island divertor, are characterized by a smaller theta pitch $ds_{\perp} / ds_{\parallel}$.

2.1.2 The Bohm criterion

The ionization mainly takes place in the sheath edge (targets), so $S_p(s_\parallel)$ is practically zero everywhere except from the targets. This means that since $Q_E \propto S_p(s_\parallel)$, it is safely set $Q_E = 0$. Additionally, the plasma flow originates near the magnetic midplane, yielding zero velocity above the X-point [3]. From the other side, at the sheath-edge the Debye sheath exists creating electrostatic potentials.

Due to pressure difference the plasma flows from the midplane to the targets, reaching finally the local sonic ion speed in order for the ions to be able to penetrate the Debye potentials. This is the so-called Bohm's Criterion:

$$v_{se} = c_s \equiv \sqrt{\frac{k_B(T_e + T_i)}{m_i}} \simeq \sqrt{\frac{2k_B T}{m_i}} \,. \tag{2.11}$$

This means that the particle conservation equation is satisfied identically nearly across all the SOL. Thus, at the targets, the conservation equation is dropped and instead some kind of boundary equation for the sheath-edge is used. In this framework, the sheath heat transmission coefficient γ, as the ratio of the power removed from the plasma to be received from the solid surfaces, over the power reaching the sheath edge, is defined:

$$\gamma \equiv \frac{q_{se}}{k_B T_e \Gamma_{se}} \,. \tag{2.12}$$

Typical plasma values are $\gamma \approx 7-8$. Using (2.11) and (2.12), the ***<u>Bohm's Boundary condition</u>*** is formed:

$$q_\parallel \approx q_{se} = \gamma k_B T_{se} n_{se} c_{s,se} \,. \tag{2.13}$$

Applying all above to the final simplified system for SOL plasma flow, in order to compute the temperature T, velocity v, and plasma density n, the following set of equations is obtained:

$$\boxed{\begin{aligned} & q_\parallel = \gamma k_B T_{se} n_{se} c_{s,se} \\ & \frac{d}{ds_\parallel}\left[nv^2 + 2\frac{k_B nT}{m_i} \right] = S_{mom} \equiv -v\overline{\sigma v_{in}} n_n n \\ & \frac{d}{ds_\parallel}\left[-k_{0,e} T^{5/2}\frac{dT}{ds_\parallel} - \frac{1}{\Theta^2} x_\perp n \frac{dT}{ds_\parallel} \right] = Q_R \end{aligned}} \tag{2.14}$$

The parallel flux $q_{\parallel}$ at the sheath edge is a control variable, while the connection length L_c, theta pitch Θ and sheath heat transmission coefficient γ are specified constants.

2.2 Two-Point Model (TPM) of the SOL

The system of simplified 1D fluid equations (2.14) is computationally expensive to be solved and in order to have a detailed parametric analysis of the SOL further simplifications are applied.

2.2.1 Tokamak Two-Point Model (TTPM)

The most well-known approach to this problem has been proposed by Stangeby [3]. The idea is to neglect the perpendicular conduction, as well as heat and momentum losses ($S_{mom}=0$, $Q_R=0$), and then integrate the resulting homogeneous ODEs of (2.14) between two specific points to correlate the desired upstream and downstream plasma properties (n,v,T). The most convenient approach is to select the two points to be an upstream point (u) and the sheath-edge or target point (t).

Start by setting the momentum losses $S_{mom}=0$ and then integrate the plasma momentum ODE:

$$\int_u^t \frac{d}{ds_{\parallel}}\left[nv^2+2\frac{k_B nT}{m_i}\right]ds_{\parallel}=0 \Rightarrow nv^2+2\frac{k_B nT}{m_i}\bigg|_u^t=0 \quad . \tag{2.15}$$

Using the approximation $v_u \approx 0$ [3], and Bohm's criterion (2.11), Eq. (2.15) can be written in the form:

$$2n_tT_t=n_uT_u \, . \tag{2.16}$$

Next, by setting $Q_R=0$ and $q_{\perp}=0$ the energy conservation equation is written as

$$\frac{dq_{\parallel}}{ds_{\parallel}}=\frac{d}{ds_{\parallel}}\left(-k_{0,e}T^{5/2}\frac{dT}{ds_{\parallel}}\right)=0 \, . \tag{2.17}$$

By taking the indefinite integral of (2.17) with respect to $ds_{\parallel}$ it becomes

$$q_{\parallel}=-k_{0,e}T^{5/2}\frac{dT}{ds_{\parallel}}=\text{const} \, . \tag{2.18}$$

The variables are separated and Eq. (2.70) is integrated as

$$\int_{T_u}^{T_t} k_{0,e} T^{5/2} dT = -\int_0^{L_c} q_{\parallel} ds_{\parallel} . \tag{2.19}$$

Considering that $q_{\parallel} = \text{const}$ from Eq. (2.18), the solution can be written as

$$T_t^{7/2} - T_u^{7/2} = -\frac{7}{2} \frac{q_{\parallel} L_c}{k_{0,e}} . \tag{2.20}$$

So, the tokamak two-point model (TTPM) system is:

$$\begin{aligned} & q_{\parallel} = \gamma k_B T_t n_t c_{s,t} \\ & T_t^{7/2} - T_u^{7/2} = -\frac{7}{2} \frac{q_{\parallel} L_c}{k_{0,e}} \\ & 2 n_t T_t = n_u T_u \end{aligned} \tag{2.21}$$

This system must be solved for T_u, T_t, n_t, having as control parameters n_u and $q_{\parallel}$.

Further simplification is possible by assuming that plasma cools significantly as it flows towards the targets and therefore $T_t \ll T_u$. This means that these quantities raised to the power of $7/2$ will experience an even larger difference and it is assumed that $T_u^{7/2} - T_t^{7/2} \approx T_u^{7/2}$.

Now, Eq. (2.20) can be solved for the *upstream temperature* T_u:

$$\boxed{T_u = \left(\frac{7}{2} \frac{q_{\parallel} L_c}{k_{0,e}} \right)^{2/7}} . \tag{2.22}$$

Combining (2.13) and (2.16), closed form solution for the *target temperature* T_t and *target density* n_t are obtained:

$$\boxed{T_t = \frac{m_i}{2e} \frac{4 q_{\parallel}^2}{\gamma^2 e^2 n_u^2} \left(\frac{7}{2} \frac{q_{\parallel} L_c}{k_{0,e}} \right)^{-4/7}} \tag{2.23}$$

$$\boxed{n_t = \frac{n_u^3}{q_{\parallel}^2} \left(\frac{7}{2} \frac{q_{\parallel} L_c}{k_{0,e}} \right)^{6/7} \frac{\gamma^2 e^3}{4 m_i}} \tag{2.24}$$

2.2.2 Stellarator Two-Point Model (STPM)

Although the TTPM is capable of providing computationally cheap and accurate results (at least within an order of magnitude) for tokamaks, this is not the case for stellarator devices. As shown in Figure 2.1, dropping the perpendicular conduction heat flux term can be a highly inaccurate assumption for island divertor geometries. In [13], Feng expanded on this idea. Since the STPM differs from the TTPM only in the energy conservation equation, a new energy equation is derived.

Start with the general semi-1D energy conservation equation

$$\frac{dq_{\parallel}}{ds_{\parallel}}+\frac{1}{\Theta}\frac{dq_{\perp}}{ds_{\parallel}}=\frac{d}{ds_{\parallel}}\left(-k_{0,e}T^{5/2}\frac{dT}{ds_{\parallel}}\right)+\frac{d}{ds_{\parallel}}\left(-\frac{1}{\Theta^2}x_{\perp}n\frac{dT}{ds_{\parallel}}\right)=0\,, \tag{2.25}$$

and take the indefinite integral of Eq. (2.25) with respect to $ds_{\parallel}$, assuming the perpendicular heat flux to be independent of the $s_{\parallel}$:

$$q_{\parallel}+\frac{1}{\Theta}\int\cancel{\frac{dq_{\perp}}{ds_{\parallel}}}ds_{\parallel}=-k_{0,e}T^{5/2}\frac{dT}{ds_{\parallel}}-\frac{1}{\Theta^2}x_{\perp}n\frac{dT}{ds_{\parallel}}=c\,.$$

Thus, the constant of integration c is again the heat flux parallel to the magnetic field lines, i.e., $c=q_{\parallel}$. Then, the simplified ODE is integrated as follows:

$$\int_{T_u}^{T_t}\left(k_{0,e}T^{5/2}+\frac{1}{\Theta^2}x_{\perp}n\right)dT=-\int_0^{L_c}q_{\parallel}ds_{\parallel}\,. \tag{2.26}$$

Approximating the perpendicular heat conduction integral, by taking the average value of the particle density $(\bar{n})$, yields the final result:

$$T_u^{7/2}=T_t^{7/2}+\frac{7}{2}\frac{q_{\parallel}L_c}{k_{0,e}}-\frac{7}{4}\frac{x_{\perp}(n_u+n_t)}{k_{0,e}\Theta^2}\left(T_u-T_t\right). \tag{2.27}$$

Therefore, the final <u>Stellarator Two-Point Model system</u> is given by:

$$\boxed{\begin{aligned}&q_{\parallel}=\gamma k_B T_t n_t c_{s,t}\\ &T_u^{7/2}=T_t^{7/2}+\frac{7}{2}\frac{q_{\parallel}L_c}{k_{0,e}}-\frac{7}{4}\frac{x_{\perp}(n_u+n_t)}{k_{0,e}\Theta^2}\left(T_u-T_t\right)\\ &2n_tT_t=n_uT_u\end{aligned}} \tag{2.28}$$

Equations (2.28) do not have a known closed form solution.

2.2.3 Extended Two-Point Model

In order to derive the two-point model for both tokamaks and stellarators, various important physical phenomena are ignored. To improve the prediction capabilities of this simplified model, specific loss parameters may be used. This approach has been firstly implemented in [3] for tokamaks and recently for stellarators [5]. The employed loss parameters include:

Convective power fraction (f_{conv}):

In the 2nd moments of Boltzmann's equation (energy conservation) it is assumed that all power is carried by perpendicular conduction and bi-normal diffusion. The abandonment of the convection terms of course is valid, but only on an order of magnitude basis. To account for this heat transfer mode, the *convective power* fraction is introduced:

$$1 - f_{conv} \equiv \frac{q_{\|cond}}{q_{\|}} . \tag{2.29}$$

Target-localized power dissipation fraction (f_{cool}):

The heat flux prediction at the targets is improved, by taking into account, three important physical phenomena.

Impurity radiation which happens when hot electrons from the plasma collide with neutrals or impurity ions and excite the bound electrons in these atoms. When those bound electrons drop back, they emit photons, effectively redistributing the energy.

Ionization losses which take place when electrons collide with a neutral with enough energy to tear away an electron from the atom. This process ionizes the neutral atom using energy from the plasma.

Charge exchange which occurs when a hot ion collides with a cold neutral and steals its electron. The resulting hot neutral is not affected any more by the magnetic field lines and escapes the core taking power in the form of kinetic energy.

All above are formulated using the target-localized power dissipation fraction:

$$1 - f_{cool} \equiv \frac{q_{\|se}}{q_{\|}} . \tag{2.30}$$

Momentum loss factor (f_{mom}):

Following the same procedure as before, the plasma momentum losses are modeled using a momentum loss factor that measures the pressure drop along the SOL:

$$1 - f_{mom} \equiv \frac{p_t^{tot}}{p_u^{tot}} . \tag{2.31}$$

The power parameters defined in (2.29) and (2.30) are capable of modeling physical phenomena for both tokamak and stellarator SOL transport, since they are independent of the divertor geometry. Thus, no distinction was made. The momentum loss parameter, defined in (2.83), is more complicated since tokamaks and stellarators experience momentum loss differently [5]. Tokamak are mainly due to plasma interactions with neutrals, while stellarator momentum losses are mainly originated from the friction between counter streaming flows in the island divertor.

The closed form expression of f_{mom} derived for the stellarator island divertor is given by [5], [14]:

$$1 - f_{mom} = \left(1 + \frac{\alpha}{\sqrt{T_t}}\right)^{-1} , \tag{2.32}$$

where α is the *strength of momentum loss* and is a free parameter.

Substituting the above definitions into the Stellarator TPM (2.28) yields the ***Extended Stellarator Two-Point Model***:

$$\boxed{\begin{aligned} q_{\parallel} &= \frac{\gamma k_B T_t n_t c_{s,t}}{1 - f_{cool}} \\ T_u^{7/2} &= T_t^{7/2} + \frac{7}{2}\frac{(1 - f_{conv})q_{\parallel} L_c}{k_{0,e}} - \frac{7}{4}\frac{x_{\perp}(n_u + n_t)}{k_{0,e}\Theta^2}\left(T_u - T_t\right) \\ n_u T_u &= \frac{2 n_t T_t}{1 - f_{mom}} = 2 n_t T\left(1 + \frac{\alpha}{\sqrt{T_t}}\right) \end{aligned}} \tag{2.33}$$

This system of equations is solved numerically and a parametric analysis is performed. The system is solved with respect to the unknown variables T_u, T_t, n_t and then, the solutions are plotted with respect to the upstream particle density n_u. The control parameters are the ones in [5] and their values are:

$$\alpha = 2,\ f_{conv} = 0,\ f_{cool} = 0,\ \Theta = 10^{-3},$$
$$x_{\perp} = 1.5\,[m^2/s],\ L_c = 180\,[m],\ q_{\parallel} = 100\ [MW/m^2] \tag{2.34}$$

Due to the fact that the target density is in the order of about 10^{19}, while the maximum temperature is in the order of about 10^2, the Jacobian is ill-conditioned. In addition, the system is highly non-linear and a typical iterative solution is not straightforward. It is proposed to algebraically manipulate the system (2.33) in a simpler form. Start by solving the Bohm's criterion for the T_t and using the relation $eT\,[eV] = k_B T\,[K]$ to express the temperature to eV as

$$T_t = \left[\frac{1}{2} A \frac{1}{n_t^2}\left(1 - f_{cool}\right)^2\right]^{1/3}, \tag{2.35}$$

where $A \equiv \dfrac{q_{\parallel}^{\,2} m_i}{\gamma^2 e^3}$. Then, substitute Eq. (2.35) to the pressure balance and solve for T_u to obtain:

$$T_u = \left[4A\frac{n_t}{n_u^{\,3}}\left(1 - f_{cool}\right)^2\right]^{1/3}\left\{1 + \alpha\left[\frac{1}{2}A\frac{1}{n_t^2}\left(1 - f_{cool}\right)^2\right]^{-1/6}\right\}. \tag{2.36}$$

Finally, use (2.35) and (2.36) in the energy conservation, and form the final equation to be solved:

$$\boxed{\begin{aligned}&\left(\left[4A\frac{n_t}{n_u^{\,3}}\left(1 - f_{cool}\right)^2\right]^{1/3}\left\{1 + \alpha\left[\frac{1}{2}A\frac{1}{n_t^2}\left(1 - f_{cool}\right)^2\right]^{-1/6}\right\}\right)^{7/2} = \left[\frac{1}{2}A\frac{1}{n_t^2}\left(1 - f_{cool}\right)^2\right]^{7/6} + B\left(1 - f_{conv}\right)\\ &- C\left(n_u + n_t\right)\left(\left[4A\frac{n_t}{n_u^{\,3}}\left(1 - f_{cool}\right)^2\right]^{1/3}\left\{1 + \alpha\left[\frac{1}{2}A\frac{1}{n_t^2}\left(1 - f_{cool}\right)^2\right]^{-1/6}\right\} - \left[\frac{1}{2}A\frac{1}{n_t^2}\left(1 - f_{cool}\right)^2\right]^{1/3}\right)\end{aligned}} \tag{2.37}$$

In Eq. (2.37), $B \equiv \dfrac{7}{2}\dfrac{q_{\parallel}L_c}{k_{0,e}}$ and $C \equiv \dfrac{7}{4}\dfrac{x_{\perp}e}{k_{0,e}\Theta^2}$. Equation (2.37) is solved for the target density n_t and then the target and upstream temperatures are estimated from (2.35) and (2.36). It is noted that the same numerical procedure may also be employed in the TTPM if the simplification $T_u \gg T_t$ is not introduced.

Plots of T_u, T_t, n_t with respect to n_u, with varying control parameter values, are shown in Figure 2.2-2.6. In all cases, as the upstream density n_u increases, the target plasma density n_t, initially increases linearly and then cubically, before finally saturates at high n_u values. Also, at low n_u

values, the upstream and target temperatures are very close to each other and then, as n_u increases, the target temperature drops rapidly, while the upstream temperature stays relatively constant. Finally, at high n_u both temperatures keep decreasing but in a very slow pace. These behaviors characterize the operational regimes, as discussed in Section 2.2.4.

More specifically, Figure 2.2 demonstrates that for $n_u > 10^{19}\, m^{-3}$, increasing the momentum loss strength parameter α mitigates the density buildup and elevates both target and upstream temperatures. Furthermore, in Figure 2.3, a higher convection fraction f_{conv} diminishes the density buildup, and decreases the temperature difference by lowering the upstream temperature and raising the target temperature.

On the contrary, Figure 2.4 demonstrates that increasing the power dissipation fraction f_{cool} raises the target density only at small upstream density values. Additionally, the upstream temperature stays relatively unaffected, while the target temperature plummets. As shown in Figure 2.5, increasing the field line pitch Θ, elevates the target density and upstream temperature, while decreases the target temperature, but only for medium to large values of upstream density, i.e. for $n_u > 3-4\times 10^{19}\, m^{-3}$.

Finally, Figure 2.6 indicates that increasing the parallel heat flux $q_{\|}$ always increases both target and upstream temperatures, but its effect on the target density is nonmonotonic. At small upstream density values the effect on the target density is negligible, at medium values of n_u the target density drops, but at larger values, increasing $q_{\|}$ increases the n_u.

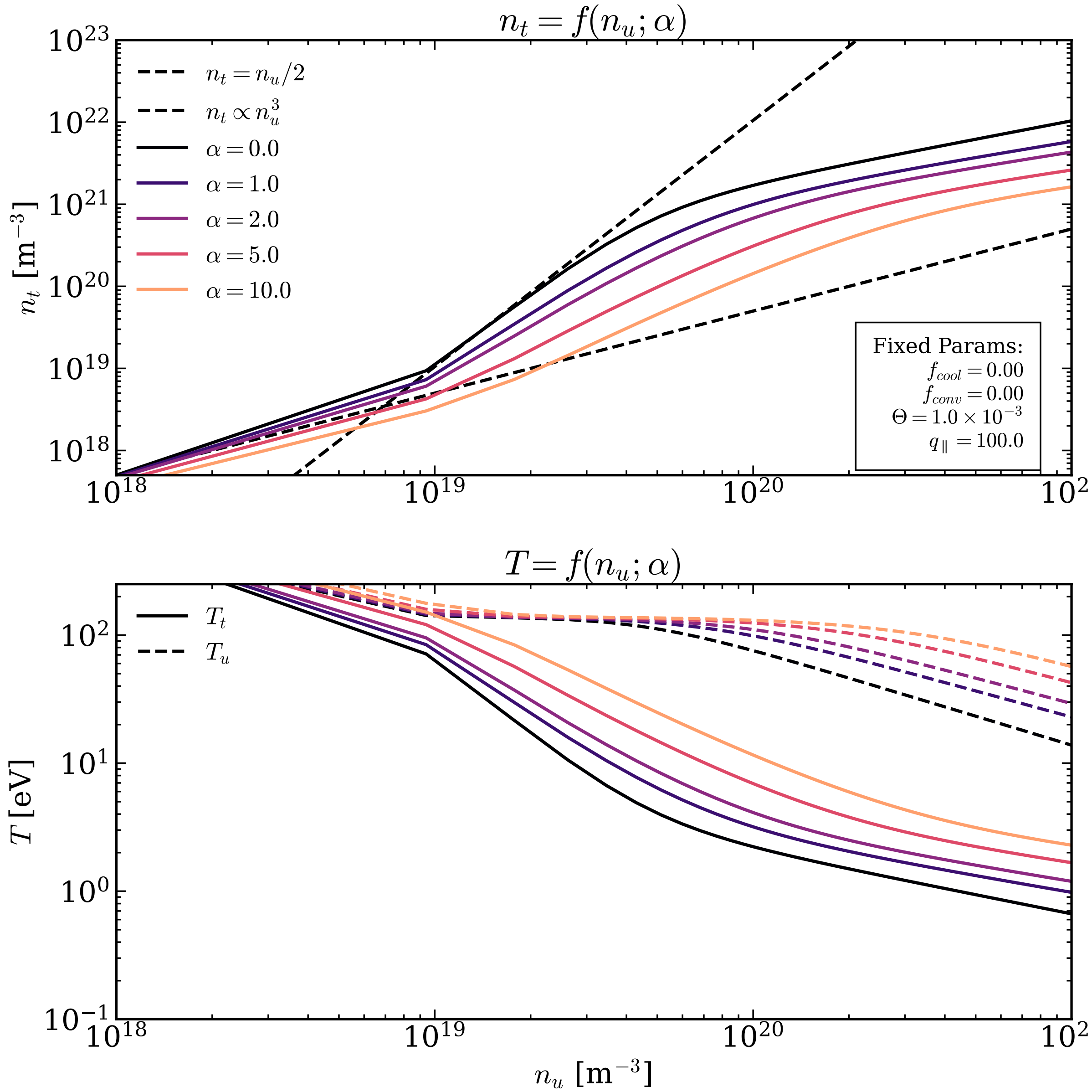

Figure 2.2: Target density n_t (up) and target T_t and upstream T_u temperatures (down) vs upstream density n_u for various momentum loss strength parameters α via the STPM.

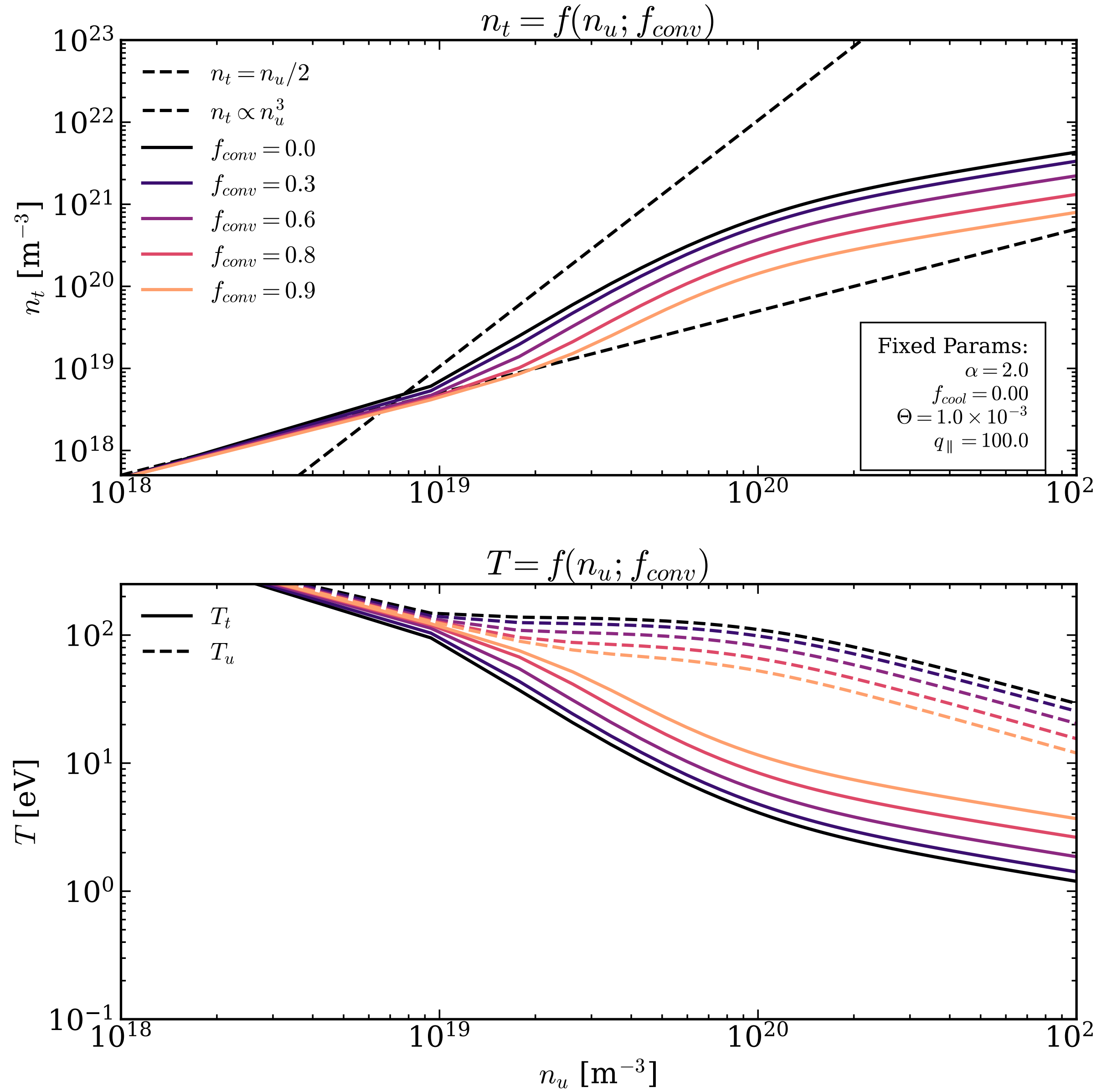

Figure 2.3: Target density n_t (up) and target T_t and upstream T_u temperatures (down) vs upstream density n_u for various convective power fractions f_{conv} via the STPM.

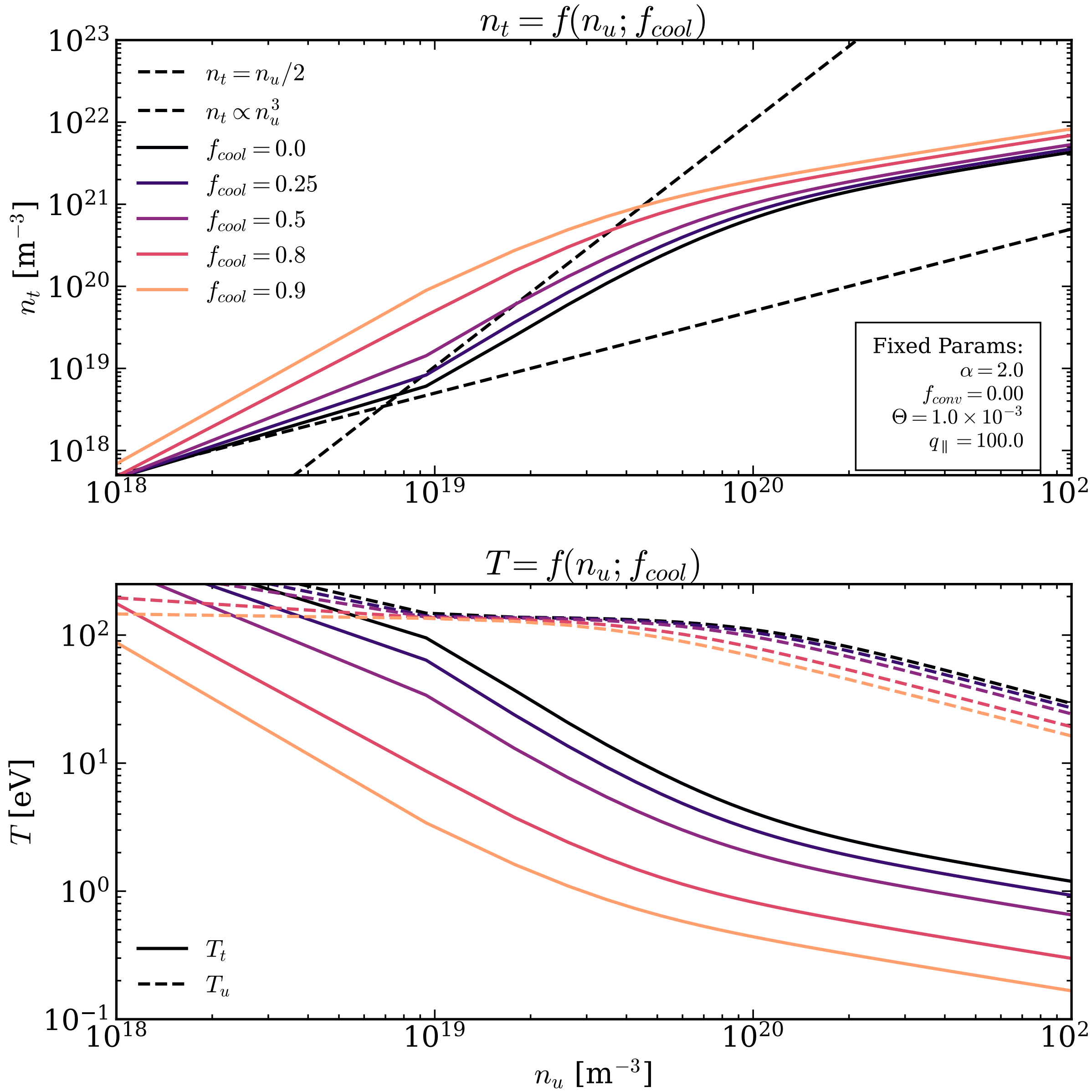

Figure 2.4: Target density n_t (up) and target T_t and upstream T_u temperatures (down) vs upstream density n_u for various target-localized power dissipation fractions f_{cool} via the STPM.

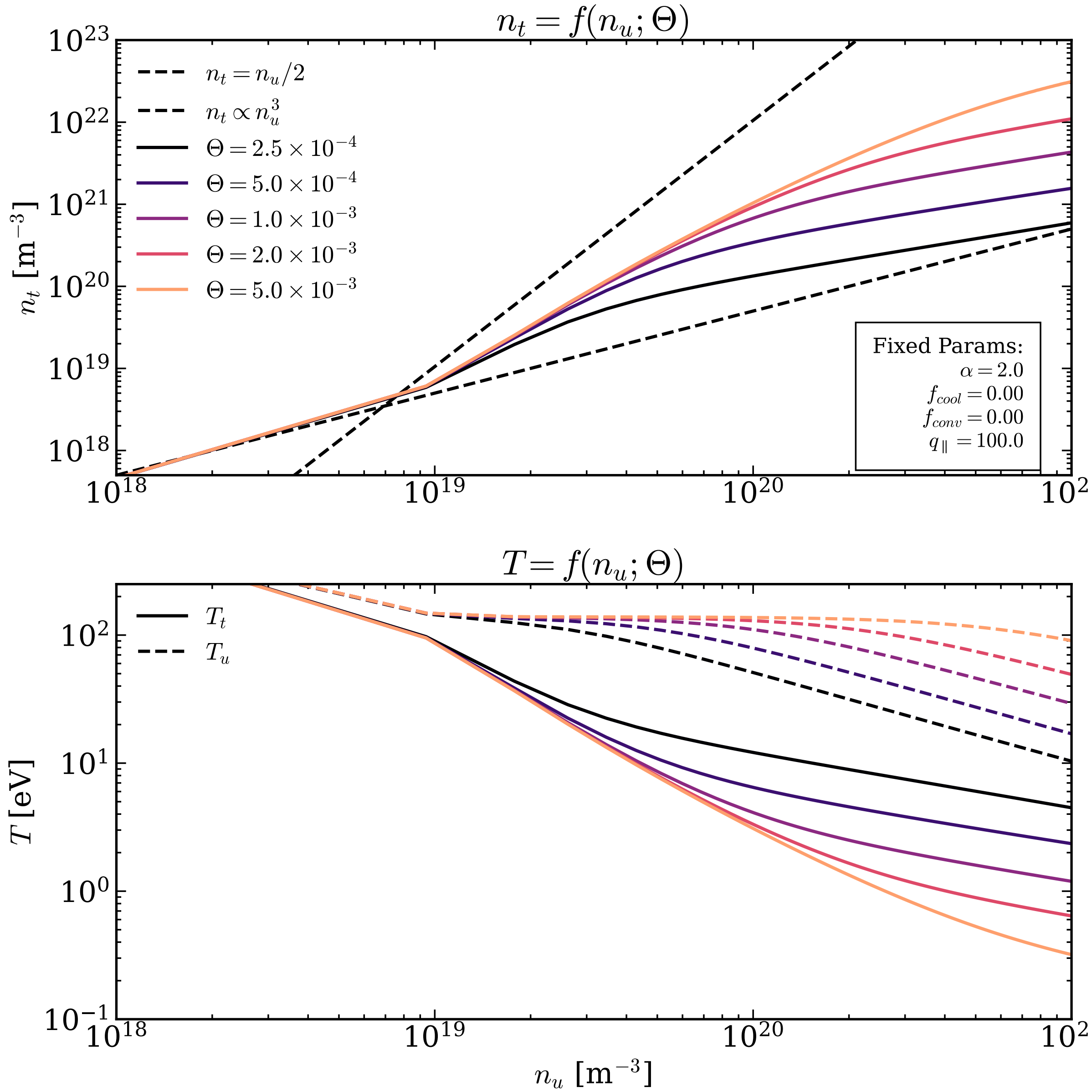

Figure 2.5: Target density n_t (up) and target T_t and upstream T_u temperatures (down) vs upstream density n_u for various theta pitch Θ via the STPM.

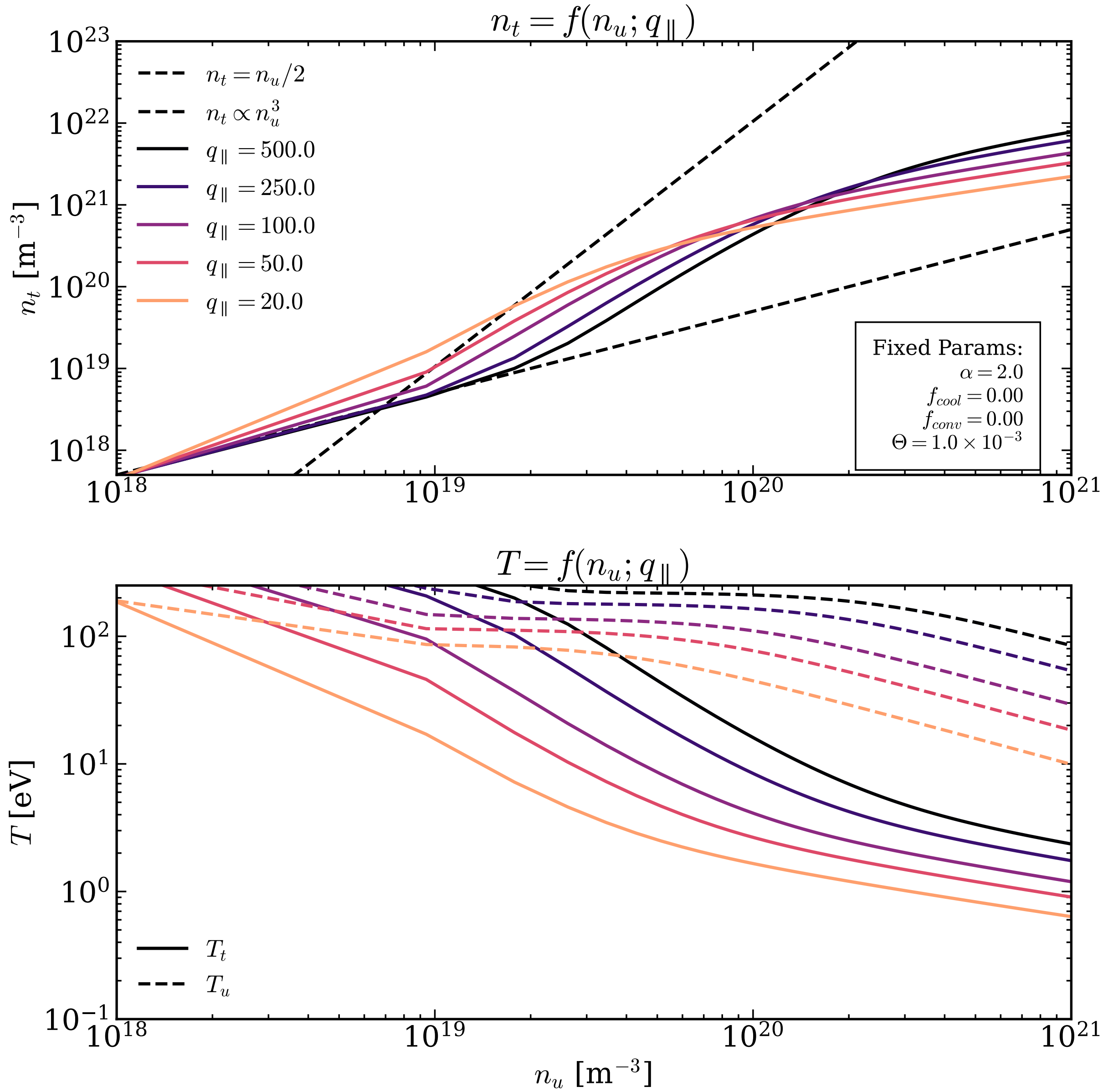

Figure 2.6: Target density n_t *(up) and target* T_t *and upstream* T_u *temperatures (down) vs upstream density* n_u *for various parallel heat fluxes* $q_\parallel$ [Wm^{-2}] *via the STPM.*

2.2.4 Differences between TTPM and STPM, and operational regimes

As it is previously stated, the main difference between the TTPM and STPM lies in the absence or not of the perpendicular heat conduction term. It is shown in Figure 2.1, that in stellarators, especially close to the targets where the temperature drops, the operation point can drop well below the unit fraction lines. This, reflects to the governing equations of the two models (2.21) and (2.28), which differ only in the energy equation. More specifically, the STPM contains the extra term

$$\frac{7}{4}\frac{x_{\perp}(n_u + n_t)}{k_{0,e}\Theta^2}\left(T_u - T_t\right).$$

Generally, in magnetically confined fusion devices three plasma operation regimes have been observed. The sheath-limited regime, the conduction-limited regime and the diffusion-limited regime [5]. The latter one is perceived only in stellarators. The most important distinguishing property that categorizes them, is the existence or not of a significant temperature drop along the SOL.

Sheath-limited regime

When heat conduction is powerful, the plasma becomes essentially isothermal along each flux tube $T_u \approx T_t$. In this regime the sheath-edge is the only important part of the SOL, since it influences the particle and heat transport from the core to the targets. Stangeby [3] proves the governing equations for the sheath-limited regime, using the isothermal assumption. The simplicity of the model allowed the following 1D analytical solutions:

$$\begin{aligned} \frac{dM(x)}{dx} &= \frac{S_p}{n(x)c_s}\frac{1+M^2(x)}{1-M^2(x)}, \quad & V(x) &= -\frac{k_B T_e}{e}\ln(1+M^2(x)) \\ n(x) &= \frac{n_0}{1+M^2(x)} & v(x) &= M(x)c_s \end{aligned} \tag{2.38}$$

Here, $M(x)$ is the local Mach number of plasma and $V(x)$ is the electric potential, while the target plasma density is calculated using the boundary condition $M(L_c) = 1$ as

$$n(L) = \frac{n_0}{2} \Rightarrow n_t = \frac{n_u}{2}. \tag{2.39}$$

These results can be observed in the extended STPM in Figures 2.2-2.5. For low upstream particle densities $n_u < 10^{19}\,[m^{-3}]$, nearly all the target density lines follow the slope $n_t = n_u / 2$, and the upstream temperature is comparable, and sometimes equal, with the target one.

Conduction-limited regime

This regime is characterized by a significant parallel temperature drop along each flux tube $\nabla_{\|} \neq 0$. Now, the edge transport is not only influenced by the sheath, but also from the heat conduction. The conduction-limited regime is the most typical regime of tokamak operation and under its assumptions the TTPM is deduced. This means that the temperatures are given by Eqs. (2.22) and (2.23) as

$$T_u = \left(\frac{7}{2}\frac{q_{\|}L_c}{k_{0,e}}\right)^{2/7}, \quad T_t = \frac{m_i}{2e}\frac{4q_{\|}^{\;2}}{\gamma^2 e^2 n_u^{\;2}}\left(\frac{7}{2}\frac{q_{\|}L_c}{k_{0,e}}\right)^{-4/7},$$

which implies that the upstream temperature stays constant while the target one drops, as n_u increases. The particle density is given by

$$n_t = \frac{n_u^{\;3}}{q_{\|}^{\;2}}\left(\frac{7}{2}\frac{q_{\|}L_c}{k_{0,e}}\right)^{6/7}\frac{\gamma^2 e^3}{4m_i}.$$

Thus, the particle density at the targets is proportional to the cube of upstream one:

$$n_t \propto n_u^{\;3}. \tag{2.40}$$

This rapid increase of the target density can be justified by the fact that under these assumptions, pressure must be conserved ($p \sim nT$). As plasma flows to the targets the temperature drops significantly, thus the particle density must increase drastically in order to keep the product constant. These results are directly visible in Figures 2.2-2.5, since for $n_u > 10^{19}\,[m^{-3}]$, the temperature gradient starts to significantly increase and the target density lines follow the $n_u^{\;3}$ slope. Additionally, the upstream temperature stays relatively constant.

Diffusion-limited regime

Finally, when the reactor operates under conditions of large upstream densities, the target temperature drops below a critical temperature $T_{t,crit}$, and the cross-field transport dominates the edge transport (Figure 2.1). This leads to a milder temperature gradient, with the upstream temperature slightly dropping and the target one flattening out, as upstream density increases. Due to the drop in the upstream temperature as the upstream density increases, the upstream pressure saturates ($p \sim nT$). Thus, the target pressure also saturates and since the target temperature flattens, target density flattens as well.

The corresponding results are also observable in Figures 2.2-2.5 for upstream densities $n_u > 10^{20}$. A mitigation of the temperature gradient is observed due to the decrease of upstream temperature and the saturation of the target one with a significant flattening tendency in the target density.

The characteristics of the operation regimes are summarized in the Table 2.1.

Table 2.1: Characteristics of operation regimes.

Regime	Upstream Temperature Scaling (T_u)	Behavior as n_u increases	Critical fraction (ξ_{crit}) for transition
Sheath-limited	$T_u \simeq T_t$	$n_t \propto \frac{n_u}{2}$	
			1.84
Conduction-limited	$\simeq \left(\frac{7 q_{\parallel} L_c}{2 k_{0,e}}\right)^{2/7}$	$T_u \simeq \text{const}$ $n_t \propto n_u^3$	
			0.16
Diffusion-limited	$\sim \left(\frac{7}{4} \frac{x_{\perp}(n_u + n_t)}{k_{0,e}\Theta^2}(T_t - T_u)\right)^{2/7}$	T_u decreases T_t, n_t saturate	

As previously mentioned, the diffusion-limited regime is observed only in stellarators. One way to observe clearly how each reactor behaves with the increase of upstream density, and where these regimes appear is to compare the solution of the TTPM (2.21) (without the simplification of $T_u^{7/2} - T_t^{7/2} \approx T_u^{7/2}$), the STPM with no loss parameters (2.28), and the STPM with only the momentum losses (2.33). The comparison is performed in Figure 2.7, where the characteristics of each regime, also summarized in Table 2.1, are clearly seen. All three solutions provide similar results for the sheath-limited and the conduction-limited regime, but start to distinguish in the diffusion-limited regime.

More specifically, it is seen that the TTPM solution (orange line) never enters the diffusion-limited regime. The upstream temperature keeps scaling proportional to n_u^3, the upstream temperature stays constant, and the target temperature continues decreasing with the same slope as in conduction-limited regime. Furthermore, it is observed that the presence of non-zero momentum losses (pink line) delays the critical upstream density $n_{u,crit}$ at which the diffusion-limited regime starts. Finally, considering the adverse effects of the upstream temperature, and target density reduction on the confinement inside the separatrix and divertor performance, it is seen that in high upstream densities the stellarators underperform compared to tokamaks. Thus, it is crucial to be able to predict the diffusion-limited regime in order to avoid operating the reactor in this regime.

Next, it is shown that the regime changes are mainly driven by varying the upstream density, and that the critical point that this shift happens is vague. One very logical question is how to effectively predict the regime change for specific control variables. To solve this problem, a scale analysis can be obtained in the most general TPM energy conservation equation, namely the one in the extended STPM [5]:

$$T_u^{7/2} = T_t^{7/2} + \frac{7}{2}\frac{(1-f_{conv})q_{\parallel}L_c}{k_{0,e}} - \frac{7}{4}\frac{x_{\perp}(n_u+n_t)}{k_{0,e}\Theta^2}\left(T_u - T_t\right).$$

As shown in Table 2.1, each regime is characterized by the dominance of one of the three RHS terms of the energy conservation equation. The first term dominates in the sheath-limited regime, the second one in the conduction-limited regime, and the third one in the diffusion-limited regime.

Start by investigating the transition between conduction-limited to diffusion-limited regime. The energy conservation, approximately, becomes

$$T_u^{7/2} \simeq \frac{7}{2}\frac{(1-f_{conv})q_{\parallel}L_c}{k_{0,e}} - \frac{7}{4}\frac{x_{\perp}(n_u+n_t)}{k_{0,e}\Theta^2}\left(T_u - T_t\right), \tag{2.41}$$

since the term $T_t^{7/2}$ is only present in the sheath-limited. Divide Eq. (2.41) with $(7/2)(1-f_{conv})q_{\parallel}L_c \,/\, k_{0,e}$ and define the parameter ξ_{c-d} as the ratio of the third to the second term of the energy equation:

$$\xi_{c-d} \equiv \frac{1}{2}\frac{x_\perp (n_u + n_t)(T_u - T_t)}{(1-f_{conv}) q_\parallel \Theta^2 L_c}. \tag{2.42}$$

The ratio ξ_{c-d} is used as an indicator of the transition from conduction-limited to diffusion-limited regimes. When a critical value $\xi_{c-d,crit}$ is reached, which means that a critical set of $n_{u,crit}, T_{t,crit}$ is defined, then the Eq. (2.42) becomes

$$T_{t,crit} = T_u - \xi_{c-d,crit} \frac{2(1-f_{conv}) q_\parallel \Theta^2 L_c}{x_\perp (n_{u,crit} + n_t)}.$$

One way of estimating the $\xi_{c-d,crit}$ is by substituting (2.42) in (2.41) and then applying an empirical minimum reduction number of T_u, that after this value, the diffusion-limited regime dominates. This leads to

$$\frac{T_u}{T_{u,c}} \simeq (1-\xi_{c-d})^{2/7} \Rightarrow \xi_{c-d} = 1 - \left(\frac{T_u}{T_{u,c}}\right)^{7/2}, \tag{2.43}$$

where $T_{u,c}$ is the conduction-limited expression for the upstream temperature, given in Table 2.1. In [5], the percentage that triggers the diffusion-limited regime is chosen to be 5% reduction of T_u , so $(T_u - T_{u,c}) / T_u = 0.05$. Using (2.43) it is found that

$$\xi_{c-d,crit} \simeq 0.16 .$$

One can easily extend the above analysis made in [5] to estimate the transition from sheath-limited to conduction-limited regimes. The energy equation, in the transition conditions, becomes

$$T_u^{7/2} \simeq T_t^{7/2} + \frac{7}{2}\frac{(1-f_{conv}) q_\parallel L_c}{k_{0,e}}. \tag{2.44}$$

Defining the ratio $\xi_{s-c} \equiv \dfrac{(7/2)(1-f_{conv}) q_\parallel L_c / k_{0,e}}{T_t^{7/2}}$ it is readily deduced that

$$\xi_{s-c} = 1 + \left(\frac{T_u}{T_{u,s}}\right)^{7/2}. \tag{2.45}$$

Then, for a 5% T_u reduction of T_u it is estimated that $\xi_{s-c,crit} \simeq 1.84$.

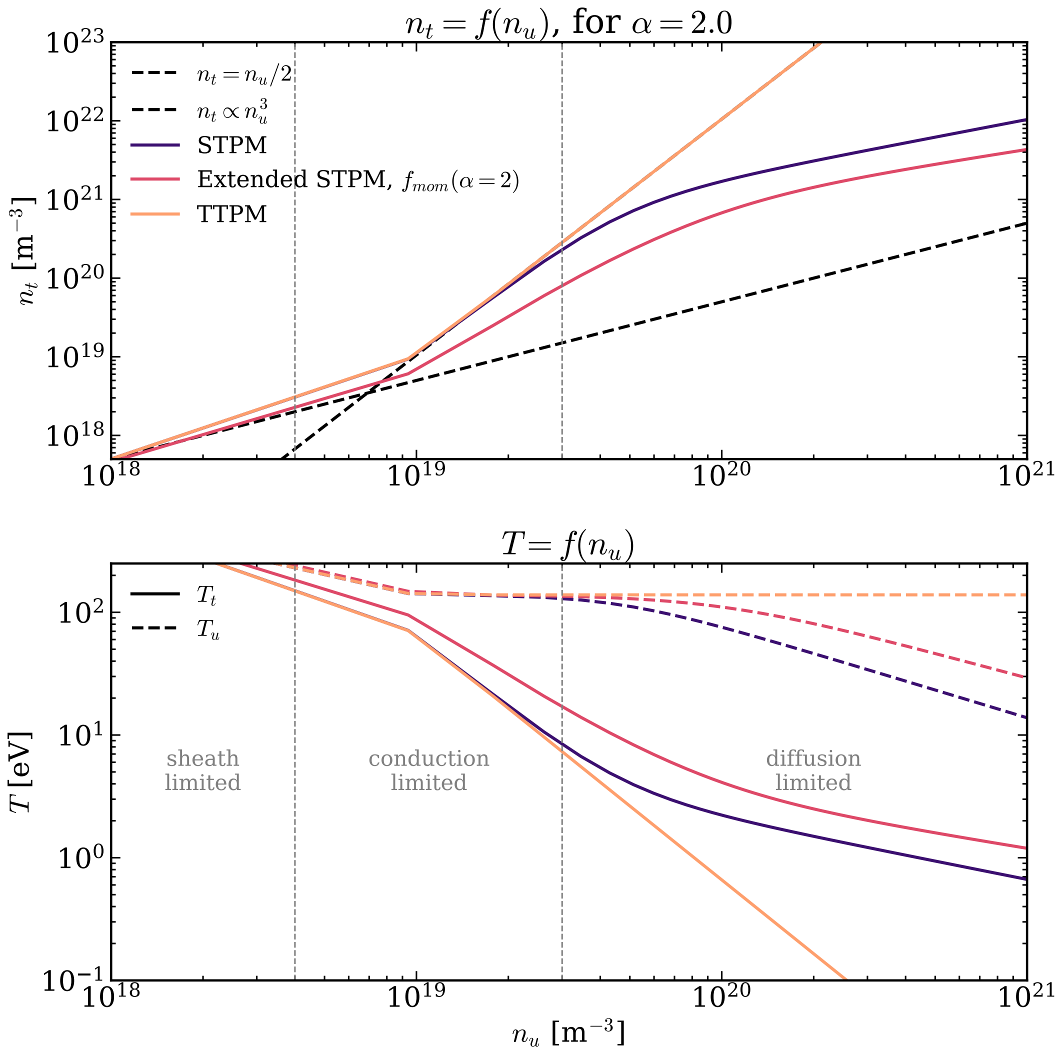

Figure 2.7: Comparison of target density n_t (up) and target T_t and upstream T_u temperatures (down) vs upstream density n_u based on the TTPM, STPM with no losses, and STPM with momentum loss strength parameter $\alpha = 2$ [14].

Closing this chapter some main remarks are stated. Following the formulation of the tokamak and stellarator two-point models some computationally inexpensive parametric results correlating the upstream and target parameters have been presented. Loss parameters have been introduced to account for the neglected physical phenomena neglected in the basic 1D derivation.

While these correction factors manage to quantify some of the losses and improve the predictive capabilities of the model, describing 3D momentum losses through a single free parameter and omitting the mass conservation equation limit its physical completeness [15].

Furthermore, identifying operational regimes and deriving critical points for regime shifts enables the prediction of the reactor's state, which is critical for optimizing overall performance. The connection of the divertor target quantities is still missing and this issue will be the topic of the next chapter.

3 Neutral gas pressure modeling in the divertor region

3.1 Neutral gas flow

Once the ionized particles collide with the divertor targets they are neutralized and are reflected back as neutral particles. One of the most important parameters that drives the pumping efficiency is the pressure of the neutral gas in the divertor and subdivertor chambers. In order to estimate the neutral gas pressure, mathematical models that describe the gas flow through the pumping gaps and inside the subdivertor region have been developed. There are high-fidelity neutral gas dynamics codes for fusion reactors, such as the Divertor Gas Simulator (DIVGAS), that are able to provide accurate results but remain computationally expensive and prohibit parametric investigations, as well as simplified approximated models. This work focuses on the latter ones. More specifically, simplified approaches will be presented for the neutral gas pressure in the divertor region close to the targets and the subdivertor region, as well as for the interconnection between the divertor and subdivertor regions. In the literature, common approaches for simplified closed-form solutions for the neutral gas pressure (p_{sub}) include conservation of mass (or throughput) principles between the divertor and subdivertor chambers ([4], [7], [16], [17]), and regression models ([18], [19]). In the present thesis only conservation-based models are examined.

Consider two control volumes (CVs), as shown in Figure 3.1. The divertor CV is the area directly in front of the targets (see Figure 1.9), while the subdivertor CV contains the area under the targets, i.e. from the pumping gap all the way to the pumping ports. By convention, the pumping gap area, where the particle flux Γ_{in} passes through, belongs to the divertor region.

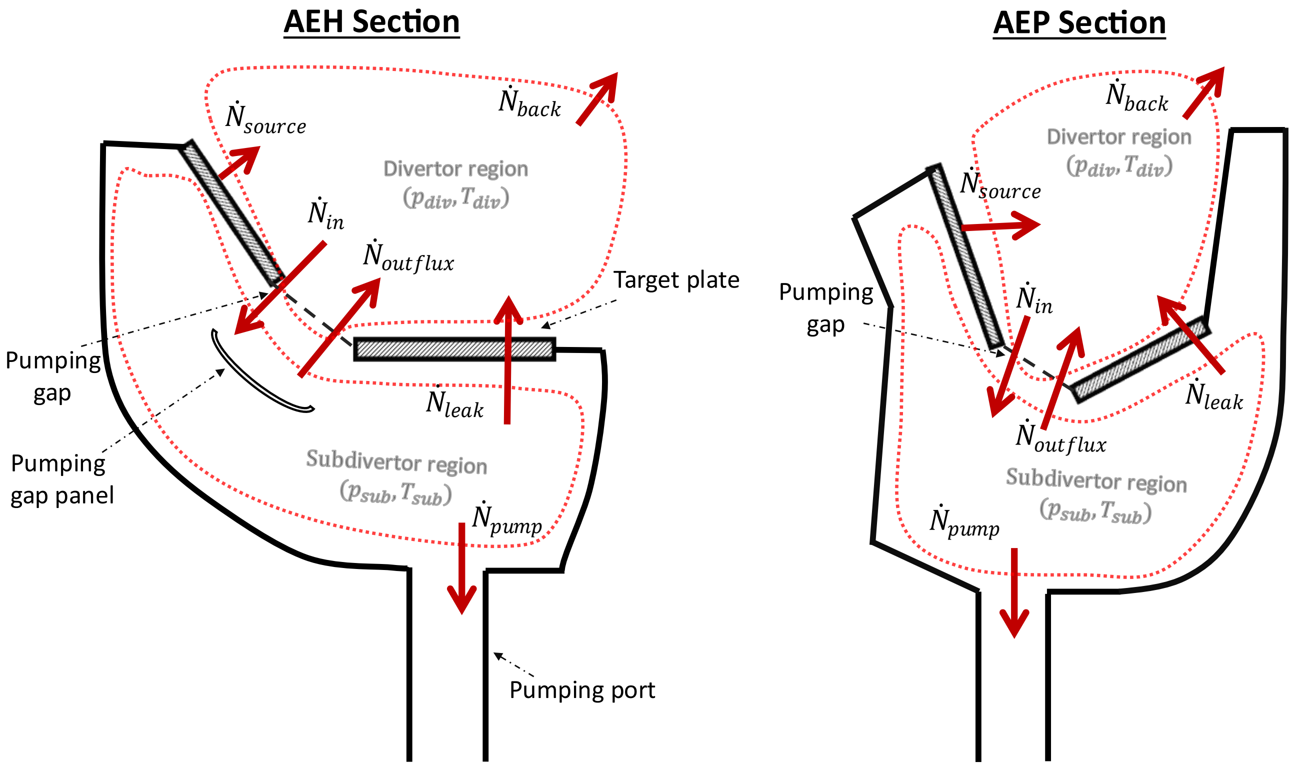

Figure 3.1: Simplified 2D sketch of the AEH (left) and AEP (right) sections, including the target plates, the pumping port, the pumping gap, and the pumping gap panel that exists only in AEH section. The divertor and subdivertor control volumes, and the particle fluxes are shown [8].

Start by assuming that the plasma particle flux $\Gamma_{pl} \equiv \Gamma_t \left[\text{plasma particles}/m^2/s\right]$, originated in the SOL plasma, is reaching the targets region, where it strikes the targets and then is reflected back into the divertor CV. This particle flow, denoted by $\dot{N}_{source}$, is considered as a source term, which is split into the particle flow $\dot{N}_{in}$ going through the pumping gap into the subdivertor and the particle flow $\dot{N}_{back}$ returning back to the main vessel. Then, the particle flow $\dot{N}_{in}$ in the subdivertor CV is either pumped by the turbomolecular pumps ($\dot{N}_{pump}$), or returns to the divertor CV through the pumping gap ($\dot{N}_{outflux}$), or slips through the leakage openings ($\dot{N}_{leak}$) again back to the divertor CV. The particle balance equation for the divertor region is

$$\dot{N}_{source} + \dot{N}_{leak} + \dot{N}_{outflux} = \dot{N}_{in} + \dot{N}_{back}, \tag{3.1}$$

while for the subdivertor region

$$\dot{N}_{in} = \dot{N}_{pump} + \dot{N}_{leak} + \dot{N}_{outflux}. \tag{3.2}$$

Furthermore, it is assumed that each control volume is in homogenous conditions, i.e., operates at its own constant pressure and temperature. The objective is to estimate the divertor pressure

p_{div} and the subdivertor pressure p_{sub}, while the associated temperatures T_{div}, T_{sub} are specified parameters.

The pressure p_{sub} may be related to the particle flow $\dot{N}$ as follows: From kinetic theory of gases, the particle flux for a non-stationary gas (accounting for a bulk velocity) is given by [16],

$$\Gamma = \frac{B}{4} n v_{th} \, [m^{-2} s^{-1}] \; , \tag{3.3}$$

where $v_{th} = \sqrt{\dfrac{8 k_B T}{\pi m}}$ is the average thermal speed. The parameter B is a bulk flow correction factor and is given by

$$B = \exp\left(-s^2 \cos\theta^2\right) + \sqrt{\pi} s \cos\theta \left[1 + \operatorname{erf}(s\cos\theta)\right], \tag{3.4}$$

where the parameters s, θ are taken as $s = 0.5 / \sqrt{\pi}$, and $\theta = 0^\circ$ [16]. The particle flow $\dot{N}$ is obtained by multiplying the particle flux in Eq. (3.3) by some effective area A_{eff} ($\dot{N} = \Gamma A_{eff}$). Then, using the ideal gas law ($p = n k_B T$) it is readily deduced that

$$\boxed{\dot{N} = \frac{B}{4} \frac{p v_{th}}{k_B T} A_{eff}} \; [s^{-1}] \, . \tag{3.5}$$

Equation (3.5) can be also written using the definition of throughput [20]

$$q \equiv \frac{d(pV)}{dt} = \dot{N} k_B T \; \left[Pa \cdot m^3 / s\right], \tag{3.6}$$

in the form

$$\dot{N} = \frac{q}{k_B T} \, . \tag{3.7}$$

Also, using Eqs. (3.6) and (3.5) the throughput may be expressed as a function of pressure as

$$q = \frac{B}{4} p v_{th} A_{eff} \, . \tag{3.8}$$

The above description with the related basic definitions is very useful in the models presented next, in Sections 3.2 and 3.3.

3.2 Subdivertor model

The particle flux Γ_{in} (or the particle flow $\dot{N}_{in}$) is of paramount importance in the computation of the subdivertor pressure p_{sub}. Unfortunately, this quantity is not part of the model and must be provided as given data. Thus, for the estimation of p_{sub} [7], [8], [16], it is commonly assumed that this flux is a known constant, which is either given by high-fidelity codes, like EIRENE, or treated as an input parameter in codes like DIVGAS. Here, again it is assumed that Γ_{in} is known and a simplified subdivertor model that predicts the subdivertor neutral gas pressure is constructed.

It is assumed that all particles that enter the subdivertor (Figure 3.1) have been cooled enough to undergo volumetric recombination; thus, a single-species gas is considered with negligible impurities and the mass balance is equivalent to the particle balance according to Eq. (3.2).

Each term of the balance equation is considered separately starting with the pumping flow $\dot{N}_{pump}$. Based on vacuum gas dynamics, the effective pumping speed [8] generated by the turbomolecular pumps at the cross-section of the ports, by assuming that the distribution function in the subdivertor is Maxwellian and the gas is collisionless, is given by

$$S_{eff} = \frac{1}{4}\xi A\sqrt{\frac{8k_B T_{vv}}{\pi m}}. \tag{3.9}$$

Here, ξ is the capture coefficient, $A = \pi d^2/4$ with $d = 0.4[m]$ is the cross-sectional area and $T_{vv} = 303[K]$ the vacuum vessel temperature. Also, $\xi = 0.06$ for the AEH and $\xi = 0.0264$ for the ASP section [8]. Therefore, the particle flow due to pumping is:

$$\dot{N}_{pump} = \frac{S_{eff}}{k_B T_{sub}} p_{sub}. \tag{3.10}$$

To find the particle flow through the leakages it is assumed that the subdivertor gas slipping through the openings is at temperature T_{sub} and pressure p_{sub}. Thus,

$$\dot{N}_{leak} = \frac{B}{4} A_{leak} \frac{v_{th,sub}}{k_B T_{sub}} p_{sub}. \tag{3.11}$$

The particle outflux $\dot{N}_{outflux}$ is due to particles that manage to pass the pumping gap but fail to enter deep into the subdivertor, so they return back (e.g. they are redirected by the pumping gap panel). For this reason, these particles are modeled as having the subdivertor pressure p_{sub} (since the flow is pressure driven), but retaining the divertor temperature T_{div} (since the heat conductivity of gases is low). Based on these assumptions,

$$\dot{N}_{outflux} = \frac{B}{4} A_{pg} \frac{v_{th,div}}{k_B T_{div}} p_{sub} . \tag{3.12}$$

Substituting Eqs. (3.10)-(3.12) into the balance expression (3.2), the following closed-form expression for the subdivertor pressure is deduced:

$$\boxed{p_{sub} = \frac{\dot{N}_{in}}{\frac{B}{4}\left(A_{leak} \frac{v_{th,sub}}{k_B T_{sub}} + A_{pg} \frac{v_{th,div}}{k_B T_{div}} \right) + \frac{S_{eff}}{k_B T_{sub}}}} \tag{3.13}$$

In Eq. (3.13), p_{sub} is in Pa, $\dot{N}_{in}$ is in s^{-1}, while A_{pg} and A_{leak} are the pumping gap area and leakage openings area, respectively. The values of both areas are given in Table 1.1 for each AEH and AEP port.

3.2.1 Comparison of the subdivertor model with experimental and modeling data

The proposed model for the subdivertor neutral gas pressure is validated by comparisons with corresponding experimental and numerical data available in the literature. In [7], experimental measurements for H_2 gas are provided, for two different magnetic configurations (standard and high iota). These configurations connect the influx for both AEH and AEP ports, with the corresponding divertor and subdivertor pressures. The results are summarized in Table 3.1. Additionally, some models approximating the neutral gas pressure given the particle flow that passes through the pumping gaps are available. Here, two of two of these models, namely the ones by Varoutis et al [18] and Litovoli et al [16] are considered. Overall, the subdivertor neutral gas pressure obtained by the present model as well as by the ones in [16] and [18] are compared with experimental data.

Table 3.1: Experimental data for the subdivertor pressures (working gas: H_2) [7].

Discharge	Section	$\dot{N}_{in}\,[s^{-1}]$	$p_{div}\,[mbar]$	$p_{sub}\,[mbar]$
Standard Configuration (.008)	AEH	3.23×10^{20}	2.6×10^{-4}	1.4×10^{-4}
	AEP	2.66×10^{19}	1.5×10^{-4}	8.2×10^{-5}
High-iota Configuration (.031)	AEH	1.15×10^{20}	3.5×10^{-5}	2.4×10^{-5}
	AEP	2.78×10^{20}	5.9×10^{-4}	4.1×10^{-4}

In [18], Varoutis et al, employed the DIVGAS high-fidelity code to estimate the subdivertor pressure p_{sub} for various values of the particle flow $\dot{N}_{in}$ in the pumping gaps. A nearly linear relation was found between these two quantities for both AEH and AEP ports:

$$\begin{aligned} p_{sub,AEH}^{Var} &= 5\times10^{-25}\dot{N}_{in}^{\;1.0747} \\ p_{sub,AEP}^{Var} &= 4\times10^{-24}\dot{N}_{in}^{\;1.0779} \end{aligned} \tag{3.14}$$

Here, p_{sub} is given in $[Pa]$, and the particle flow is $\dot{N}_{in}\,[s^{-1}]$. The above expressions have been obtained via classical regression methods with a coefficient of determination of $R^2 = 0.999$ and $R^2 = 0.998$, respectively.

In [16], Litovoli et al, developed a throughput conservation model of the subdivertor control volume. The subdivertor region has been divided into "reservoirs", and based on CAD geometric distances, the inter-reservoir conductances have been estimated using a modified Knudsen expressions for all the flow regions [21].

The simplest model contains only two reservoirs, the AEH and AEP sections, and is shown in Figure 3.2. The throughput balance gives the system of equations

$$\begin{aligned} q_{in,AEH} - q_{outflux,AEH} - q_{leakage,AEH} - C_{1,2}\left(p_{AEH} - p_{AEP}\right) - q_{pump,AEH} &= 0 \\ q_{in,AEP} - q_{outflux,AEP} - q_{leakage,AEP} + C_{1,2}\left(p_{AEH} - p_{AEP}\right) - q_{pump,AEP} &= 0 \end{aligned} \tag{3.15}$$

The system is coupled due to the conductance term $C_{1,2}$. In order to compare the particle balance model, derived in the present thesis, with the conductance model in [16], it is needed to neglect the intersection conductance between the AEH and AEP ports and solve for each pressure. Thus, for $C_{1,2} = 0$, the system of equations yields the single equation

$$q_{in} - q_{outflux} - q_{leakage} - q_{pump} = 0 \, , \tag{3.16}$$

for both AEH and AEP.

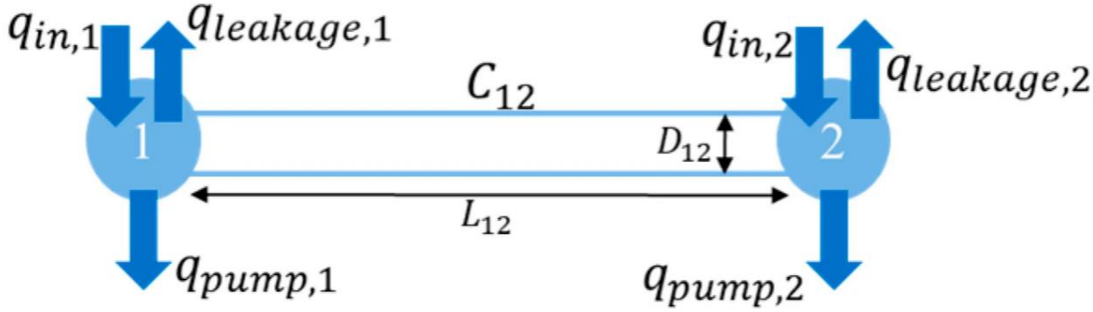

Figure 3.2: Two reservoir model; Vessels 1 and 2 represent the AEH and AEP sections respectively [16].

Substituting Eq. (3.8) for the throughput into Eq. (3.16) and solving the resulting equation for the subdivertor pressure it is readily deduced that

$$p_{sub}^{L-H} = \frac{\dot{N}_{in} k_B T_{div}}{\frac{B}{4}\left(v_{th,div} A_{pg} + v_{th,sub} A_{leak}\right) + S_{eff}} \, . \tag{3.17}$$

All above expressions are in the form $p_{sub} = c_1 \left(\dot{N}_{in}\right)^{c_2}$, with the analytical models (3.13) and (3.17) yielding a perfectly linear relationship between the neutral gas pressure and the incoming particle flow ($c_2 = 1$), while the regression model (3.14) is approximately linear.

Looking at the aforementioned models, one can note that they are completely independent of the magnetic configuration. This happens due to the simplification that the particle flow ($\dot{N}_{in}$) is taken as an externally specified parameter. The particle flow that passes through the pumping gaps, inherently hides information about the configuration, since by changing the magnetic lines, the strike lines may move closer or more far away from the gaps and the incoming particle flux changes as well. Thus, these simplified models hide the important information about how the neutral gas pressure is affected by the magnetic configuration and provide just a first insight view.

In Figure 3.3 a comparison between all three models and the experimental data is provided for $T_{sub} = 300[K]$ and $T_{div} = 600[K]$. It is seen that the subdivertor pressures based on the models are close to the experimental results. Tabulated data are also provided in Table 3.2. In general, all models are able to predict the order of magnitude of the subdivertor pressure.

More precisely, as illustrated in Figure 3.3, the Varoutis et al model [18] scales slightly different than the other two models. This is expected since it is slightly nonlinear, while the Litovoli et al [16] and particle balance models are linear and behave almost identically. Furthermore, from Table 3.2 it is observed that all models manage to predict some discharges with good accuracy but fail in others (e.g., the particle balance model accurately predicts the .008 discharge in the AEH section but not in the AEP section). Considering all these and that the uncertainty of the experimental data is unknown, no conclusive remarks on the validity of the models may be drawn and further investigation is needed once more experimental data are available.

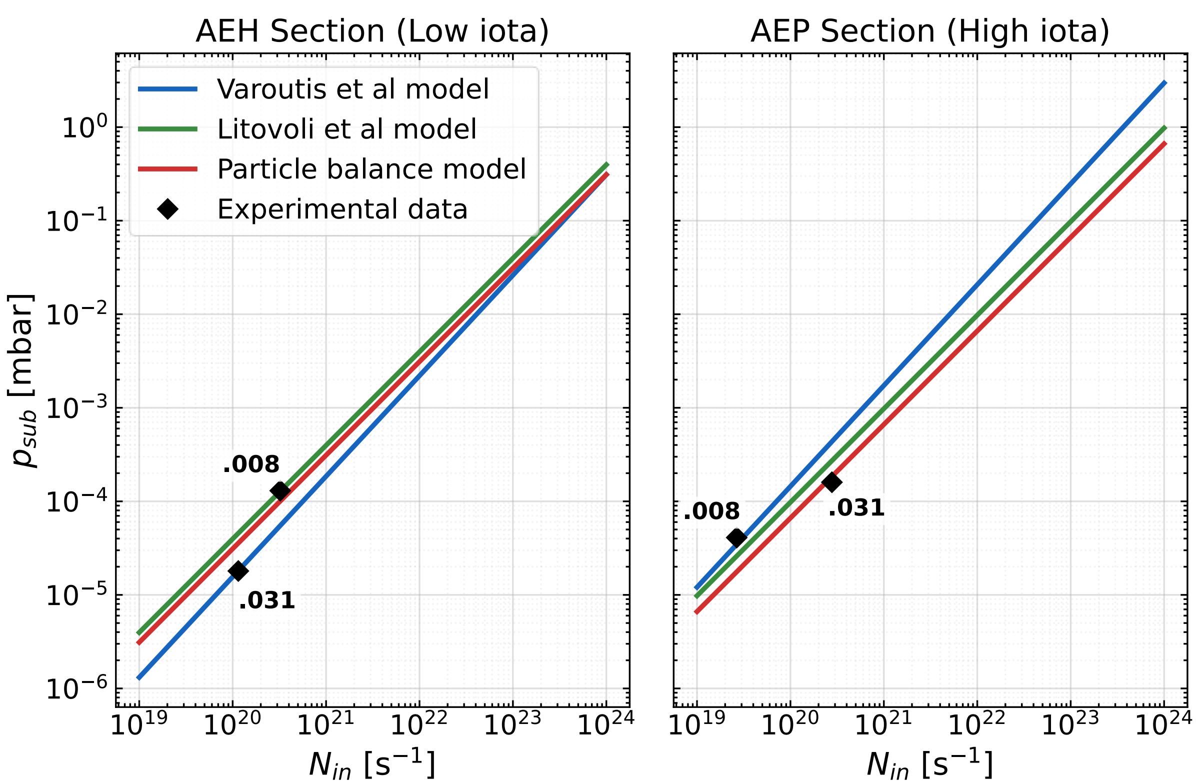

Figure 3.3: Comparison of the subdivertor pressure by the present particle balance model, as well as by Varoutis et al [16] and Litovoli et al [13] with corresponding experimental data [7].

Table 3.2: Experimental data from [7] with the corresponding predictions of the models.

Region	$\dot{N}_{in}\ [s^{-1}]$	Subdivertor Pressure $p_{sub}\ [mbar]$			
		Experimental	Varoutis et al (3.14)	Litovoli et al (3.17)	Particle balance (3.13)
AEH (.008)	3.23×10^{20}	1.40×10^{-4}	5.50×10^{-5}	1.28×10^{-4}	1.00×10^{-4}
AEP (.008)	2.66×10^{19}	8.20×10^{-5}	3.47×10^{-5}	2.60×10^{-5}	1.76×10^{-5}
AEH (.031)	1.15×10^{20}	2.40×10^{-5}	1.81×10^{-5}	4.56×10^{-5}	3.58×10^{-5}
AEP (.031)	2.78×10^{20}	4.10×10^{-4}	4.35×10^{-4}	2.72×10^{-4}	1.84×10^{-4}

It is useful to note that Eqs. (3.17) and (3.13) look almost identical. Such a similarity is expected since they both rely on a mass balance in the subdivertor. However, their significant difference lies on the fact that the model in [16] is based on a throughput conservation, while the present work on particle conservation.

It has been explained why the particle conservation is valid under the assumptions of negligible impurity concentration and single-species gas. On the other hand, the throughput conservation is also valid under the above assumptions, plus the assumption of isothermal flow all the way from the divertor to the subdivertor. This means that $T_{div} = T_{sub}$ must hold. This is an additional, very restrictive constraint, which is not needed and often not a valid one.

In [8], Table 2 provides a detailed list of all the AEP and AEH gaps, along with their corresponding cross-sectional areas and associated toroidal/poloidal leakages. Therefore, once the subdivertor pressure (p_{sub}) is known, it is possible to estimate the particle flows, or throughputs, passing through each of these openings. Specifically, if j is an opening of interest with an area A_j (e.g., the D1 leakage opening in the AEH port [8]), then the particle flow and throughput through this opening can be calculated using Eqs. (3.5) and (3.8), respectively, by setting $A_{eff} = A_j$, $p = p_{sub}$ and $T = T_{sub}$.

This procedure is followed for three different particle inflows, namely $\dot{N}_{in} = \left[10^{20}, 10^{21}, 10^{22}\right] s^{-1}$, and the results for the Litovoli et al model [16] and the present particle balance model, are divided with the corresponding inflow: q_j / q_{in} , or $\dot{N}_j / \dot{N}_{in}$. This provides information on how the particle flow that enters the pumping gaps is distributed in the openings. Finally, the sum of the throughput and particle flow fractions, given by

$$\frac{q_{out} + q_{leak} + q_{pump}}{q_{in}}, \quad \frac{\dot{N}_{out} + \dot{N}_{leak} + \dot{N}_{pump}}{\dot{N}_{in}},$$

is evaluated to examine whether each quantity is conserved. The results are shown in Figure 3.4 and the following interesting observations are stated.

The leak fraction of $\dot{N}_{in}$ at each opening does not depend on the influx $\dot{N}_{in}$. This is due to the linear relationship between the subdivertor pressure and the incoming particle flow in these analytical simplified models. Consider, again, some gap or leakage particle flow $\dot{N}_j$ from some opening j . Since the temperatures are externally defined, it is seen from Eq. (3.5) that $\dot{N}_j \sim p_{sub}$, and therefore the fraction becomes constant:

$$\frac{\dot{N}_j}{\dot{N}_{in}} \sim \frac{p_{sub}}{c_1 p_{sub}} \sim \text{const} .$$

Most importantly, it is seen that the sum of the particle flow fractions in the throughput model [16] is greater than one, which shows that throughput is not conserved. In each case, the exact value of the sum of the fractions is obtained and is shown in Figure 3.4. Suppose that the throughput is conserved:

$$\frac{q_{out} + q_{leak} + q_{pump}}{q_{in}} = 1 . \tag{3.18}$$

Using the equation (3.7), the expression (3.18) can be written as:

$$\frac{\dot{N}_{out} T_{div} + \dot{N}_{leak} T_{sub} + \dot{N}_{pump} T_{sub}}{\dot{N}_{in} T_{div}} = 1 \Rightarrow \frac{\dot{N}_{out}}{\dot{N}_{in}} \left[\frac{T_{div}}{T_{sub}} - 1 \right] + \underbrace{\frac{\dot{N}_{out} + \dot{N}_{leak} + \dot{N}_{pump}}{\dot{N}_{in}}}_{\equiv \delta} = \frac{T_{div}}{T_{sub}} ,$$

and then δ may be written as

$$\delta = \frac{T_{div}}{T_{sub}} + \frac{\dot{N}_{out}}{\dot{N}_{in}}\left[1 - \frac{T_{div}}{T_{sub}}\right]. \tag{3.19}$$

Thus, for the throughput and particle conservation to hold together, δ must equal one. Since $\dot{N}_{out} \neq \dot{N}_{in}$, this is satisfied only if $T_{div} = T_{sub}$. It is also noted that the solution $\dot{N}_{out} = \dot{N}_{in}$ is a contradiction. For $T_{sub} = 300[K]$ and $T_{div} = 600[K]$, the sum of particle flow fractions δ becomes

$$\delta = 2 - \frac{\dot{N}_{out}}{\dot{N}_{in}} = 2 - \frac{\frac{B}{4} A_{pg} v_{th,div}}{\frac{B}{4}\left(A_{pg} v_{th,div} + A_{leak} v_{th,sub} \right) + S_{eff}}. \tag{3.20}$$

The ratio in expression (3.20) follows from Eqs. (3.12) and (3.17). Evaluating the ratio for AEH and AEP data, it is concluded that for the model by Litovoli et al [14]

$$\delta_{AEH} = 1.274\,, \quad \delta_{AEP} = 1.474\,. \tag{3.21}$$

Thus, it is shown that the throughput cannot be conserved unless the control volumes are isothermal.

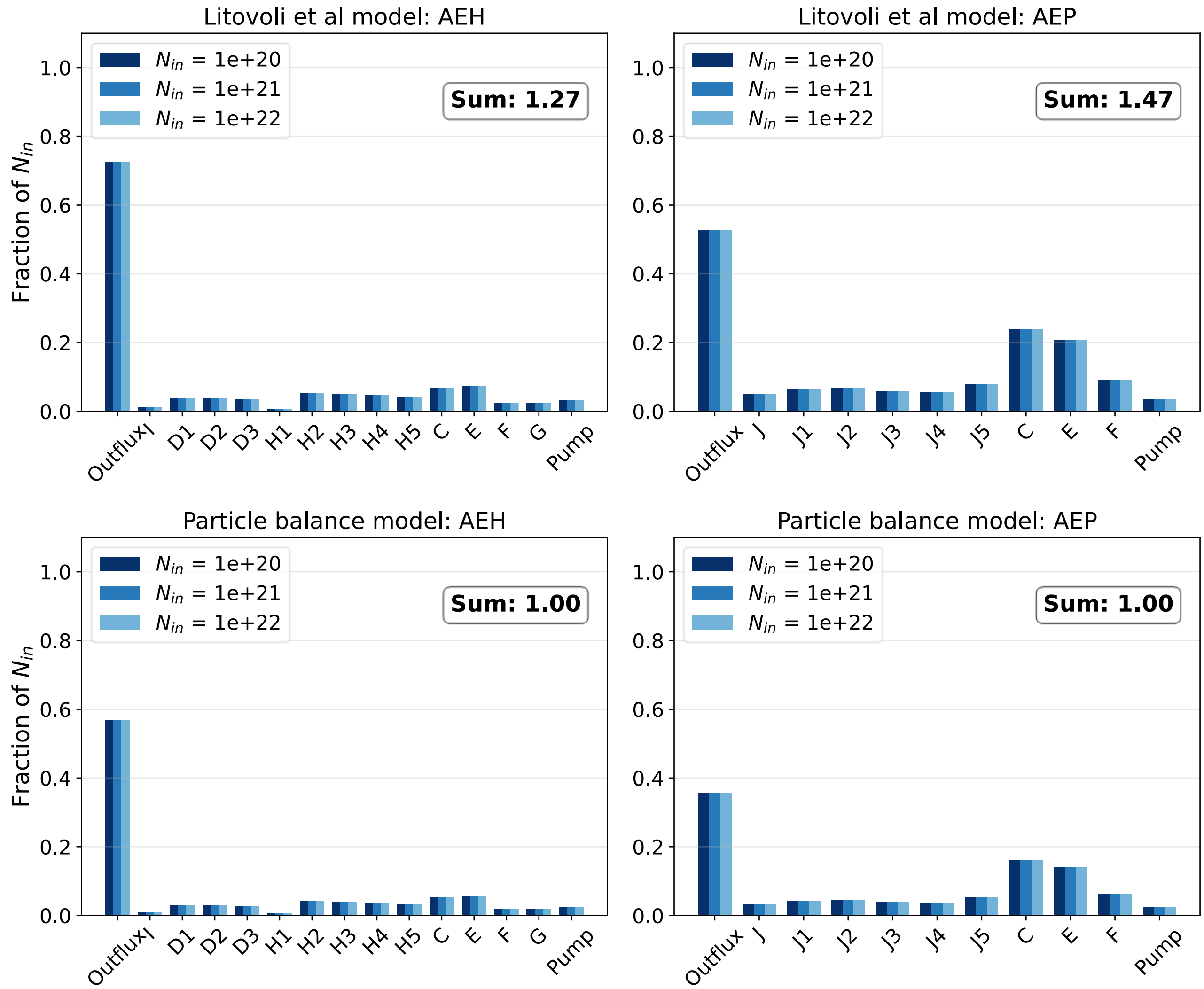

Figure 3.4: Fraction of the influx that passes through the leakages or gaps, for various particle influxes by the present particle balance model and [13]; the summation of all the fractions is shown.

Figure 3.4 is presented in this format to allow comparison with the numerical study conducted in [8], where the DIVGAS code is used for various values of the particle inflow with the corresponding fractions shown in Figure 3.5. It is clearly seen, by comparing the two figures, that the analytical models show qualitative agreement with the numerical ones. In Figure 3.4 the fractions are the same for all values of the incoming particle fluxes, while in Figure 3.5, the percentages are slightly different. This is due to the non-linear relation between the neutral gas pressure and the particle inflow. Additionally, it is seen that the particle balance model qualitatively agrees with the results in Figure 3.5, but predicts a smaller outflux percentage for the AEH port (first bar). This modest discrepancy is expected, since the actual subdivertor geometry is a complex 3D object that cannot be easily described by a simple control volume. Specifically, as seen in Figure 3.1, the AEH port contains the pumping gap panel which is an object placed to protect the subdivertor region from large heat fluxes. The AEP port is not equipped with such a panel since it has a small gap area. This pumping gap panel results in some particles hitting it and reflecting back into the divertor region, thus increasing the outflux. Such interactions are not accounted in the present simplified model.

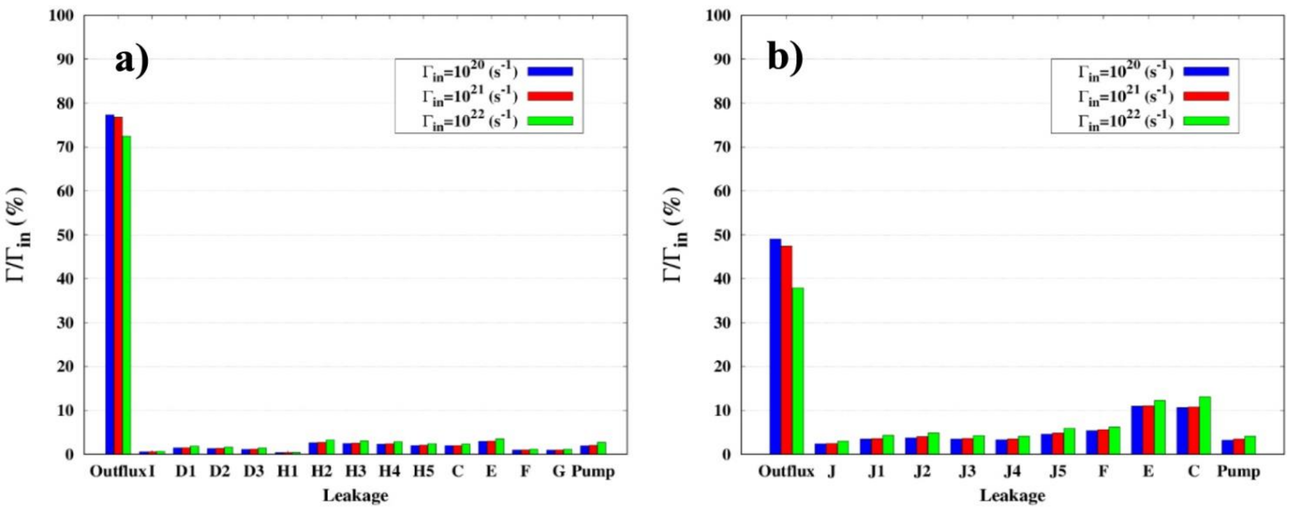

Figure 3.5: Percentage of total influx passing through the openings [8].

3.3 Divertor-subdivertor model

In the subdivertor model derived in Section 3.2, the neutral gas flow passing through the pumping gaps $\dot{N}_{in}$ is assumed to be an externally defined parameter, or should be given as an input obtained from some high-fidelity codes like EIRENE. This approach, however, partially contradicts the main concept of providing integrated simplified semi-analytical solutions that enable computationally inexpensive parametric studies of the SOL and divertor physics. For this reason, in this section a relationship that estimates the neutral gas flux recycling from the targets, given the plasma flux striking the targets, is proposed, allowing for a full divertor-subdivertor simplified model that connects the upstream plasma parameters directly to the divertor and subdivertor pressure.

Start by assuming that all particles directly above the target plates are of the same species and have the same molecular mass. This simplification is made to justify the conservation of particles in the two control volumes. The validity of this assumption is somewhat controversial in the divertor CV, since it implicitly assumes that no plasma particles and no impurity atoms enter or are present in the divertor region.

For the divertor CV and under steady-state conditions, the particle balance is given by Eq. (3.1). In order to express the particle flux $\dot{N}_{back}$ that returns back to the core using Eq. (3.5) an effective area needs to be selected. However, this is not straightforward considering the complexity of the divertor geometry. Instead, an empirical parameter, the *pumping gap particle collection efficiency*, is defined as

$$\varepsilon \equiv \frac{\dot{N}_{in}}{\dot{N}_{source} + \dot{N}_{leak} + \dot{N}_{outflux}} . \tag{3.22}$$

The fraction ε of neutral particles passing through the pumping gaps and entering the subdivertor is measured to be well below 10%. For instance, in [22], this fraction is mentioned to be around 4%.

The subdivertor particle balance is the same with the ones used in Section 3.2 and is given by $\dot{N}_{in} = \dot{N}_{pump} + \dot{N}_{leak} + \dot{N}_{outflux}$ (Eq. (3.2)). Consequently, the system that needs to be solved with respect to the divertor and subdivertor pressures is defined by Eqs. (3.2) and (3.22), provided that the fraction ε is known.

In Section 3.2, the neutral particle flows $\dot{N}_{leak}$, $\dot{N}_{outflux}$, and $\dot{N}_{pump}$ are expressed with respect to the local pressure. Following the same procedure, the influx can be calculated, assuming the gas pressure and temperature in the divertor conditions as

$$\dot{N}_{in} = \frac{B}{4} A_{pg} \frac{v_{th,div}}{k_B T_{div}} p_{div} . \tag{3.23}$$

The incoming SOL plasma is modeled as a source term originating from the targets after the recombination, since strictly attached conditions have been assumed. The task of finding an expression for this flux is complex. In the present work, a simplified approach is proposed.

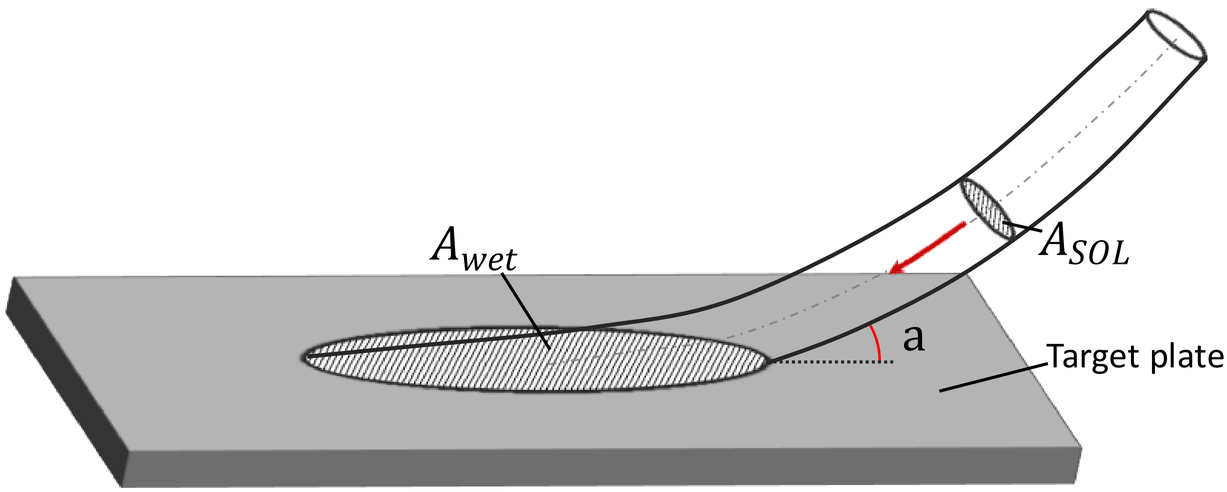

Figure 3.6: Schematic of the strike line carried by SOL, the angle of incidence a and the effective areas A_{SOL} and A_{wet}.

To derive the $\dot{N}_{source}$ term, it is assumed that the plasma operates under <u>attached conditions</u>, and that all the volumetric recombination processes required to turn the incoming plasma particles into neutrals, occur at the targets. Stangeby [3], states that in attached conditions the plasma flux reaching at the sheath edge is approximately equal to the flux striking the walls, so

$$\Gamma_{se} \equiv \Gamma_t \approx \Gamma_{wall} . \tag{3.24}$$

At the wall, all the plasma particles undergo the recombination reaction ($2H^+ \rightarrow H_2$). Some of them return to the divertor CV, while a small fraction, especially before steady-state conditions have been established, is absorbed in the metals lattice [23]. The fraction of particles that recycle, is defined as the *recycling coefficient* [3]:

$$\bar{R} \equiv \frac{\text{particles recycling}}{\text{total particles}} \tag{3.25}$$

Considering that, based on the recombination reaction, for every two plasma particles one neutral particle is formed, the equation

$$\dot{N}_{H_2} = \frac{1}{2}\dot{N}_{H^+}, \tag{3.26}$$

holds true. If $\dot{N}_{H^+}$ represents the particle flow striking the walls that originates from the SOL, while $\dot{N}_{H_2}$ represents the recycled particle flow ($\dot{N}_{H_2} \equiv \dot{N}_{source}$), then combining Eqs. (3.26) and (3.25) yields

$$\dot{N}_{H_2}\Big|_{rec} = \frac{\bar{R}}{2}\dot{N}_{H^+}. \tag{3.27}$$

To express the particle flows $\dot{N}$ as particle fluxes Γ, the effective area through which each flux is transferred needs to be selected (see Figure 3.6). The plasma flux that strikes the targets $\dot{N}_{H^+}$ travels through the SOL, so the effective area is the parallel SOL cross-section $A_{SOL\parallel}$. Thus, it is given by

$$\dot{N}_{H^+} = \Gamma_t\, A_{SOL\parallel} \tag{3.28}$$

The recycling flux $\dot{N}_{H_2}\Big|_{rec}$, originates from the targets and has an effective area equal to the target wetted area A_{wet}. Thus, it is given by

$$\dot{N}_{H_2}\Big|_{rec} = \Gamma_{rec} A_{wet}. \tag{3.29}$$

Substituting Eqs. (3.28) and (3.29) into expression (3.27) yields a closed-form expression for the recycling flux:

$$\Gamma_{rec} = \frac{\bar{R}}{2}\frac{A_{SOL\parallel}}{A_{wet}}\Gamma_t. \tag{3.30}$$

Looking at Figure 3.6, a geometric correlation of the two effective areas can be found using the geometric angle of incidence a, which is the angle of the magnetic field lines striking the wall relative to the target plane and has typical values of $a = 1^\circ \sim 3^\circ$ [24], [25]. Thus, the ratio of the two effective areas is given by the geometric expression

$$\frac{A_{SOL\parallel}}{A_{wet}} = \sin(a). \tag{3.31}$$

The wetted area for the W7-X reactor varies with heating power (P_{ECRH}), but a good approximation is $A_{wet} \approx 1.5[m^2]$, as discussed in [26], [27], and [28]. By substituting Eq. (3.31) into (3.30), the expression for the recycling particle flux is reached:

$$\boxed{\Gamma_{rec} = \frac{\bar{R}}{2}\Gamma_t \sin(a)}. \tag{3.32}$$

Substituting all the expressions for the individual particle flows into the system defined by the Eqs. (3.2) and (3.22), the final 2x2 linear algebraic system for p_{div} and p_{sub} is deduced:

$$\begin{aligned} \frac{B}{4}A_{pg}\frac{v_{th,div}}{k_B T_{div}}p_{div} &= \varepsilon\left[\Gamma_{rec}A_{wet} + \frac{B}{4}A_{leak}\frac{v_{th,sub}}{k_B T_{sub}}p_{sub} + \frac{B}{4}A_{pg}\frac{v_{th,div}}{k_B T_{div}}p_{sub}\right] \\ \frac{B}{4}A_{pg}\frac{v_{th,div}}{k_B T_{div}}p_{div} &= \frac{S_{eff}}{k_B T_{sub}}p_{sub} + \frac{B}{4}A_{leak}\frac{v_{th,sub}}{k_B T_{sub}}p_{sub} + \frac{B}{4}A_{pg}\frac{v_{th,div}}{k_B T_{div}}p_{sub} \end{aligned} \tag{3.33}$$

For simplicity, the constant $c \equiv \dfrac{B}{4}\dfrac{v_{th}}{k_B T}$ is defined and the system (3.33) is rewritten as

$$\underbrace{\begin{bmatrix} A_{pg}c_{div} & -\varepsilon\left(A_{leak}c_{sub} + A_{pg}c_{div}\right) \\ A_{pg}c_{div} & -\dfrac{S_{eff}}{k_B T_{sub}} - \left(A_{leak}c_{sub} + A_{pg}c_{div}\right) \end{bmatrix}}_{\equiv K}\begin{Bmatrix} p_{div} \\ p_{sub} \end{Bmatrix} = \begin{Bmatrix} \varepsilon\,\Gamma_{rec}A_{wet} \\ 0 \end{Bmatrix} \tag{3.34}$$

In order for the system (3.34) to have a unique solution the determinant of the K matrix

$$\det(K) = A_{pg}c_{div}\left[(\varepsilon - 1)\left(A_{leak}c_{sub} + A_{pg}c_{div}\right) - \frac{S_{eff}}{k_B T_{sub}}\right] \tag{3.35}$$

must be different than zero. The determinant is equal to zero when

$$S_{eff} = (\varepsilon - 1)\left(A_{leak}c_{sub} + A_{pg}c_{div}\right)k_B T_{sub}. \tag{3.36}$$

However, since $\varepsilon < 1$ (it represents the efficiency with which particles enter the pumping gap), the right-hand side of Eq. (3.36) is negative, which cannot be true, since the pumping speed $S_{eff} > 0$. As a result, $\det(K) \neq 0$ and the system has a unique solution, written as

$$\begin{Bmatrix} p_{div} \\ p_{sub} \end{Bmatrix} = \frac{1}{\det(K)} \begin{bmatrix} K_{22} & -K_{12} \\ -K_{21} & K_{11} \end{bmatrix} \begin{Bmatrix} \varepsilon \Gamma_{rec} A_{wet} \\ 0 \end{Bmatrix} = \frac{\varepsilon \Gamma_{rec} A_{wet}}{\det(K)} \begin{Bmatrix} K_{22} \\ -K_{21} \end{Bmatrix},$$

where K_{ij} are the elements of matrix K. The final closed form expression for the divertor and subdivertor pressures are:

$$\boxed{p_{div} = \varepsilon \frac{\bar{R}}{2} \Gamma_t \sin(a) A_{wet} \frac{\frac{B}{4}\left(A_{leak} \frac{v_{th,sub}}{k_B T_{sub}} + A_{pg} \frac{v_{th,div}}{k_B T_{div}}\right) + \frac{S_{eff}}{k_B T_{sub}}}{A_{pg} \frac{B}{4} \frac{v_{th,div}}{k_B T_{div}} \left[(1-\varepsilon) \frac{B}{4}\left(A_{leak} \frac{v_{th,sub}}{k_B T_{sub}} + A_{pg} \frac{v_{th,div}}{k_B T_{div}}\right) + \frac{S_{eff}}{k_B T_{sub}} \right]}} \quad ,(3.37)$$

$$\boxed{p_{sub} = \varepsilon \frac{\bar{R}}{2} \Gamma_t \sin(a) A_{wet} \frac{1}{(1-\varepsilon) \frac{B}{4}\left(A_{leak} \frac{v_{th,sub}}{k_B T_{sub}} + A_{pg} \frac{v_{th,div}}{k_B T_{div}}\right) + \frac{S_{eff}}{k_B T_{sub}}}} . \qquad (3.38)$$

The subdivertor and divertor pressures with respect to the upstream plasma density can be obtained by solving the STPM for the target conditions and then evaluating the target plasma particle flux as

$$\Gamma_t = n_t c_{s,t} , \qquad (3.39)$$

where $c_{s,t} = \sqrt{2 k_B T_t / m_i}$ is the plasma sound speed at the targets, and $m_i = 1.67 \times 10^{-27}\,[kg]$ is the ion (H^+ proton) mass. In general, the dependence of the divertor and subdivertor pressures from the upstream particle density is captured in the functionals $p_{div} = g_1(\Gamma_t(n_u))$, and $p_{sub} = g_2(\Gamma_t(n_u))$. Finally, having knowledge of the subdivertor pressure, the particle inflow ($\dot{N}_{in}$) can be evaluated using Eq. (3.23).

Results are provided using the STPM solution discussed in Section 2.2.3 for the fixed control parameters given in Eq. (2.34). Additionally, an inclination angle of $a = 1^\circ$, a recycling coefficient of $\bar{R} = 1$ since steady-state conditions are assumed, and a pumping gap particle collection efficiency of $\varepsilon = 3\%$, is used. The pumping gap particle collection efficiency is selected to be only 1% smaller than the reported value of 4% [22], since this model does not account for the pumping gap panel which may further decrease the total fraction of particles that enter the subdivertor.

Results for the divertor and subdivertor pressures and for the particle incoming flow are shown in Figures 3.7, 3.8, and 3.9. It is readily observed that in all cases the scaling of the divertor pressure p_{div}, the subdivertor pressure p_{sub}, and the particle influx $\dot{N}_{in}$ with the increase of the upstream particle density n_u is characterized by the same shape. Such a functional behavior is expected considering the linear relationship relating them as

$$p_{div} = c_1\Gamma_t(n_u), \quad p_{sub} = c_2\Gamma_t(n_u), \quad \dot{N}_{in} = c_3 p_{sub} = c_3'\,\Gamma_t(n_u). \tag{3.40}$$

Here, c_1 and c_2 are the constants that multiply the plasma particle flux at the targets in Eqs. (3.37) and (3.38), c_3 the constant of Eq. (3.23), while $c_3' = c_2 c_3$. Overall, a simplified model that connects the upstream plasma parameters with the subdivertor neutral gas pressure, has been effectively obtained.

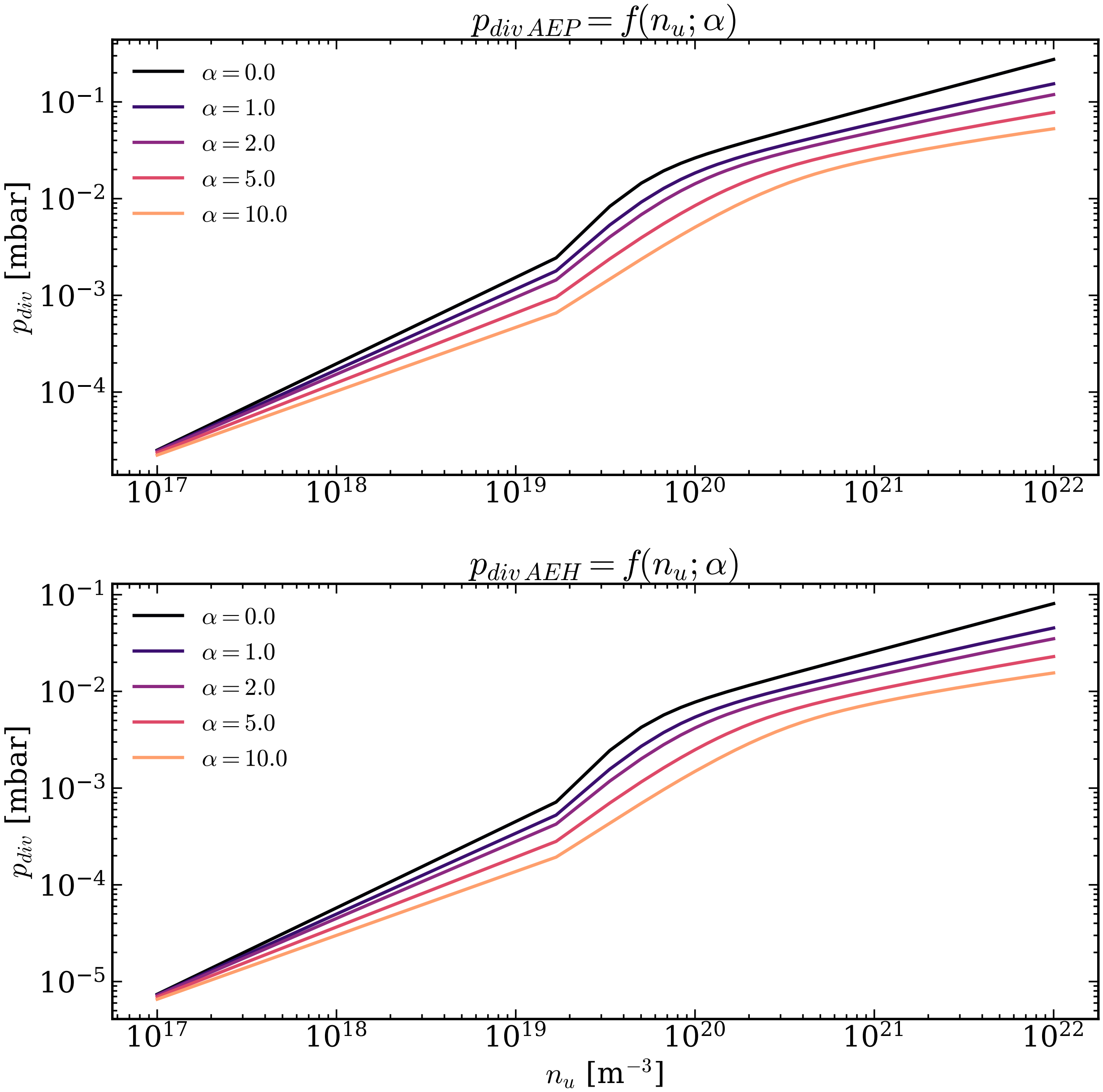

Figure 3.7: Divertor pressure p_{div} *vs upstream particle density* n_u*, for various momentum losses* α *(*$\varepsilon = 3\%$, $a = 1^\circ$, $\bar{R} = 1$*).*

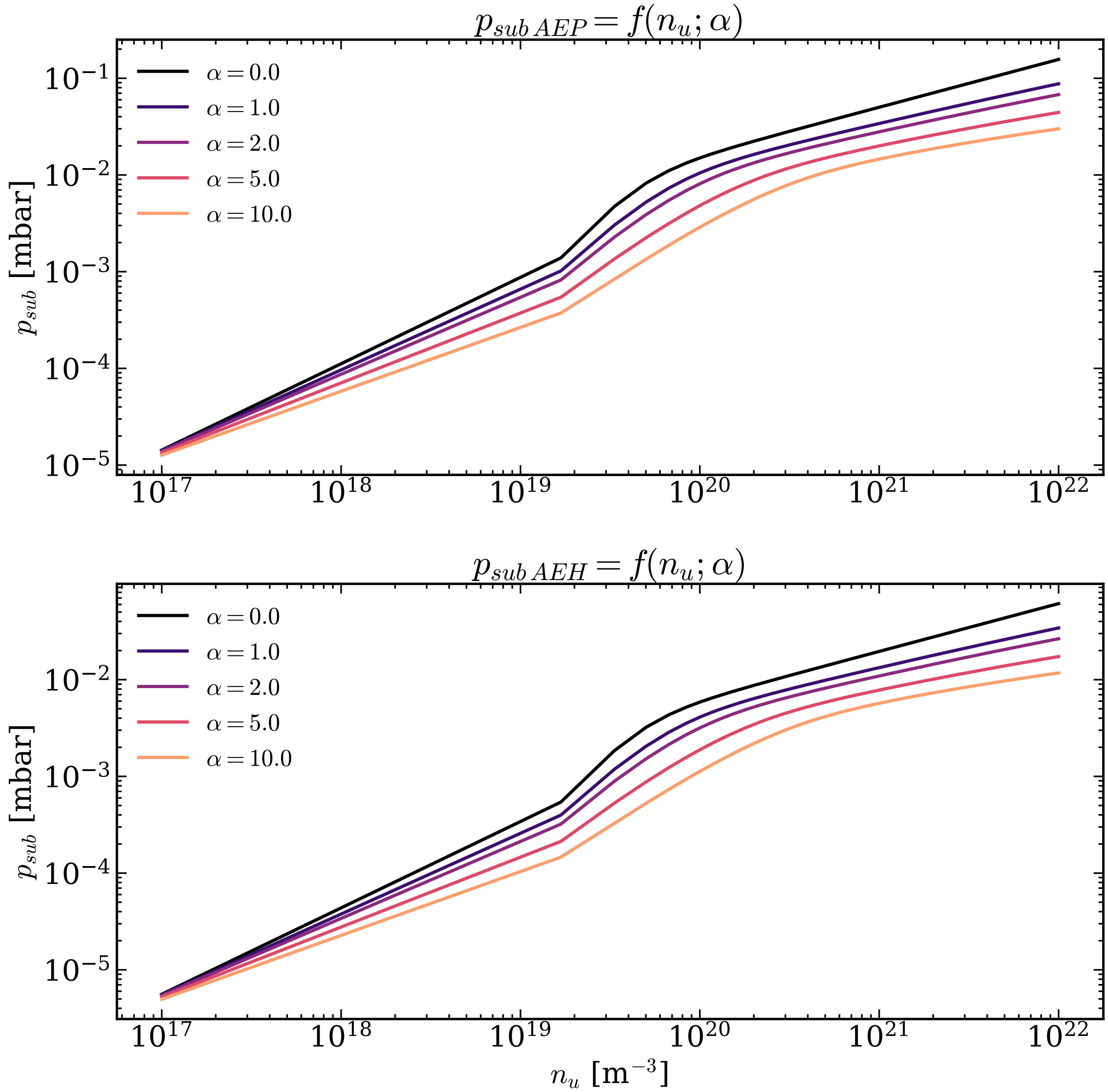

Figure 3.8: Subdivertor pressure p_{sub} vs upstream particle density n_u, for various momentum losses α ($\varepsilon = 3\%$, $a = 1^\circ$, $\bar{R} = 1$).

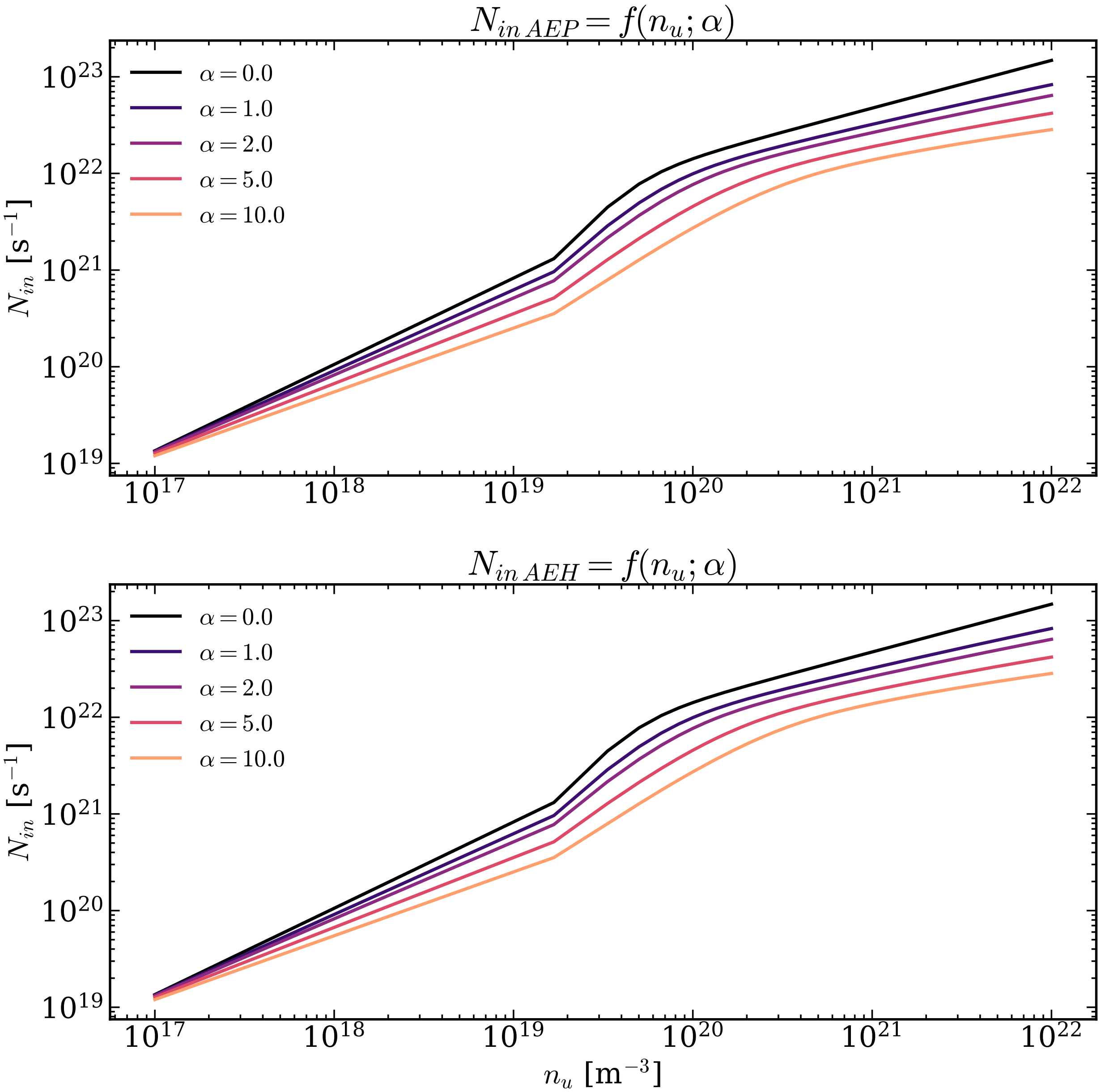

Figure 3.9: Particle influx $\dot{N}_{in}$ vs upstream particle density n_u, for various momentum losses α ($\varepsilon = 3\%$, $a = 1°$, $\bar{R} = 1$).

3.3.1 Comparison of the divertor-subdivertor model with experimental data

As shown in Section 3.3, the divertor-subdivertor particle balance model can correlate the upstream properties of the plasma entering the SOL, with the divertor and subdivertor parameters, as well as to estimate the particle flows through all openings.

Comparison of the divertor and subdivertor predictions of this model with the data provided in [7], and summarized in Table 3.1, is not as straightforward as in the corresponding comparisons of Section 3.2.1. The main reason is that now the quantity $\dot{N}_{in}$ is not specified. More specifically, in the particle balance model presented in Section 3.2, the subdivertor pressure is in the form $p_{sub} = f(\dot{N}_{in})$, allowing a direct comparison with the experimental data. Now, the integrated model is a function of the plasma flux at the targets Γ_t, or more generally of the upstream plasma density n_u. Thus, the particle influx $\dot{N}_{in}$ is no longer an input parameter, but instead it is calculated following the computations of the divertor pressure, via Eq. (3.23).

Therefore, the comparison is performed as follows: The system is solved for various values of upstream plasma density n_u to find the divertor (p_{div}) and subdivertor (p_{sub}) pressures and then, the corresponding particle influx $\dot{N}_{in}$ are estimated. The array of the deduced particle influxes is searched to find the ones which are close to the experimental inputs given in Table 3.1. The divertor and subdivertor pressure solutions that correspond to the selected $\dot{N}_{in}$ array are saved and compared with the experimental data.

Table 3.3 compares the predicted divertor and subdivertor pressures with their corresponding experimental values, using the fixed parameters defined in Eq. (2.34) and applying the methodology described. The model manages to predict the order of magnitude of both the neutral gas pressure (p_{sub}) and the divertor pressure (p_{div}) across all discharges, except for the divertor pressure in the .008 discharge in the AEP section, where an error of approximately one order of magnitude is observed.

Here, some important notes should be made. First, the selection of the parameter set defined in Eq. (2.34) does not affect the predictions shown in Table 3.3. Changing these parameters will change the pressures and the corresponding particle influx. However, the ration of the divertor (p_{div}) and subdivertor (p_{sub}) pressures with the particle influx ($\dot{N}_{in}$) remains constant, which can be justified by evaluating the ratio of the subdivertor pressure to the particle influx, using Eq. (3.40) , as follows:

$$\frac{p_{div}}{\dot{N}_{in}} = \frac{c_1 \Gamma_t(n_u;\alpha)}{c_3' \Gamma_t(n_u;\alpha)} = \frac{c_1}{c_3'} = \text{const}\,, \tag{3.41}$$

and

$$\frac{p_{sub}}{\dot{N}_{in}} = \frac{c_2 \Gamma_t(n_u;\alpha)}{c_3' \Gamma_t(n_u;\alpha)} = \frac{c_2}{c_3'} = \text{const}\,. \tag{3.42}$$

Second, the experimental values for the divertor pressure provided in [7] have been obtained from pressure measurements at the entrance of the subdivertor. Therefore, they are equivalent to the uniform divertor CV pressure (p_{div}) only as a first approximation. Finally, this comparison relies on only two sets of experimental data. Considering the statistical noise and the significant difficulty of obtaining accurate experimental measurements in fusion devices, the model agreement with the data is only an indication of good predictive capabilities, and further investigation is needed.

Table 3.3: Comparison of the full model with the experimental divertor and subdivertor pressures, for $\alpha = 2$ *,* $f_{cool} = f_{conv} = 0$ *,* $\Theta = 10^{-3}$ *, and* $q_{\parallel} = 100\,[MW\,m^{-2}]$*.*

		Divertor Pressure $p_{div}\,[mbar]$		Subdivertor Pressure $p_{sub}\,[mbar]$	
Region	$\dot{N}_{in}\,[s^{-1}]$	Experimental	Predictions	Experimental	Predictions
AEH (.008)	3.23×10^{20}	2.60×10^{-4}	1.77×10^{-4}	1.40×10^{-4}	1.34×10^{-4}
AEP (.008)	2.66×10^{19}	1.50×10^{-4}	4.94×10^{-5}	8.20×10^{-5}	2.81×10^{-5}
AEH (.031)	1.15×10^{20}	3.50×10^{-5}	6.28×10^{-5}	2.40×10^{-5}	4.76×10^{-5}
AEP (.031)	2.78×10^{20}	5.90×10^{-4}	5.17×10^{-4}	4.10×10^{-4}	2.94×10^{-4}

4 Bayesian inference

In the two-point model and its extensions, both in tokamaks and stellarators, several free parameters are introduced, to approximate, as much as possible, the actual complex phenomena taking place. All these parameters are externally defined. However, some of these parameters, such as the momentum loss strength parameter α, the convection fraction f_{conv} and the cooling losses f_{cool}, as well as the pumping gap particle collection efficiency ε, introduced in the present work, are not easily estimated, since they depend on the operating conditions, such as the magnetic configuration, the gas mixture, and the core conditions. For these reasons, here, a statistical analysis is presented that uses experimental data to infer the values of such parameters. This statistical analysis is the well-known Bayesian parameter estimation.

4.1 N-dimensional baseline Maximum a Posteriori (MAP) parameter estimation model

Whether the two-point model, the subdivertor model, or a joint upstream to subdivertor model is used, the mathematical formulation can be written in the general form using the operator $\mathbf{g}:\mathbb{R}^n \rightarrow \mathbb{R}^m$:

$$\hat{\mathbf{Y}} = \mathbf{g}(\mathbf{X};\boldsymbol{\beta}), \tag{4.1}$$

In Eq. (4.1), $\mathbf{X} \in \mathbb{R}^n$ is the input vector, $\hat{\mathbf{Y}} \in \mathbb{R}^m$ the output model's prediction vector, and $\boldsymbol{\beta} \in \mathbb{R}^k$ is the free parameter vector containing the k parameters of interest. For instance, in the STPM $\mathbf{X} = [n_u]$, $\hat{\mathbf{Y}} = [n_t, T_t, T_u]^T$, and $\boldsymbol{\beta} = [a, f_{conv}, f_{cool}]^T$.

Now, suppose that the correlation of the input parameters with the output parameters has been measured from N observations. The set of data is given by:

$$D = \{(\mathbf{X}_1, \mathbf{Y}_1), (\mathbf{X}_2, \mathbf{Y}_2), \ldots, (\mathbf{X}_N, \mathbf{Y}_N)\} \equiv \{(\mathbf{X}_i, \mathbf{Y}_i)\}_{i=1}^{N}, \tag{4.2}$$

where $\mathbf{X}_i$ is the input parameter vector of the measurement and $\mathbf{Y}_i$ is the output vector observed, for $i = 1,\ldots,N$. No error in the input measurements is considered in this work. Additionally, since each measurement originates from separate experimental pulses, the data can be safely assumed to be noncorrelated.

The difference between the prediction of the model $\hat{\mathbf{Y}}_i$ and the experimental measurement $\mathbf{Y}_i$ is defined as the *prediction error:*

$$\mathbf{e}_i = \mathbf{Y}_i - \mathbf{g}(\mathbf{X}_i; \boldsymbol{\beta}), \text{ for } i = 1, \ldots, N. \tag{4.3}$$

Usually, this error is assumed to follow a multivariate Gaussian (normal) distribution

$$\mathbf{e}_i \sim N(\mathbf{0}, \Sigma_{e,i}), \tag{4.4}$$

with zero mean and a covariance matrix of $\Sigma_{e,i} \in \mathbb{R}^{m \times m}$ for the i-th observation. The value of the covariance matrix is usually an unknown, but for this simple analysis it is considered as given.

4.1.1 Bayesian framework and posterior distribution formulation

Bayes theorem is a fundamental theorem of statistics and expresses the idea that knowledge is not fixed, but rather updated when new information becomes available. Using Bayes' theorem [29], [30], the posterior distribution of the free parameters $\boldsymbol{\beta}$ given the set of data D, is given by

$$P(\boldsymbol{\beta} | D) = \frac{P(D | \boldsymbol{\beta}) P(\boldsymbol{\beta})}{P(D)}, \tag{4.5}$$

where $P(\boldsymbol{\beta} | D)$ is the ***posterior*** probability distribution function (PDF) and quantifies how plausible is each possible set of free parameter values, taking into account the information provided by the observations. Also, $P(\boldsymbol{\beta})$ is the ***prior*** probability and expresses the belief for the distribution of parameters $\boldsymbol{\beta}$ before data are utilized and $P(D | \boldsymbol{\beta})$ is the ***likelihood*** probability and provides the probability to observe the data given a possible set of model's free parameter values, while $P(D)$ is the ***evidence*** and acts as normalization term so the posterior integrates to one.

Prior PDF

The prior distribution is externally defined since it represents the current belief for the state of the parameter's distribution. If prior knowledge and data about the typical values expected the free parameters to take exists, then the prior distribution can be assumed to be Gaussian with a mean of the typical values. In the case that no knowledge exists or no value is to be favored, the prior can be assumed to follow the Uniform distribution.

In this analysis all the free parameters are assumed to be normally distributed around typical values so the prior is written as a multivariate Gaussian distribution

$$P(\boldsymbol{\beta}) = N(\boldsymbol{\beta} \mid \boldsymbol{\mu}_\beta, \Sigma_\beta), \tag{4.6}$$

with a mean of $\boldsymbol{\mu}_\beta \in \mathbb{R}^k$ and a covariance matrix $\Sigma_\beta \in \mathbb{R}^{k \times k}$. Thus,

$$P(\boldsymbol{\beta}) = \frac{1}{\sqrt{(2\pi)^k \det(\Sigma_\beta)}} \exp\left\{-\frac{1}{2}\left(\boldsymbol{\beta} - \boldsymbol{\mu}_\beta\right)^T \Sigma_\beta^{-1} \left(\boldsymbol{\beta} - \boldsymbol{\mu}_\beta\right)\right\}. \tag{4.7}$$

Likelihood distribution

The likelihood distribution of the data is written as

$$P(D \mid \boldsymbol{\beta}) = P\left(\{\mathbf{Y}_i\}_{i=1}^N \mid \{\mathbf{X}_i\}_{i=1}^N, \boldsymbol{\beta}\right). \tag{4.8}$$

Using the assumption that the observations are statistically independent, the product rule from probabilities $P(a,b) = P(b \mid a)P(a)$, becomes $P(a,b) = P(b)P(a)$ [31] and thus Eq. (4.8) yields

$$P(D \mid \boldsymbol{\beta}) = \prod_{i=1}^{N} P\left(\mathbf{Y}_i \mid \mathbf{X}_i, \boldsymbol{\beta}\right). \tag{4.9}$$

By the definition of predictive error (4.3), the i-th observation vector is expressed as

$$\mathbf{Y}_i = \mathbf{e}_i + \mathbf{g}(\mathbf{X}_i; \boldsymbol{\beta}). \tag{4.10}$$

Since $\mathbf{g}(\mathbf{X}_i; \boldsymbol{\beta})$ is deterministic, by adding it to the zero-mean Gaussian error $\mathbf{e}_i \sim N(\mathbf{0}, \Sigma_{e,i})$, the mean shifts to $\mathbf{g}(\mathbf{X}_i; \boldsymbol{\beta})$. Therefore, the PDF of the i-th observation is

$$P\left(\mathbf{Y}_i \mid \mathbf{X}_i, \boldsymbol{\beta}\right) = N\left(\mathbf{Y}_i \mid \mathbf{g}(\mathbf{X}_i; \boldsymbol{\beta}), \Sigma_{e,i}\right), \tag{4.11}$$

or written in a more detailed manner

$$P\left(\mathbf{Y}_i \mid \mathbf{X}_i, \boldsymbol{\beta}\right) = \frac{1}{\sqrt{(2\pi)^m \det(\Sigma_{e,i})}} \exp\left\{-\frac{1}{2}\left(\mathbf{Y}_i - \mathbf{g}(\mathbf{X}_i; \boldsymbol{\beta})\right)^T \Sigma_{e,i}^{-1} \left(\mathbf{Y}_i - \mathbf{g}(\mathbf{X}_i; \boldsymbol{\beta})\right)\right\}. \tag{4.12}$$

By substituting Eq. (4.12) into Eq. (4.9) and applying product rules, the likelihood becomes:

$$P(D \mid \boldsymbol{\beta}) = \prod_{i=1}^{N} \left(\frac{1}{\sqrt{(2\pi)^m \det(\Sigma_{e,i})}}\right) \exp\left\{-\frac{1}{2}\sum_{i=1}^{N}\left(\mathbf{Y}_i - \mathbf{g}\left(\mathbf{X}_i; \boldsymbol{\beta}\right)\right)^T \Sigma_{e,i}^{-1} \left(\mathbf{Y}_i - \mathbf{g}\left(\mathbf{X}_i; \boldsymbol{\beta}\right)\right)\right\}. \tag{4.13}$$

Evidence

The evidence distribution is constant with respect to the free parameter vector, acting solely as a normalization term. For this reason, it is usually justified to neglect it and state that the posterior is proportional to the product of the prior and the likelihood. This simplifies the computations required to formulate the posterior PDF, but results in an unnormalized density function.

Posterior PDF

Using Bayes' theorem (4.5) the posterior distribution can be written as

$$P(\boldsymbol{\beta}|D) \propto P(D|\boldsymbol{\beta})P(\boldsymbol{\beta}). \tag{4.14}$$

By substituting Eqs. (4.7) and (4.13), into expression (4.14) the final form of the posterior PDF is

$$P(\boldsymbol{\beta}|D) \propto \lambda \exp\left\{-\frac{1}{2}\sum_{i=1}^{N}\left[\left(\mathbf{Y}_i - \mathbf{g}(\mathbf{X}_i;\boldsymbol{\beta})\right)^T \Sigma_{e,i}^{-1}\left(\mathbf{Y}_i - \mathbf{g}(\mathbf{X}_i;\boldsymbol{\beta})\right)\right] - \frac{1}{2}\left(\boldsymbol{\beta} - \boldsymbol{\mu}_\beta\right)^T \Sigma_\beta^{-1}\left(\boldsymbol{\beta} - \boldsymbol{\mu}_\beta\right)\right\}. \tag{4.15}$$

The function λ has been defined as

$$\lambda\left(\Sigma_\beta, \Sigma_{e,1}, \ldots, \Sigma_{e,N}\right) \equiv \frac{1}{\sqrt{(2\pi)^k \det(\Sigma_\beta)}} \prod_{i=1}^{N}\left(\frac{1}{\sqrt{(2\pi)^m \det(\Sigma_{e,i})}}\right), \tag{4.16}$$

and even though it is assumed constant in this simplified model, it is not dropped just for the case that $\Sigma_{e,i}$ and Σ_β need to be considered as unknowns.

4.1.2 Posterior maximization and MAP

The posterior PDF shows the distribution of the possible free parameter values $\boldsymbol{\beta}$, taking into account the data set. For this reason, the set of values $\boldsymbol{\beta}^{opt}$ that maximizes the posterior is the most probable one. The set $\boldsymbol{\beta}^{opt}$ is called the Maximum A Posteriori (MAP), and is given by the solution of the non-linear constraint optimization problem

$$\begin{aligned} &\max_{\boldsymbol{\beta}} \left\{P(\boldsymbol{\beta}|D)\right\} \\ &\text{s.t } \boldsymbol{\beta}_{\min} \le \boldsymbol{\beta} \le \boldsymbol{\beta}_{\max} \end{aligned} \tag{4.17}$$

with $\boldsymbol{\beta}_{\min}, \boldsymbol{\beta}_{\max} \in \mathbb{R}^k$ representing the vector containing the minimum and maximum allowable values of each free parameter, respectively.

Observing that the posterior is an exponential function and that most of the optimization algorithms work for minimization problems, a common approach is not to maximize the posterior PDF but rather to minimize its negative logarithm. The negative natural logarithm of the posterior is defined as

$$L(\boldsymbol{\beta}) \equiv -\ln\left\{P(\boldsymbol{\beta} \mid D)\right\}, \tag{4.18}$$

or by utilizing Eq. (4.15), is rewritten as

$$L(\boldsymbol{\beta}) = -\ln(\lambda) + \frac{1}{2}\sum_{i=1}^{N}\left[\left(\mathbf{Y}_i - \mathbf{g}(\mathbf{X}_i;\boldsymbol{\beta})\right)^T \Sigma_{e,i}^{-1}\left(\mathbf{Y}_i - \mathbf{g}(\mathbf{X}_i;\boldsymbol{\beta})\right)\right] + \frac{1}{2}\left(\boldsymbol{\beta} - \boldsymbol{\mu}_\beta\right)^T \Sigma_\beta^{-1}\left(\boldsymbol{\beta} - \boldsymbol{\mu}_\beta\right). \tag{4.19}$$

Thus, the final constrained optimization problem is defined as

$$\boxed{\begin{gathered}\min_{\boldsymbol{\beta}} L(\boldsymbol{\beta}) = -\ln(\lambda) + \frac{1}{2}\sum_{i=1}^{N}\left[\left(\mathbf{Y}_i - \mathbf{g}(\mathbf{X}_i;\boldsymbol{\beta})\right)^T \Sigma_{e,i}^{-1}\left(\mathbf{Y}_i - \mathbf{g}(\mathbf{X}_i;\boldsymbol{\beta})\right)\right] + \frac{1}{2}\left(\boldsymbol{\beta} - \boldsymbol{\mu}_\beta\right)^T \Sigma_\beta^{-1}\left(\boldsymbol{\beta} - \boldsymbol{\mu}_\beta\right) \\ \text{s.t } \boldsymbol{\beta}_{\min} \le \boldsymbol{\beta} \le \boldsymbol{\beta}_{\max}\end{gathered}} \tag{4.20}$$

where the most probable loss parameters are given by

$$\begin{gathered}\boldsymbol{\beta}_{opt} = \arg\min_{\boldsymbol{\beta}} L(\boldsymbol{\beta}) \\ \text{s.t } \boldsymbol{\beta}_{\min} \le \boldsymbol{\beta} \le \boldsymbol{\beta}_{\max}\end{gathered} \tag{4.21}$$

4.2 Reduction to the 1-dimensional parameter space

In this section, the simplification of the MAP problem defined in (4.20) is presented, for the case of a single input ($n=1$) and single output ($m=1$) model and observations, and one free parameter of interest ($k=1$).

4.2.1 The 1D MAP problem

By assuming a 1D free parameter vector, $\boldsymbol{\beta} \equiv \beta \in \mathbb{R}$, the minimization problem of the negative log-posterior distribution is significantly simplified. First, the prior distribution is characterized by a mean of $\boldsymbol{\mu}_\beta \equiv \mu_\beta$, while the covariance matrix reduces to a single variance variable $\Sigma_\beta \equiv {\sigma_\beta}^2$, which is treated as an externally defined known parameter of the problem.

Further simplifications can be obtained if the predictive model and the corresponding dataset are restricted to a single variable input and single variable output. Suppose that measurements relating an output parameter $Y \in \mathbb{R}$ to an input parameter $X \in \mathbb{R}$ exist. If the mathematical model linking them is defined as $g : \mathbb{R} \to \mathbb{R}$, then

$$Y = g(X;\beta). \tag{4.22}$$

Similar to the free parameter vector, the covariance matrix of each prediction error (4.4) yields

$$\Sigma_{e,i} = \left[{\sigma_{e,i}}^2 \right] \tag{4.23}$$

where ${\sigma_{e,i}}^2$ represents the independent variance of the i-th observation. Thus, the negative log-posterior distribution becomes

$$L(\beta) = -\ln(\lambda) + \sum_{i=1}^{N} \frac{1}{2{\sigma_{e,i}}^2} \left(Y_i - g(X_i;\beta) \right)^2 + \frac{1}{2{\sigma_\beta}^2} \left(\beta - \mu_\beta \right)^2, \tag{4.24}$$

with the λ function reduced to

$$\lambda\left(\sigma_\beta, \sigma_{e,1}, \ldots, \sigma_{e,N}\right) = \frac{1}{\sqrt{2\pi}\sigma_\beta} \prod_{i=1}^{N} \frac{1}{\sqrt{2\pi}\sigma_{e,i}}. \tag{4.25}$$

As a result of the aforementioned simplifications, the MAP estimation of a single free parameter β, given N observations $D = \left\{ \left(X_i, Y_i \right) \right\}_{i=1}^{N}$, is given by the optimization problem

$$\beta_{opt} = \underset{\beta}{\arg\min}\left[-\ln(\lambda) + \sum_{i=1}^{N}\frac{1}{2{\sigma_{e,i}}^2}\left(Y_i - g\left(X_i;\beta\right)\right)^2 + \frac{1}{2{\sigma_\beta}^2}\left(\beta-\mu_\beta\right)^2\right] \quad (4.26)$$
$$\text{s.t } \beta_{\min} \le \beta \le \beta_{\max}$$

4.3 Indicative test cases in tokamaks

For illustration purposes, a simple numerical example of the 1D MAP problem is presented. The Stangeby Extended TTPM [3] is considered as the predictive model and more specifically the analytical expression relating the upstream density with the target density under momentum loss conditions is selected:

$$n_t = f_{mom}^3 \frac{{n_u}^3}{{q_\parallel}^2}\left(\frac{7}{2}\frac{q_\parallel L_c}{k_{0,e}}\right)^{6/7}\frac{\gamma^2 e^3}{4m_i} \quad (4.27)$$

Here, n_t is in $[m^{-3}]$ and represents the model prediction for the target plasma density, while f_{mom} is the momentum loss parameter, scaling from 0 (100% momentum losses) to 1 (negligible losses). Considering the extremely large numerical values that the particle densities attain ($\sim 10^{19}$), for numerical purposes, it is useful to nondimensionalize the upstream and target densities with a parameter of $10^{19}\,[m^{-3}]$, as follows:

$$\tilde{n}_t = \frac{n_t}{10^{19}}, \quad \tilde{n}_u = \frac{n_u}{10^{19}}. \quad (4.28)$$

Therefore, using expressions (4.28), the predictive model (4.27) yields

$$\tilde{n}_t \equiv g\left(\tilde{n}_u; f_{mom}\right) = f_{mom}^3\, {\tilde{n}_u}^3\, \frac{10^{38}}{{q_\parallel}^2}\left(\frac{7}{2}\frac{q_\parallel L_c}{k_{0,e}}\right)^{6/7}\frac{\gamma^2 e^3}{4m_i}. \quad (4.29)$$

To generate a realistic pseudo-dataset D of N observations, the model prediction (4.29) is evaluated for N upstream nondimensionalized densities ${\tilde{n}_u}^{(i)}$, setting a specific value $f_{mom} = f_0$ for the momentum losses, and adding Gaussian noise $\varepsilon_i \sim N(0, {\sigma_{\varepsilon,i}}^2)$ to each output, as follows:

$$y_i = g\left(\tilde{n}_{u,i}; f_0\right) + \varepsilon_i\,,$$

or

$$y_i = f_0^3 \, \tilde{n}_{u,i}{}^3 \, \frac{10^{38}}{q_{\|}{}^2} \left(\frac{7}{2} \frac{q_{\|} L_c}{k_{0,e}} \right)^{6/7} \frac{\gamma^2 e^3}{4 m_i} + \varepsilon_i \ , \ \text{for } i = 1, \ldots, N \, . \tag{4.30}$$

Given a single input and single output model with only one free parameter, the 1D Maximum a Posteriori problem is to be solved. The optimal momentum loss parameter given the dataset D is determined by solving

$$\begin{gathered} f_{mom,opt} = \arg\min_{f_{mom}} \left[\sum_{i=1}^{N} \frac{1}{2\sigma_{e,i}{}^2} \left(y_i - g\left(\tilde{n}_{u,i}; f_{mom} \right) \right)^2 + \frac{1}{2\sigma_f{}^2} \left(f_{mom} - \mu_f \right)^2 \right] \\ \text{s.t } \ 0 \le f_{mom} \le 1 \end{gathered} \tag{4.31}$$

where μ_f and $\sigma_f{}^2$ are the mean and variance of the momentum loss prior distribution, respectively. Additionally, the $\ln(\lambda)$ term has been dropped from the optimization problem, since the prediction errors $\sigma_{e,i}{}^2$ and the prior variance $\sigma_\beta{}^2$ are considered as known parameters.

For this simple illustrative example, the standard deviations of the prediction errors $\sigma_{e,i}$ and the noise added to the data-generation process $\sigma_{\varepsilon,i}$, are assumed to be equal. Their values are estimated by multiplying the model's prediction (4.29), using the real value of the momentum loss $f_{mom} = f_0$, with a constant noise percentage Z_{noise}:

$$\sigma_{e,i} = \sigma_{\varepsilon,i} = Z_{noise} \, g\left(\tilde{n}_{u,i}; f_0 \right) . \tag{4.32}$$

The following numerical test cases use a tokamak connection length $L_c = 1m$, proton mass, parallel flux $q_{\|} = 50 \, MW \, m^{-2}$, and N nondimensional upstream densities linearly spaced between the values of 1 and 4×10. Finally, the dataset is generated using a momentum loss factor $f_0 = 0.5$. For the rest of the parameters, two cases are examined as summarized in Table 4.1. Case A represents the optimal scenario, while Case B is a worst-case scenario assuming both untrustworthy data and a wrong prior belief for the momentum loss factor.

Table 4.1: Parameters used for test Cases A and B.

Case	Number of data N	Noise percentage Z_{noise}	Prior mean μ_f	Prior variance σ_f
A	20	10%	0.54	0.05
B	4	80%	0.80	0.05

Case A

In test Case A, the optimal conditions are considered. A robust dataset of 20 points with a relatively low noise margin of 10% yields a very strong likelihood. Additionally, the prior distribution is highly informed, as its mean of 0.54 is very close to the actual value $f_0 = 0.5$ and its small standard deviation of 0.05 demonstrates high certainty. Solving the MAP optimization problem (4.31) yields an optimal parameter estimate of $f_{mom}^{opt} = 0.499$, which is extremely accurate, as expected. The density build-up for the optimal value of 0.499 is calculated using the nondimensionalized Eq. (4.29). The predicted target density, plotted against the dataset D used for the Bayesian optimization problem, is shown in Figure 4.1, demonstrating an excellent agreement.

To evaluate the uncertainty around this prediction, the spread of the posterior PDF around its mean needs to be considered. The posterior distribution in known only up to a constant, thus the Markov Chain Monte Carlo (MCMC) algorithm [31] may be used to produce samples. The resulting histogram of the posterior distribution samples, calculated using the “emcee” Python library designed for Bayesian parameter estimation [32], is presented in Figure 4.2. Furthermore, the prior distribution, the MAP estimation, and the true value of momentum losses are also shown. Figure 4.2 clearly shows that the model is highly confident regarding the optimal prediction $f_{mom}^{opt} = 0.499$, since the spread of the posterior is extremely small. Additionally, as already mentioned, the MAP prediction (black dashed line) is almost identical to the true momentum losses (red line). Consequently, the highly confident prior close to the real value, coupled with the high-quality large dataset, makes the parameter inference accurate.

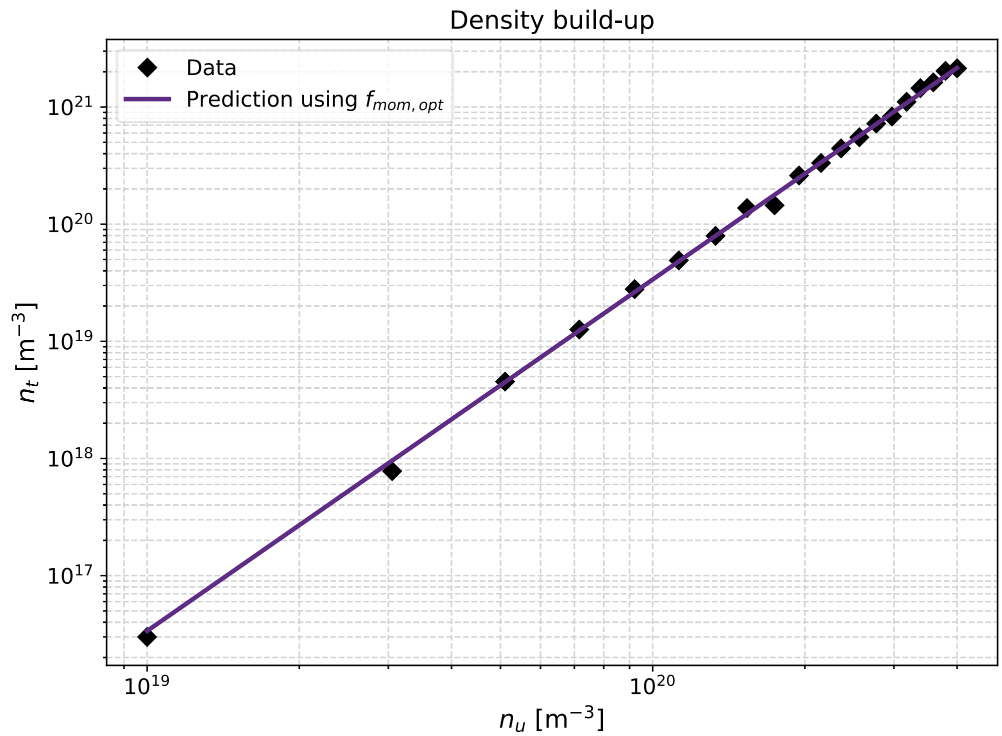

Figure 4.1: Predicted target density n_t, evaluated for the most probable momentum loss value of test Case A, against the upstream density n_u.

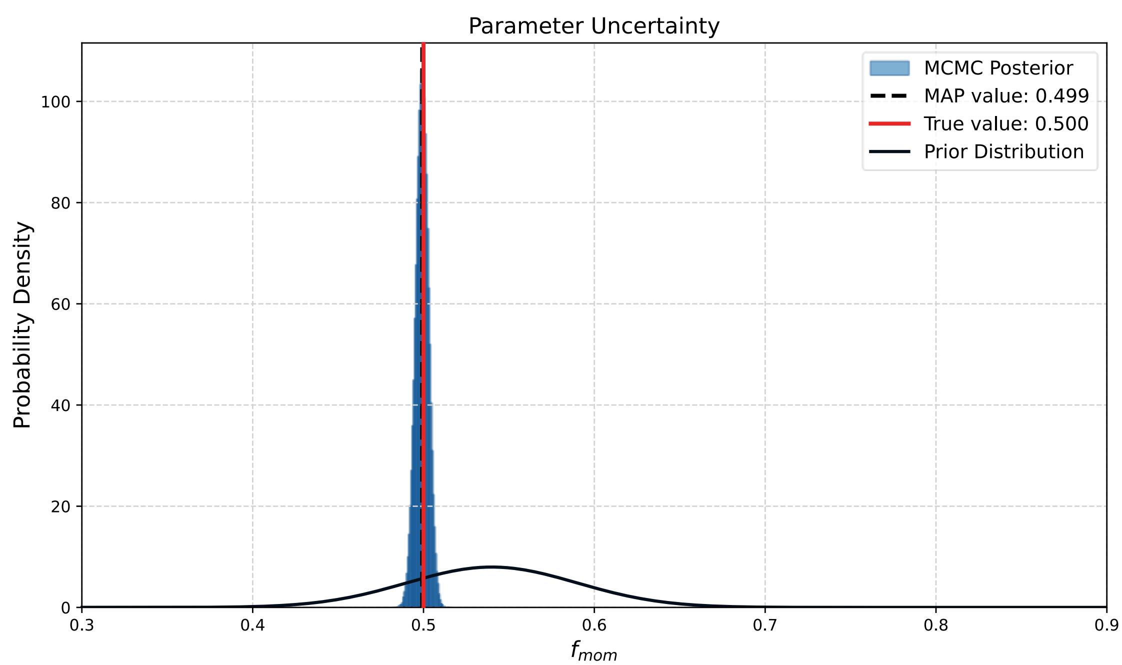

Figure 4.2: The posterior distribution histogram of test Case A with its maximum (MAP) value in dashed black. The real value of the momentum losses is in red, while the posterior PDF in black solid.

Case B

In test Case B, a set of non-optimal conditions is considered. More specifically, the dataset is chosen to be extremely small, consisting of only four data points, while exhibiting an extremely high uncertainty of 80%, which is common in fusion experimental data. Moreover, the prior belief for the distribution of the momentum loss factor is set to be highly inaccurate, with a mean of 0.8 and a standard deviation of 0.05. The prediction for the momentum losses, based on the dataset, is $f_{mom}^{opt} = 0.645$, which is obviously inaccurate compared to the real value of 0.5.

Following the same procedure as in Case A, the target density is calculated for the most probable value of the momentum losses and is plotted in Figure 4.3 and compared to the four observed data points.

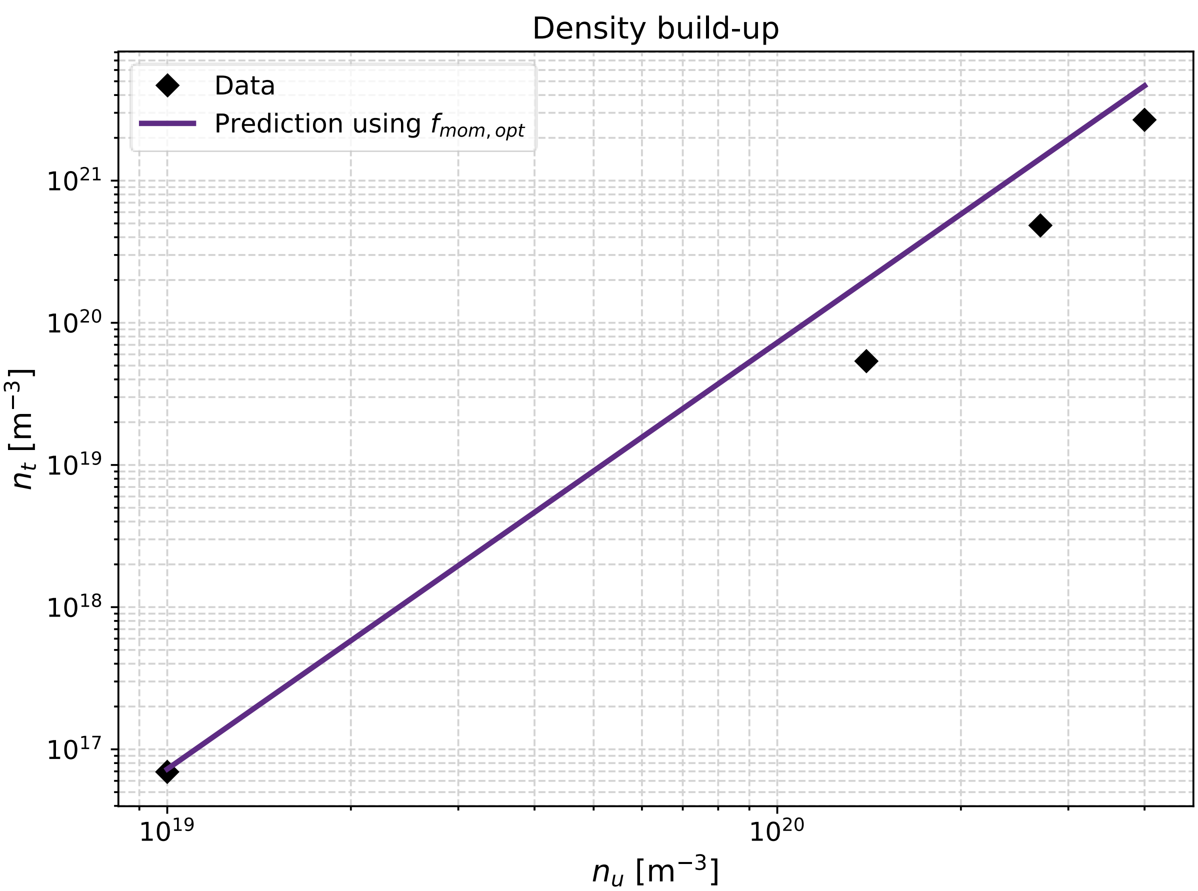

Figure 4.3: Predicted target density n_t, evaluated for the most probable momentum loss value of Case B, against the upstream density n_u.

The posterior distribution is again evaluated using the MCMC algorithm from the emcee library, and is plotted in Figure 4.4 along with the prior and the true value of momentum losses. The inaccurate prior belief provides a bad initial condition that the data try to correct. However, due to the high uncertainty in the dataset and the small number of data points, the posterior does not manage to completely overcome the wrong prior, moving only slightly towards the true momentum losses (red line). Additionally, it is observed that the spread around the posteriors mean increases drastically, indicating the model uncertainty regarding this prediction.

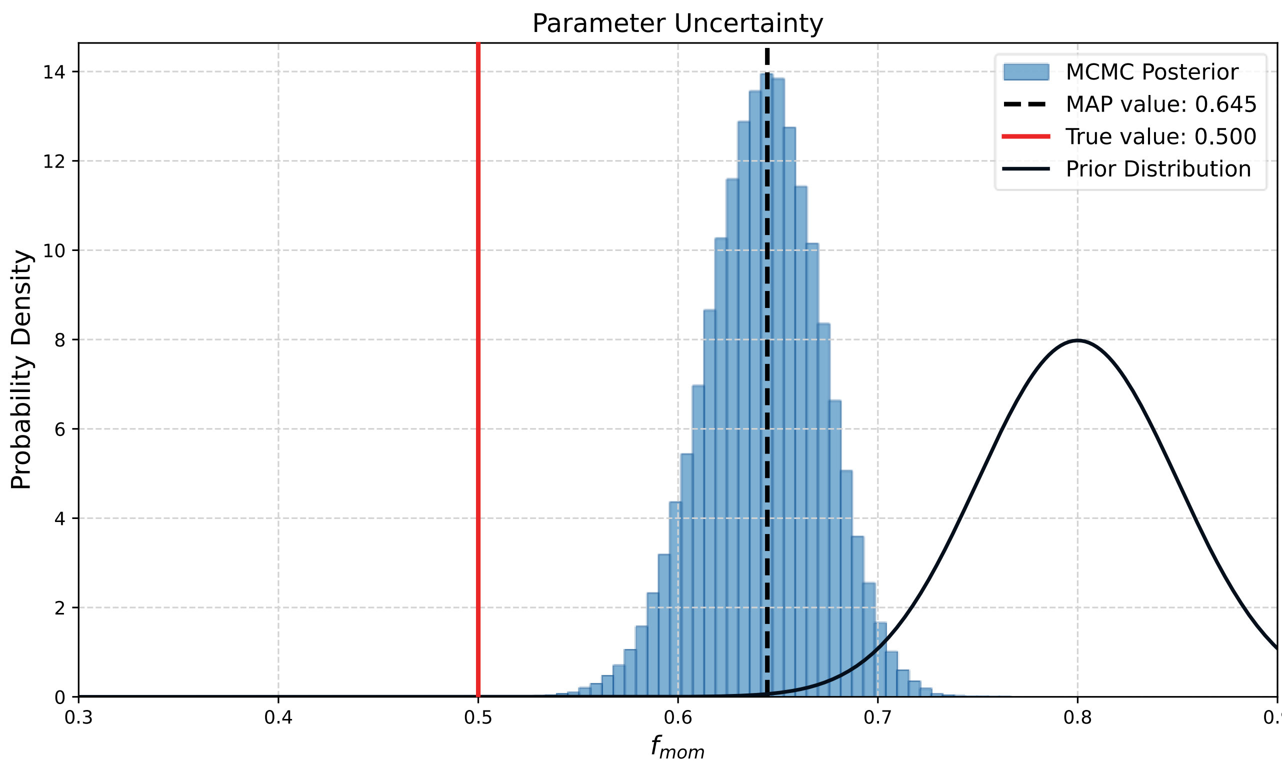

Figure 4.4: The posterior distribution histogram of Case B with its maximum (MAP) value in dashed black. The real value of the momentum losses is in red, while the posterior PDF in black solid.

In conclusion, the two simple test cases demonstrate that given high-quality data in sufficient quantity and a robust estimation of the prior value, then the free parameters introduced in this thesis may be evaluated. Of course, in reality, many challenges arise, which are overseen, due to the introduced simplifications. For instance, the variances of the prediction errors ${\sigma_{e,i}}^2$ are unknown parameters that must be estimated from the Bayesian inference. Additionally, the evaluation of model parameters using synthetic datasets is an inaccurate process and is used strictly for demonstrative purposes of the model implementation.

5 Concluding remarks

5.1 Main conclusions

In the present work, simplified models for the edge plasma transport and neutral gas exhaust of thermonuclear magnetically confined fusion reactors have been analyzed under steady-state conditions. Although both tokamaks and stellarators are considered the work is focused on the latter ones.

For the edge plasma analysis, the simplified two-point model is presented and correction parameters accounting for convection, momentum losses, and target-localized power dissipation, are implemented, improving the predictive capabilities of the model. It becomes evident that in tokamak reactors, parallel convection dominates perpendicular convection for nearly the entire temperature range. In contrast, in stellarators, perpendicular convection is of paramount importance, especially close to the targets where the temperature drops, necessitating the usage of a stellarator two-point model accounting for the perpendicular convection. Another important difference between tokamaks and stellarators is shown to be the absence of a diffusion-limited regime in stellarators. In this regime, cross-field transport dominates, reducing the upstream temperature and target density, and increasing the target temperature. These conditions degrade both the confinement and exhaust processes. Therefore, to effectively avoid operating in this regime, critical thresholds that predict the transition from conduction-limited to diffusion-limited regime have been presented.

Regarding the particle exhaust system, a simplified particle balance model that predicts the neutral gas pressure in the subdivertor is derived, considering the neutral gas flow that passes through the pumping gaps as an externally defined parameter. The model assumes that the neutral gas entering the subdivertor either returns back to the vessel through the pumping gap or the leakage openings, or is pumped out. The model follows the throughput balance model of Litovoli et al [16], but instead of throughput balance, which is shown not to be conserved in non-isothermal conditions, a consistent well-conserved particle balance is employed. The model is compared with experimental data and satisfactory agreement is obtained.

As it is well known, the connection of the two-point model outputs, with the flux passing through the pumping gaps (i.e., providing a model that directly connects the upstream plasma density with the neutral gas pressure) is not straightforward. Here, a simplified approximation is proposed that assumes that all plasma flux striking the targets recombines at the targets, and then returns as a source term in the region right above the targets. This allows the definition of the divertor control volume and of the particle collection efficiency parameter. In the divertor control volume since the plasma is recombined it is treated as single species, while the particle collection efficiency parameter measures the fraction of particles that manage to pass through the pumping gaps. Based on these assumptions and definitions a simplified model connecting the two-point model with the particle exhaust system is proposed. Additionally, this simplified model estimates the pressure in front of the targets, i.e., the divertor control volume pressure. The model prediction of the divertor and subdivertor pressures is compared with some experimental data, and shows satisfactory agreement. However, this is not a solid proof, since the dataset is limited and its uncertainty is unknown and therefore further validation is needed.

Finally, a Bayesian MAP framework is derived in order to address the challenge of estimating the free parameter values. The prior distribution is considered as Gaussian and the likelihood is modeled via a prediction error, assuming a known covariance matrix.

Overall, this Diploma Thesis attempts to provide approximate and computationally inexpensive models that capture the scaling of important parameters regarding the edge plasma transport and particle exhaust system of the Wendelstein 7-X stellarator reactor. Fast track estimations of the neutral gas pressure are very useful in the design and optimization of future divertor modules. Thus, approximate simplified approaches, like the presented one, allowing computationally efficient parametric studies of the involved physics are very important.

5.2 Suggestions for future work

Several features analyzed in this thesis, especially in Chapters 3 and 4, provide a foundation for further investigation. Some suggested areas for future research follow:

1. The proposed simplified model, which couples the two-point model and the subdivertor model, presented in Section 3.3, requires a more in-depth analysis. Specifically, the model does not account for the 3D location of the strike lines, which significantly affects the particle flux passing through the pumping gaps. Additionally, future works should incorporate the coupling of the particle flux returning to the vessel with the two-point model target conditions.
2. The Bayesian framework derived in Chapter 4 can be applied to a large experimental dataset to estimate the most probable values of the associated free parameters. For instance, by establishing a correlation between the upstream density and the target

density through experimental observations, this statistical analysis can yield estimations for the correction factors α, f_{conv}, and f_{cool} that hold under the experimental conditions.

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Appendices

Appendix A: Fundamentals of kinetic theory

Important concepts regarding the Kinetic Theory of Gases used in this thesis are summarized based on [33], [34], and [35].

i. *The Distribution Function and Moments:*

The motion of a particle is described from the Boltzmann equation:

$$\underbrace{\frac{\partial f}{\partial t}}_{\substack{\text{Time}\\\text{Evolution}}} + \underbrace{\mathbf{v}\cdot\nabla_{\mathbf{r}} f}_{\substack{\text{Spatial}\\\text{Advection}}} + \underbrace{\frac{q}{m}(\mathbf{E}+\mathbf{v}\times\mathbf{B})\cdot\nabla_{\mathbf{v}} f}_{\substack{\text{Acceleration due to}\\\text{Lorentz Force}}} = \underbrace{\left.\frac{\partial f}{\partial t}\right|_{coll}}_{\substack{\text{Collision}\\\text{Operator}}} + \underbrace{S(\mathbf{r},\mathbf{v},t)}_{\substack{\text{Volumetric}\\\text{Source/Sink}}} \tag{A.1}$$

This equation describes the evolution of the one-particle velocity distribution function $f(\mathbf{r},\mathbf{v},t)$. The quantity $f(\mathbf{r},\mathbf{v},t)d^3\mathbf{r}d^3\mathbf{v}$ is the expected number of particles at location $\mathbf{r}+d^3\mathbf{r}$ with velocities in the domain v_x to v_x+dv_x, v_y to v_y+dv_y and v_z to v_z+dv_z. By integrating over the entire velocity domain $\Omega_{\mathbf{v}}$ (taking the 0th moment of the distribution function), the spatial *particle number density* can be evaluated:

$$n(\mathbf{r},t)=\iiint_{\Omega_{\mathbf{v}}} f(\mathbf{r},\mathbf{v},t)d^3\mathbf{v}\,. \tag{A.2}$$

Thus, the probability of a particle being located at $(\mathbf{r},\mathbf{v},t)$, is

$$\bar{f}(\mathbf{r},\mathbf{v},t)=\frac{f(\mathbf{r},\mathbf{v},t)}{n(\mathbf{r},t)}\,. \tag{A.3}$$

Considering a microscopic velocity-dependent quantity $\mathbf{A}(\mathbf{r},\mathbf{v},t)$, the macroscopic average value of $\mathbf{A}$ is defined as

$$\langle\mathbf{A}\rangle(\mathbf{r},t)\equiv\iiint_{\Omega_{\mathbf{v}}}\mathbf{A}(\mathbf{r},\mathbf{v},t)\bar{f}(\mathbf{r},\mathbf{v},t)d^3\mathbf{v}\,. \tag{A.4}$$

By using Eq. (A.3), the final expression for the average macroscopic quantity takes the form

$$\langle \mathbf{A} \rangle (\mathbf{r},t) = \frac{1}{n(\mathbf{r},t)} \iiint_{\Omega_{\mathbf{v}}} \mathbf{A}(\mathbf{r},\mathbf{v},t) f(\mathbf{r},\mathbf{v},t) d^3\mathbf{v} . \tag{A.5}$$

Taking the first velocity moment of the distribution function (or letting $\mathbf{A} = \mathbf{v}$ in Eq. (A.5)), yields the *average bulk velocity:*

$$\bar{\mathbf{v}}(\mathbf{r},t) = \frac{1}{n(\mathbf{r},t)} \iiint_{\Omega_{\mathbf{v}}} \mathbf{v} f(\mathbf{r},\mathbf{v},t) d^3\mathbf{v} . \tag{A.6}$$

Multiplying the particle density $n(\mathbf{r},t)$ by the average velocity, given by Eq. (A.6), yields the particle flux

$$\mathbf{\Gamma} \equiv n(\mathbf{r},t)\bar{\mathbf{v}}(\mathbf{r},t) \left[\frac{\text{particles}}{m^2 s} \right] . \tag{A.7}$$

Every function can be written as the sum of a mean value and a perturbation around it. Thus, the *random thermal velocity* is defined as the difference between the actual velocity and the average bulk velocity:

$$\mathbf{v}_{rand} = \mathbf{v} - \bar{\mathbf{v}} . \tag{A.8}$$

Using the random thermal velocity, the *pressure tensor* is defined as the second velocity moment of the particle distribution function:

$$\overleftrightarrow{P}_{i,j} = \iiint_{\Omega_{\mathbf{v}}} m u_i u_j f(\mathbf{r},\mathbf{v},t) d^3\mathbf{v} = mn(\mathbf{r},t) \langle u_i u_j \rangle . \tag{A.9}$$

The pressure tensor is usually decomposed into the hydrostatic pressure tensor (diagonal elements) and the viscous stress tensor (off-diagonal elements):

$$\overleftrightarrow{P}_{i,j} = \underbrace{p\delta_{i,j}}_{\substack{\text{hydrodynamic}\\ \text{pressure}}} + \underbrace{\overleftrightarrow{\Pi}_{i,j}}_{\substack{\text{viscous}\\ \text{pressures}}} . \tag{A.10}$$

The hydrodynamic pressure p is defined as the mean of the normal stresses:

$$p \equiv \frac{\overleftrightarrow{P}_{i,i}}{3} = nm \frac{\langle {u_x}^2 \rangle + \langle {u_y}^2 \rangle + \langle {u_z}^2 \rangle}{3} \equiv nm \langle u^2 \rangle . \tag{A.11}$$

On the other hand, for the viscous pressure tensor, the following expression holds true:

$$\overleftrightarrow{\Pi}_{i,j} = mn(\mathbf{r},t)\langle u_i u_j \rangle, \quad \text{for } i \neq j\,. \tag{A.12}$$

Finally, the *heat flow vector* is defined as the third velocity moment of the particle distribution function

$$q_i = \iiint_{\Omega_v} \frac{1}{2} m u^2 u_i f(\mathbf{r},\mathbf{v},t) d^3\mathbf{v} = \frac{1}{2} mn(\mathbf{r},t) \langle (u_x{}^2 + u_y{}^2 + u_z{}^2) u_i \rangle, \tag{A.13}$$

where q_i represents the conductive heat flux moving in the i-direction.

Maxwell-Boltzmann distribution and Temperature:

For a gas (or plasma) in thermal equilibrium the most probable distribution of velocities can be found analytically. The solution is the so-called Maxwellian distribution and in 1D it is given by the expression

$$f_{Maxw}(v) = \frac{n(r,t)}{\sqrt{\pi} v_{th}} \exp\left(-\frac{v^2}{v_{th}{}^2}\right), \tag{A.14}$$

where the *random thermal speed* is defined as

$$v_{th} = \sqrt{\frac{2k_B T}{m}}\,. \tag{A.15}$$

Substituting $\mathbf{A} = K_e \equiv \frac{1}{2} mv^2$ in Eq. (A.5) and assuming an a 1D Maxwellian distribution, $f(\mathbf{r},\mathbf{v},t) = f_{Maxw}(v)$, yields the *average particle kinetic energy*:

$$\langle K_e \rangle = \frac{1}{n(r,t)} \int_{-\infty}^{\infty} \frac{1}{2} mv^2 f_{Maxw}(v) dv = \frac{m}{2\sqrt{\pi} v_{th}} \int_{-\infty}^{\infty} v^2 \exp\left(-\frac{v^2}{v_{th}{}^2}\right) dv \tag{A.16}$$

Substituting the solution from expression (B.3), with $a = 1/v_{th}{}^2$, the integral evaluates to $\sqrt{\pi} v_{th}{}^3/2$. Therefore, the average particle kinetic energy is $\langle K_e \rangle = mv_{th}{}^2/4$. Using the definition of random thermal speed given in Eq. (A.15), the final 1D result is

$$\langle K_e \rangle = \frac{1}{2} k_B T\,. \tag{A.17}$$

Finally, since the average particle kinetic energy is defined as $\langle K_e \rangle = 1/2\, m \langle v^2 \rangle$, it is deduced that the average squared velocity is

$$\left\langle v^2 \right\rangle = \frac{k_B T}{m}\,. \tag{A.18}$$

Appendix B: Gaussian integrals

Calculating macroscopic moments often requires solving of 1D Gaussian (or Euler-Poisson) integrals. The general 1D form is

$$I_D \equiv \int_{-\infty}^{\infty} x^D \exp\left(-a(x-\mu)^2\right)dx\,, \tag{B.1}$$

where D is the number of dimensions and μ the shifting term. This integral can be evaluated analytically for all values of D.

i. For odd powers of x $(D = 2n+1)$:

$$\int_{-\infty}^{\infty} x^{2n+1} \exp(-ax^2)dx = 0\,. \tag{B.2}$$

ii. For even powers of x $(D = 2n)$:

$$\int_{-\infty}^{\infty} x^{2n} \exp(-ax^2)dx = \frac{(2n-1)!!}{(2a)^n}\sqrt{\frac{\pi}{a}}\,. \tag{B.3}$$

Finally, to handle the shifting term, the u-substitution technique may be used by setting $u = x - \mu$

.

Appendix C: Derivation of the semi-1D fluid equations for plasma flow

Here, the moments of the Fokker Plank kinetic equation

$$v_x \frac{\partial f}{\partial x} + \frac{eE}{m}\frac{\partial f}{\partial v_x} = \left.\frac{\partial f}{\partial t}\right|_{coll} + S(x,\mathbf{v}), \tag{C.1}$$

provided in (2.3), are derived in order to reach to the semi-1D fluid system for plasma flow (2.5).

Conservation of Particles (0th Moment)

The operator $\iiint\limits_{\Omega_\mathbf{v}} (\cdot) d^3\mathbf{v}$, where $\Omega_\mathbf{v} \subset \mathbb{R}^3$ is applied, in the whole velocity domain, to Eq. (C.1) to obtain

$$\iiint\limits_{\Omega_\mathbf{v}} v_x \frac{\partial f}{\partial x} d^3\mathbf{v} + \iiint\limits_{\Omega_\mathbf{v}} \frac{eE}{m}\frac{\partial f}{\partial v_x} d^3\mathbf{v} = \iiint\limits_{\Omega_\mathbf{v}} \left.\frac{\partial f}{\partial t}\right|_{coll} d^3\mathbf{v} + \iiint\limits_{\Omega_\mathbf{v}} S(x,\mathbf{v}) d^3\mathbf{v}. \tag{C.2}$$

Each term is separately examined to form the particle conservation equation.

- $$\iiint\limits_{\Omega_\mathbf{v}} v_x \frac{\partial f}{\partial x} d^3\mathbf{v} = \int_{-\infty}^{\infty} v_x \frac{\partial f}{\partial x} dv_x \iint\limits_{\Omega_{v_y,v_z}} dv_y dv_z = \frac{\partial}{\partial x}\int_{-\infty}^{\infty} v_x f dv_x \iint\limits_{\Omega_{v_y,v_z}} dv_y dv_z =$$
$$= \frac{\partial}{\partial x}\iiint\limits_{\Omega_\mathbf{v}} v_x f d^3\mathbf{v} = \frac{\partial}{\partial x} n \underbrace{\iiint\limits_{\Omega_\mathbf{v}} \frac{v_x f}{n} d^3\mathbf{v}}_{\equiv \langle v_x \rangle}$$

The macroscopic mean velocity (A.6) of the particles in x-direction is effectively constructed. By defining $v \equiv \langle v_x \rangle$, the first term

$$\iiint\limits_{\Omega_\mathbf{v}} v_x \frac{\partial f}{\partial x} d^3\mathbf{v} = \frac{d}{dx}(nv). \tag{C.3}$$

- $$\iiint\limits_{\Omega_\mathbf{v}} \frac{eE}{m}\frac{\partial f}{\partial v_x} d^3\mathbf{v} = \iint\limits_{\Omega_{v_y,v_z}} \frac{eE}{m}\int_{-\infty}^{\infty} \frac{\partial f}{\partial v_x} dv_x dv_y dv_z = \iint\limits_{\Omega_{v_y,v_z}} \frac{eE}{m} f\Big|_{v_x=-\infty}^{v_x=\infty} dv_x dv_y dv_z$$

Since no particle can accelerate to infinite speeds the distribution function collapses to zero at infinite velocities, i.e., $f \to 0$ as $|\mathbf{v}| \to \infty$. So, the limiting boundary conditions are $f(v_x = \infty) = f(v_x = -\infty) = 0$. Thus, the second term yields zero:

$$\iiint_{\Omega_{\mathbf{v}}} \frac{eE}{m} \frac{\partial f}{\partial v_x} d^3\mathbf{v} = 0 \,. \tag{C.4}$$

- $\iiint_{\Omega_{\mathbf{v}}} \left.\frac{\partial f}{\partial t}\right|_{coll} d^3\mathbf{v} = 0$, since the integration is in the whole velocity domain and

whatever is lost due to collisions in one part of the distribution is gained in another.

- $\iiint_{\Omega_{\mathbf{v}}} S(x,\mathbf{v}) d^3\mathbf{v} = S_p(x) \left(\frac{\beta_n}{\pi}\right)^{3/2} \int_{-\infty}^{\infty} e^{-\beta_n (v_x - v_n)^2} d(v_x - v_n) \int_{-\infty}^{\infty} e^{-\beta_n v_y{}^2} dv_y \int_{-\infty}^{\infty} e^{-\beta_n v_z{}^2} dv_z$

Having formed three Gaussian integrals (B.1), each one is evaluated based on (B.3). All of them are equal to $(\pi / \beta_n)^{1/2}$, so the spatial source becomes:

$$\iiint_{\Omega_{\mathbf{v}}} S(x,\mathbf{v}) d^3\mathbf{v} = S_p(x) \,. \tag{C.5}$$

Finally, from Eq.(C.2), the *particle conservation equation* is deduced:

$$\boxed{\frac{d}{dx}\left[nv\right] = S_p(x)} \,. \tag{C.6}$$

Conservation of Momentum (1st Moment)

The operator $\iiint_{\Omega_{\mathbf{v}}} mv_x(\cdot)d^3\mathbf{v}$ is applied to Eq. (C.1):

$$\iiint_{\Omega_{\mathbf{v}}} mv_x{}^2 \frac{\partial f}{\partial x} d^3\mathbf{v} + \iiint_{\Omega_{\mathbf{v}}} v_x eE \frac{\partial f}{\partial v_x} d^3\mathbf{v} = \iiint_{\Omega_{\mathbf{v}}} mv_x \left.\frac{\partial f}{\partial t}\right|_{coll} d^3\mathbf{v} + \iiint_{\Omega_{\mathbf{v}}} mv_x S(x,\mathbf{v}) d^3\mathbf{v}\,. \qquad \text{(C.7)}$$

Similarly with the particle conservation equation, each term is examined separately:

- $$\iiint_{\Omega_{\mathbf{v}}} mv_x{}^2 \frac{\partial f}{\partial x} d^3\mathbf{v} = \frac{\partial}{\partial x}\int_{-\infty}^{\infty} mv_x{}^2 f dv_x \iint_{\Omega_{\mathbf{v}}} dv_y dv_z = \frac{\partial}{\partial x}\iiint_{\Omega_{\mathbf{v}}} mv_x{}^2 f d^3\mathbf{v}$$

The velocity at the x-axis can be written as the mean velocity, plus the random thermal velocity: $v_x = \langle v_x \rangle + u_x$ (A.8). Substituting this expression into the integral, the first term takes the form

$$\frac{\partial}{\partial x}\iiint_{\Omega_{\mathbf{v}}} mv_x{}^2 f d^3\mathbf{v} = \frac{\partial}{\partial x}\iiint_{\Omega_{\mathbf{v}}} m(\langle v_x \rangle + u_x)^2 f d^3\mathbf{v} =$$
$$= \frac{\partial}{\partial x}\left[mv^2 \iiint_{\Omega_{\mathbf{v}}} f d^3\mathbf{v}\right] + \frac{\partial}{\partial x}\left[m\iiint_{\Omega_{\mathbf{v}}} u_x{}^2 f d^3\mathbf{v}\right] + \frac{\partial}{\partial x}\left[2mv\iiint_{\Omega_{\mathbf{v}}} u_x f d^3\mathbf{v}\right]$$

By multiplying and dividing with the particle density, the macroscopic averages (A.5) of some velocity-dependent variables are formed.

$$\frac{\partial}{\partial x} mv^2 \underbrace{\iiint_{\Omega_{\mathbf{v}}} f d^3\mathbf{v}}_{=n} + \frac{\partial}{\partial x} mn \underbrace{\iiint_{\Omega_{\mathbf{v}}} \frac{u_x{}^2 f}{n} d^3\mathbf{v}}_{\equiv <u_x{}^2>} + \frac{\partial}{\partial x} 2mvn \underbrace{\iiint_{\Omega_{\mathbf{v}}} \frac{u_x f}{n} d^3\mathbf{v}}_{\equiv <u_x>} =$$

$$= \frac{d}{dx}\left(mnv^2\right) + \frac{d}{dx}(mn\langle u_x{}^2 \rangle) + \frac{d}{dx}(2mvn\langle u_x \rangle). \qquad \text{(C.8)}$$

Note that the average velocity, given by Eq. (A.6), is

$$\langle u_x \rangle = \frac{1}{n}\int_{-\infty}^{\infty} u_x f d^3\mathbf{v} = \underbrace{\frac{1}{n}\int_{-\infty}^{\infty} v_x f dv}_{\equiv\langle v_x \rangle} - \frac{1}{n}\int_{-\infty}^{\infty} \langle v_x \rangle f dv = \langle v_x \rangle - \langle v_x \rangle = 0\,. \qquad \text{(C.9)}$$

Furthermore, the second term of Eq. (C.8) can be expressed as the derivative of the parallel pressure along the x-axis, using Eq. (A.11)

$$p_{\parallel} = mn\langle u_x \rangle^2 . \tag{C.10}$$

Finally, the first term of the moment is

$$\iiint_{\Omega_\mathbf{v}} m{v_x}^2 \frac{\partial f}{\partial x} d^3\mathbf{v} = \frac{d}{dx}\left(p_{\parallel} + mnv^2\right) . \tag{C.11}$$

- $\iiint_{\Omega_\mathbf{v}} v_x eE \frac{\partial f}{\partial v_x} d^3\mathbf{v} = eE\iint\left[\cancelto{0}{v_x f\Big|_{-\infty}^{\infty}} - \int_{-\infty}^{\infty} f dv_x\right] dv_y dv_z$

Again, using the boundary conditions for the distribution function at infinity, the expression reduces to $-eE\iiint_{\Omega_\mathbf{v}} fd^3\mathbf{v}$. Thus, the second term yields

$$\iiint_{\Omega_\mathbf{v}} v_x eE \frac{\partial f}{\partial v_x} d^3\mathbf{v} = -eEn . \tag{C.12}$$

- $\iiint_{\Omega_\mathbf{v}} mv_x \left.\frac{\partial f}{\partial t}\right|_{coll} d^3\mathbf{v}$

This integral represents the macroscopic forces that result from microscopic particle collisions. In [3], Stangeby uses a phenomenological approach to approximate these friction forces. The derivation contains complex plasma physics calculations that exceed the scope of this thesis. The two separate expressions for ions and electrons are shown in [3] to be

$$\begin{aligned}
\left[\iiint_{\Omega_\mathbf{v}} mv_x \left.\frac{\partial f}{\partial t}\right|_{coll} d^3\mathbf{v}\right]_{\text{ion}} &= -m_i(v_i - v_n)\overline{\sigma v_{i,n}} n_n n + m_e(v_e - v_i)\nu_{i,e}^{mom} n + 0.71n\frac{d(kT_e)}{dx} \\
\left[\iiint_{\Omega_\mathbf{v}} mv_x \left.\frac{\partial f}{\partial t}\right|_{coll} d^3\mathbf{v}\right]_{\text{electron}} &= -m_e(v_e - v_i)\nu_{i,e}^{mom} n - 0.71n\frac{d(kT_e)}{dx}
\end{aligned} \tag{C.13}$$

In Eq. (C.13), the subscripts i, e, and n refer to the ions, electrons, and neutrals, respectively. The unsubscripted density n represents the main plasma density. Additionally, $\overline{\sigma v_{i,n}}$ is the rate coefficient, and $\nu_{i,e}^{mom}$ expresses the collision frequency between species a and b.

- $\iiint_{\Omega_\mathbf{v}} mv_x S(x,\mathbf{v}) d^3\mathbf{v} = \iiint_{\Omega_\mathbf{v}} mv_x \left[S_p(x)\left(\frac{\beta_n}{\pi}\right)^{3/2} e^{-\beta_n(v_x - v_n)^2} e^{-\beta_n v_y^2} e^{-\beta_n v_z^2}\right] d^3\mathbf{v}$

It has already been assumed that the source term comes from the ionization of neutrals with a drifting Maxwellian distribution. So, $S(x,\mathbf{v})$ is given by Eq. (2.4).

Using the Gaussian integrals yields

$$\iiint_{\Omega_{\mathbf{v}}} m v_x S(x,\mathbf{v}) d^3\mathbf{v} = m S_p(x) \left(\frac{\beta_n}{\pi}\right)^{3/2} \underbrace{\int_{-\infty}^{\infty} v_x e^{-\beta_n (v_x - v_n)^2} dv_x}_{\equiv I} \underbrace{\int_{-\infty}^{\infty} e^{-\beta_n v_y^2} dv_y}_{=\sqrt{\pi/\beta_n}} \underbrace{\int_{-\infty}^{\infty} e^{-\beta_n v_z^2} dv_z}_{=\sqrt{\pi/\beta_n}} .$$

For the integral I, the $u = v_x - v_n$ substitution is applied, as follows:

$$I = \int_{-\infty}^{\infty} v_x e^{-\beta_n (v_x - v_n)^2} dv_x = \cancel{\underbrace{\int_{-\infty}^{\infty} u e^{-\beta_n u^2} du}_{\text{odd}}} + v_n \underbrace{\int_{-\infty}^{\infty} e^{-\beta_n u^2} du}_{=\sqrt{\pi/\beta_n}} = v_n \sqrt{\frac{\pi}{\beta_n}} .$$

Therefore, the fourth term is deduced

$$\iiint_{\Omega_{\mathbf{v}}} m v_x S(x,\mathbf{v}) d^3\mathbf{v} = m v_n S_p(x) . \qquad \text{(C.14)}$$

Finally, the *ion momentum equation can be written as*

$$\frac{d}{dx}[m_i n v_i^{\,2} + p_{\parallel,i}] - eEn = -m_i (v_i - v_n) \overline{\sigma v_{in}} n_n n + m_e (v_e - v_i) \nu_{i,e}^{mom} n + 0.71 n \frac{d(k_B T_e)}{dx} + m_i v_n S_p(x) ,$$

(C.15)

while the *electron momentum equation, as*

$$\frac{d}{dx}\left[\cancel{m_e}^{\,0} n_e v^2 + p_e \right] + eEn = -m_e (v_e - v_i) \nu_{i,e}^{mom} n - 0.71 n \frac{d(k_B T_e)}{dx} + \cancel{m_e}^{\,0} v_n S_p(x) \Rightarrow$$

$$\Rightarrow \frac{dp_e}{dx} + eEn = -m_e (v_e - v_i) \nu_{i,e}^{mom} n - 0.71 n \frac{d(k_B T_e)}{dx} . \qquad \text{(C.16)}$$

Adding the two equations (C.15) and (C.16), the _plasma momentum equation_ is formed:

$$\boxed{\frac{d}{dx}[m_i n v_i^{\,2} + p_{\parallel,i} + p_e] = -m_i (v_i - v_n) \overline{\sigma v_{in}} n_n n + m_i v_n S_p(x)} . \qquad \text{(C.17)}$$

Parallel Conservation of Energy (2nd Moment)

The operator $\iiint_{\Omega_\mathbf{v}} \frac{1}{2} m v_x^2 (\cdot) d^3\mathbf{v}$ is applied to Eq. (C.1).

$$\iiint_{\Omega_\mathbf{v}} \frac{1}{2} m v_x^3 \frac{\partial f}{\partial x} d^3\mathbf{v} + \iiint_{\Omega_\mathbf{v}} \frac{1}{2} v_x^2 eE \frac{\partial f}{\partial v_x} d^3\mathbf{v} = \iiint_{\Omega_\mathbf{v}} \frac{1}{2} m v_x^2 \left.\frac{\partial f}{\partial t}\right|_{coll} d^3\mathbf{v} + \iiint_{\Omega_\mathbf{v}} \frac{1}{2} m v_x^2 S(x,\mathbf{v}) d^3\mathbf{v} \,. \quad \text{(C.18)}$$

As before, each term is examined separately.

- $$\iiint_{\Omega_\mathbf{v}} \frac{1}{2} m v_x^3 \frac{\partial f}{\partial x} d^3\mathbf{v} = \frac{1}{2} m \frac{\partial}{\partial x} \iiint_{\Omega_\mathbf{v}} \left(\langle v_x \rangle + u_x \right)^3 f d^3\mathbf{v} =$$

$$= \frac{1}{2} m \frac{\partial}{\partial x} \left[v^3 \iiint_{\Omega_\mathbf{v}} f d^3\mathbf{v} + 3v^2 \underbrace{\cancel{\iiint_{\Omega_\mathbf{v}} u_x f d^3\mathbf{v}}}_{=n\langle u_x \rangle = 0} + 3v \underbrace{\iiint_{\Omega_\mathbf{v}} u_x^2 f d^3\mathbf{v}}_{=n\langle u_x^2 \rangle} + \underbrace{\iiint_{\Omega_\mathbf{v}} u_x^3 f d^3\mathbf{v}}_{=n\langle u_x^3 \rangle} \right] =$$

$$= \frac{d}{dx} \left[\frac{1}{2} mnv^3 + \frac{3}{2} mnv \left\langle u_x^2 \right\rangle + \underbrace{\frac{1}{2} mn \left\langle u_x^3 \right\rangle}_{\substack{\text{ion-conducted} \\ \text{heat flux density}}} \right] \quad \text{(C.19)}$$

It is important to define the ion-conducted heat flux. This is the first term of the heat flux in x-direction, given by Eq. (A.13), and represents the parallel-to-B spatial transport of the parallel thermal energy $1/2 m_i u_x^2$ of ions:

$$q_{\parallel cond,x} \equiv \frac{1}{2} mn \left\langle u_x^3 \right\rangle . \quad \text{(C.20)}$$

Therefore, using expression (C.20), the integral evaluates to

$$\iiint_{\Omega_\mathbf{v}} \frac{1}{2} m v_x^3 \frac{\partial f}{\partial x} d^3\mathbf{v} = \frac{d}{dx} \left[\frac{1}{2} mnv^3 + \frac{3}{2} p_\parallel v + q_{\parallel cond,x} \right] . \quad \text{(C.21)}$$

- $$\iiint_{\Omega_\mathbf{v}} \frac{1}{2} v_x^2 eE \frac{\partial f}{\partial v_x} d^3\mathbf{v} = \frac{eE}{2} \iint \left[\cancel{v_x^2 f}\Big|_{-\infty}^{\infty} - 2 \underbrace{\int_{-\infty}^{\infty} v_x f dv_x}_{=n\langle v_x \rangle} \right] dv_y dv_z = -eEnv$$

- $\iiint_{\Omega_{\mathbf{v}}} \frac{1}{2} m v_x{}^2 \left.\frac{\partial f}{\partial t}\right|_{coll} d^3\mathbf{v}$

Following the same approach with all the above integrals containing the collisional operator, a phenomenological approach is followed. This term accounts for heat transfer associated with particle collisions. Including only collisions of charged particles, the following is defined:

<u>*Self-Collision Heating:*</u> This is the thermal energy between the parallel and perpendicular axes.

$$Q_{\perp\to\parallel} \equiv \frac{p_\perp - p_\parallel}{\tau_{\perp\to\parallel}} \tag{C.22}$$

Where, $\tau_{\perp\to\parallel}$ is the pressure anisotropy relaxation time. Here, it should be noted that electrons, having a significantly small mass and thus higher self-collision frequencies, tend to have isotropic distributions: $T_{\perp,e} \approx T_{\parallel,e}$. This makes $Q_{\perp\to\parallel,e} = 0$.

<u>*Joule Heating:*</u> This term accounts for the dissipation heating caused by the net drift of the electrons against the collisional forces R and is given by

$$Q_R \equiv -R(v_e - v_i)\,. \tag{C.23}$$

During a collision, nearly all the energy is transferred to electrons due to their lower mass. This leads the analogous heating term for ions to be practically zero: $Q_{R,i} = 0$.

<u>*Thermal Equilibrium Collisions:*</u> The collisions of electrons and ions in thermal equilibrium gives the heating term:

$$Q_{eq\parallel} = \frac{3m_e}{m_i} n \nu_{eq} (k_B T_e - k_B T_i)\,, \tag{C.24}$$

where ν_{eq} is the collision frequency under equilibrium conditions.

The two different expressions for ions and electrons, are:

$$\begin{aligned} \left[\iiint_{\Omega_{\mathbf{v}}} \frac{1}{2} m v_x{}^2 \left.\frac{\partial f}{\partial t}\right|_{coll} d^3\mathbf{v}\right]_{\text{ion}} &= Q_{\perp\to\parallel,i} + Q_{eq\parallel,i} \\ \left[\iiint_{\Omega_{\mathbf{v}}} \frac{1}{2} m v_x{}^2 \left.\frac{\partial f}{\partial t}\right|_{coll} d^3\mathbf{v}\right]_{\text{electron}} &= Q_R - Q_{eq\parallel,e} \end{aligned} \tag{C.25}$$

- $$\iiint_{\Omega_{\mathbf{v}}} \frac{1}{2} m v_x^{\,2} S(x,\mathbf{v}) d^3\mathbf{v} = \iiint_{\Omega_{\mathbf{v}}} \frac{1}{2} m v_x^{\,2} \left[S_p(x) \left(\frac{\beta_n}{\pi} \right)^{3/2} e^{-\beta_n (v_x - v_n)^2} e^{-\beta_n v_y^2} e^{-\beta_n v_z^2} \right] d^3\mathbf{v} =$$

$$= \frac{1}{2} m S_p(x) \left(\frac{\beta_n}{\pi} \right)^{3/2} \left[\underbrace{\int_{-\infty}^{\infty} u^2 e^{-\beta_n u^2} du}_{\frac{1}{2}\sqrt{\frac{\pi}{\beta_n}}} + 2 v_n \cancel{\int_{-\infty}^{\infty} u e^{-\beta_n u^2} du} + v_n^{\,2} \underbrace{\int_{-\infty}^{\infty} e^{-\beta_n u^2} du}_{=\sqrt{\pi/\beta_n}} \right] \underbrace{\int_{-\infty}^{\infty} e^{-\beta_n v_y^{\,2}} dv_y}_{=\sqrt{\pi/\beta_n}} \underbrace{\int_{-\infty}^{\infty} e^{-\beta_n v_z^{\,2}} dv_z}_{=\sqrt{\pi/\beta_n}} =$$

$$= \frac{1}{2} m S_p(x) \left(\frac{\beta_n}{\pi} \right)^{3/2} \left[\frac{1}{2} \sqrt{\frac{\pi}{\beta_n^{\,3}}} + v_n^{\,2} \sqrt{\frac{\pi}{\beta_n}} \right] \sqrt{\frac{\pi}{\beta_n}}^{\,2} = \frac{1}{4} m S_p(x) \frac{1}{\beta_n} + \frac{1}{2} m v_n^{\,2} S_p(x) =$$

$$= \frac{m_i}{2} S_p(x) \left(\frac{k_B T_n}{m_n} + v_n^{\,2} \right)$$

But, $m_n = m_i + m_e \simeq m_i$, so the term becomes:

$$\underbrace{\frac{1}{2} k_B T_n S_p(x)}_{\substack{\text{ionization source} \\ \text{of Maxwellian neutrals}}} + S_p(x) \underbrace{\frac{1}{2} m_n v_n^{\,2}}_{\substack{\text{average kinetic} \\ \text{energy of neutrals}}} .$$

The neutral mass is extremely small and additionally, neutrals are not affected by the magnetic field, so their motion is isotropic, meaning that their average velocity is very small too. All the above, make the average kinetic energy term negligible compared to the source.

Thus, the last term is equal to the ionization source of Maxwellian neutrals:

$$\iiint_{\Omega_{\mathbf{v}}} \frac{1}{2} m v_x^{\,2} S(x,\mathbf{v}) d^3\mathbf{v} \equiv Q_{E\|} = \frac{1}{2} k_B T_n S_p(x) . \qquad \text{(C.26)}$$

By combining all the above onto Eq. (C.18), the *__parallel energy conservation equations for ions and electrons__*, respectively, are obtained

$$\boxed{\begin{aligned} \frac{d}{dx}\left[\frac{1}{2} m_i n v_i^{\,3} + \frac{3}{2} p_{\|i} v_i + q_{\|cond,x,i} \right] &= eEnv_i + Q_{\perp\to\|} + Q_{eq\|} + Q_{E\|,i} \\ \frac{d}{dx}\left[\frac{3}{2} p_{\|e} v_e + q_{\|cond,x,e} \right] &= -eEnv_e + Q_R - Q_{eq\|} + Q_{E\|,e} \end{aligned}} \qquad \text{(C.27)}$$

Perpendicular Conservation of Energy (2nd Moment)

Now, the operator $\iiint_{\Omega_v} \frac{1}{2} m\left(v_y^2 + v_z^2\right)(\cdot) d^3\mathbf{v}$ is applied to Eq. (C.1).

$$\iiint_{\Omega_v} \frac{1}{2} m\left(v_y^2 + v_z^2\right) v_x \frac{\partial f}{\partial x} d^3\mathbf{v} + \iiint_{\Omega_v} \frac{1}{2}\left(v_y^2 + v_z^2\right) eE \frac{\partial f}{\partial v_x} d^3\mathbf{v} = \iiint_{\Omega_v} \frac{1}{2} m\left(v_y^2 + v_z^2\right) \left.\frac{\partial f}{\partial t}\right|_{coll} d^3\mathbf{v} + $$
$$+ \iiint_{\Omega_v} \frac{1}{2} m\left(v_y^2 + v_z^2\right) S(x,\mathbf{v}) d^3\mathbf{v}$$

(C.28)

Starting with the first term

- $\iiint_{\Omega_v} \frac{1}{2} m\left(v_y^2 + v_z^2\right) v_x \frac{\partial f}{\partial x} d^3\mathbf{v}$

Expanding all the velocity terms using the (A.8), and applying the assumption of 1D flow $\left(\langle v_y\rangle = \langle v_z\rangle = 0\right)$, the following expressions hold:

$$\begin{aligned} v_x &= \langle v_x\rangle + u_x \\ v_y &= \cancel{\langle v_y\rangle} + u_y = u_y \\ v_z &= \cancel{\langle v_z\rangle} + u_z = u_z \end{aligned} \qquad \text{(C.29)}$$

Substituting expressions (C.29) to the integral, results to

$$\frac{1}{2} m \frac{\partial}{\partial x} \iiint_{\Omega_v} \left(\langle v_x\rangle u_y^2 + \langle v_x\rangle u_z^2 + u_x u_y^2 + u_x u_z^2\right) f d^3\mathbf{v} =$$

$$= \frac{\partial}{\partial x}\left[\frac{1}{2} nm\langle v_x\rangle\left(\langle u_y^2\rangle + \langle u_z^2\rangle\right) + \frac{1}{2} nm\langle u_x u_y^2\rangle + \frac{1}{2} nm\langle u_x u_z^2\rangle\right]. \qquad \text{(C.30)}$$

Using $\langle u_y^2\rangle = \langle u_z^2\rangle = \frac{k_B T_\perp}{m} \equiv \langle u_\perp^2\rangle$, Eq. (C.30) can be written as

$$\frac{\partial}{\partial x}\left[p_\perp v + \frac{1}{2} nm\langle u_x u_y^2\rangle + \frac{1}{2} nm\langle u_x u_z^2\rangle\right].$$

The last two terms represent the parallel-to-B spatial transport of the perpendicular thermal energy $1/2m\left(u_y^2 + u_z^2\right)$:

$$q_{\|cond,y-z} \equiv \frac{1}{2} nm\left(\left\langle u_x u_y{}^2 \right\rangle + \left\langle u_x u_z{}^2 \right\rangle\right). \tag{C.31}$$

Finally, using $\left\langle u_y{}^2 \right\rangle = \left\langle u_z{}^2 \right\rangle = \frac{k_B T_\perp}{m} \equiv \left\langle u_\perp{}^2 \right\rangle$, the first term yields

$$\iiint_{\Omega_\mathbf{v}} \frac{1}{2} m\left(v_y{}^2 + v_z{}^2\right) v_x \frac{\partial f}{\partial x} d^3\mathbf{v} = \frac{d}{dx}\left[p_\perp v + q_{\|cond,y-z}\right]. \tag{C.32}$$

- $\iiint_{\Omega_\mathbf{v}} \frac{1}{2}\left(v_y{}^2 + v_z{}^2\right) eE \frac{\partial f}{\partial v_x} d^3\mathbf{v} = \frac{eE}{2} \int_{-\infty}^{\infty} \cancel{\frac{\partial f}{\partial v_x} dv_x} \iint \left(v_y{}^2 + v_z{}^2\right) dv_y dv_z = 0$
- $\iiint_{\Omega_\mathbf{v}} \frac{1}{2} m\left(v_y{}^2 + v_z{}^2\right) \left.\frac{\partial f}{\partial t}\right|_{coll} d^3\mathbf{v}$

A same treatment with the parallel energy conservation is implemented here, but the Joule Heating term is dropped, since it is negligible in the perpendicular direction [3]. The collision integral can be evaluated as:

$$\begin{aligned} \left[\iiint_{\Omega_\mathbf{v}} \frac{1}{2} m\left(v_y{}^2 + v_z{}^2\right) \left.\frac{\partial f}{\partial t}\right|_{coll} d^3\mathbf{v}\right]_{\text{ion}} &= -Q_{\perp\to\|} + Q_{eq\perp} \\ \left[\iiint_{\Omega_\mathbf{v}} \frac{1}{2} m\left(v_y{}^2 + v_z{}^2\right) \left.\frac{\partial f}{\partial t}\right|_{coll} d^3\mathbf{v}\right]_{\text{electron}} &= -Q_{eq\perp} \end{aligned} \tag{C.33}$$

- $\iiint_{\Omega_\mathbf{v}} \frac{1}{2} m\left(v_y{}^2 + v_z{}^2\right) S(x,\mathbf{v}) d^3\mathbf{v}$

$$= \frac{m}{2} S_p(x) \left(\frac{\beta_n}{\pi}\right)^{3/2} \underbrace{\int_{-\infty}^{\infty} e^{-\beta_n (v_x - v_n)^2} dv_x}_{\sqrt{\pi/\beta_n}} \left[\int_{-\infty}^{\infty} e^{-\beta_n v_z{}^2} \int_{-\infty}^{\infty} v_y{}^2 e^{-\beta_n v_y{}^2} dv_y dv_z + \int_{-\infty}^{\infty} v_z{}^2 e^{-\beta_n v_z{}^2} \int_{-\infty}^{\infty} e^{-\beta_n v_y{}^2} dv_y dv_z\right]$$

After some simple calculations, the integral evaluates to

$$\iiint_{\Omega_\mathbf{v}} \frac{1}{2} m\left(v_y{}^2 + v_z{}^2\right) S(x,\mathbf{v}) d^3\mathbf{v} \equiv Q_{E\perp} = k_B T_n S_p(x) \tag{C.34}$$

Combining the above and substituting to Eq. (C.28), leads to the *perpendicular energy conservation equations for ions and electrons*:

$$\boxed{\begin{aligned} \frac{d}{dx}\left[p_{\perp i} v_i + q_{\|cond,y-z,i} \right] &= -Q_{\perp\rightarrow\|} + Q_{eq\perp} + Q_{E\perp,i} \\ \frac{d}{dx}\left[p_{\perp e} v_e + q_{\|cond,y-z,e} \right] &= -Q_{eq\perp} + Q_{E\perp,e} \end{aligned}} \tag{C.35}$$

Total Conservation of Energy

By assuming pressure isotropy in ions and electrons, and temperature isotropy only on ions, the expressions

$$p_{\|e} = p_{\perp e} \equiv p_e \quad , \quad p_{\|i} = p_{\perp i} \equiv p_i \quad , \quad T_{\|i} = T_{\perp i} \equiv T_i \, , \tag{C.36}$$

hold. If the electron and ion parallel and perpendicular equations, (C.27) and (C.35), are combined, the equation for total conservation of energy can be derived.

Total Ion Conservation of Energy:

Substituting the corresponding expressions for parallel and perpendicular conservation yields

$$\frac{d}{dx}\left[\frac{1}{2} m_i n v_i^3 + \frac{5}{2} p_i v_i + \underbrace{q_{\|cond,x,i} + q_{\|cond,y-z,i}}_{\equiv q_{\|cond,i}} \right] = eEnv_i + \underbrace{\left(Q_{eq\|} + Q_{eq\perp} \right)}_{\equiv Q_{eq}} + \underbrace{\left(Q_{E\|,i} + Q_{E\perp,i} \right)}_{\equiv Q_{E,i}} .$$

Therefore, the final expression for the *ion conservation of energy* is

$$\frac{d}{dx}\left[q_{\|conv,i} + q_{\|cond,i} \right] = eEnv_i + Q_{eq} + Q_{E,i} \, , \tag{C.37}$$

where the parallel convection flux is defined as

$$q_{\|conv,i} \equiv \frac{1}{2} m_i n v_i^3 + \frac{5}{2} n k_B T_i v_i \, , \tag{C.38}$$

and the ion temperature T_i is given in $[K]$.

Total Electron Conservation of Energy:

Following the same procedure for electrons yields

$$\frac{d}{dx}\left[\frac{5}{2}p_e v_e + \underbrace{q_{\|cond,x,e} + q_{\|cond,y-z,e}}_{\equiv q_{\|cond,e}}\right] = -eEnv_e + Q_R - \underbrace{\left(Q_{eq\|} + Q_{eq\perp}\right)}_{\equiv Q_{eq}} + \underbrace{\left(Q_{E\|,e} + Q_{E\perp,e}\right)}_{\equiv Q_{E,e}}.$$

Consequently, the total *electron conservation of energy* is

$$\frac{d}{dx}\left[q_{\|conv,e} + q_{\|cond,e}\right] = -eEnv_e - Q_{eq} + Q_R + Q_{E,e}, \tag{C.39}$$

where, the parallel convection for electrons is

$$q_{\|conv,e} \equiv \frac{5}{2}nk_B T_e v_e, \tag{C.40}$$

and T_e is given in $[K]$.

For strong collisionality, the problem may be further simplified by assuming ion-electron thermal equilibrium $T_e = T_i = T$. Additionally, local ambipolarity $(v_e = v_i)$ is assumed. Then, the two energy equations can be combined as

$$\frac{d}{dx}\left[q_\|\right] = Q_R + \underbrace{\left(Q_{E,i} + Q_{E,e}\right)}_{\equiv Q_E},$$

where $q_\| \equiv \left(q_{\|conv,i} + q_{\|conv,e}\right) + \left(q_{\|cond,i} + q_{\|cond,e}\right)$ is the total heat flux that is transported in the x-direction.

It should be noted that the conduction term is given by (A.13). Springer, suggested a formulation using the Fourier law of heat conduction. It was proved that the thermal conductivity coefficient for collisions between charged particles is $K = k_0 T^{5/2}$, where k_0 is the Spitzer-Härm conductivity constant [3]. So, using Fourier's law the parallel conduction fluxes become:

$$\begin{aligned} q_{\|,e,cond} &= -k_{0,e} T_e^{5/2}\frac{dT_e}{dx}, \quad \text{with } k_{0,e} \approx 2000\left[Wm^{-1}eV^{-7/2}\right] \\ q_{\|,i,cond} &= -k_{0,i} T_i^{5/2}\frac{dT_i}{dx}, \quad \text{with } k_{0,i} \approx 60\left[Wm^{-1}eV^{-7/2}\right] \end{aligned} \tag{C.41}$$

where T is given in $[eV]$, and $q_\|$ in $\left[W/m^2\right]$. Therefore, the total parallel conduction flux is given by

$$q_\| = \frac{1}{2}m_i n v_i^3 + \frac{5}{2}nk_B T_i v_i + \frac{5}{2}nk_B T_e v_e - k_{0,i}T_i^{5/2}\frac{dT_i}{dx} - k_{0,e}T_e^{5/2}\frac{dT_e}{dx}$$

and after the assumptions, it reduces to

$$q_{\parallel} = \frac{1}{2} m_i n v^3 + 5 n k_B T v - k_{0,e} T^{5/2} \frac{dT}{dx}. \tag{C.42}$$

The parallel conduction terms have been effectively simplified by applying the condition $k_{0,i} \ll k_{0,e}$

The current approach is purely 1-dimensional, which inherently neglects all the perpendicular-to-B transport. It can be shown that the perpendicular heat conduction can be important in many cases, especially in stellarator conditions (Figure 2.1); thus, it is mandatory to add a cross-field heat diffusion term. This task, however, would require taking the moments of the 2D Boltzmann equation, which is rather arduous. A simpler method is to start directly from the *continuum energy conservation* equation:

$$\frac{\partial(\rho\varphi)}{\partial t} + \nabla \cdot \mathbf{J} = \dot{R}'''_{source}, \tag{C.43}$$

with $\varphi = \frac{E}{m} = e + \frac{u^2}{2} + gz \equiv \varepsilon$ and $\mathbf{J} = \rho\varepsilon\mathbf{u} + \mathbf{q} + \mathbf{w}$. Substituting into Eq. (C.43), yields

$$\frac{\partial(\rho\varepsilon)}{\partial t} + \nabla \cdot \left(\underset{\text{convection}}{\rho\varepsilon\mathbf{u}} + \underset{\text{conduction}}{\mathbf{q}} + \underset{\substack{\text{diffusion of} \\ \text{mech. work}}}{\mathbf{w}} \right) = \dot{R}'''_{source}. \tag{C.44}$$

Assuming steady-state, 2D flow, and neglecting of the diffusion of the mechanical work, Eq. (C.44) reduces to

$$\nabla \cdot \left(\mathbf{q}_{conv} + \mathbf{q}_{cond} \right) = \dot{R}'''_{source}. \tag{C.45}$$

Analyzing the convection and conduction terms along the parallel and perpendicular directions, Eq. (C.45) can be written as follows:

$$\frac{\partial}{\partial s_{\parallel}} \left[q_{\parallel conv} + q_{\parallel cond} \right] + \frac{\partial}{\partial s_{\perp}} \left[q_{\perp conv} + q_{\perp cond} \right] = \dot{R}'''_{source}. \tag{C.46}$$

Therefore, the parallel conduction and convection have been added to the system of equations.

The calculation of the perpendicular heat conduction requires the estimation of the perpendicular diffusion coefficient $D_{\perp}$, which is an extremely challenging task [3]. For this reason, Stangeby approximates this flux using an anomalous cross-field heat conduction term $D_{\perp} \approx x_{\perp}$. This anomalous perpendicular thermal diffusivity takes values of $x_{\perp}^{e,i} \simeq 3\ m^2/s$, [4]. Therefore, for electrons and ions:

$$q_{\perp cond}{}^{e,i} = -x_{\perp}{}^{e,i} n \frac{d(k_B T_{e,i})}{ds_{\perp}}, \tag{C.47}$$

where T is in $[K]$. For convenience, $s_{\parallel} \equiv x$ is used for the parallel-to-B direction and $s_{\perp}$ for the perpendicular direction.

A similar approach is followed in [3] for the perpendicular heat convection, where for electrons and ions:

$$q_{\perp conv}{}^{e,i} = -\frac{5}{2} k_B T D_{\perp} \frac{dn_{e,i}}{ds_{\perp}}, \tag{C.48}$$

where T is also in $[K]$.

Grouping all the above heat fluxes yields the simplified *plasma energy conservation* equation:

$$\frac{\partial}{\partial s_{\parallel}}\left[q_{\parallel conv} + q_{\parallel cond}\right] + \frac{\partial}{\partial s_{\perp}}\left[q_{\perp conv} + q_{\perp cond}\right] = Q_R + Q_E. \tag{C.49}$$

The heat flux terms and their units are:

$$\begin{aligned}
q_{\parallel conv} &= \frac{1}{2} m_i n v^3 + 5 n k_B T v, \quad \text{with } T[K] \\
q_{\parallel cond} &= -k_{0,e} T^{5/2} \frac{dT}{ds_{\parallel}}, \quad \text{with } T[eV] \\
q_{\perp conv} &= -\frac{5}{2} k_B T D_{\perp} \frac{dn}{ds_{\perp}}, \quad \text{with } T[K] \\
q_{\perp cond} &= -x_{\perp} n \frac{d(k_B T)}{ds_{\perp}}, \quad \text{with } T[K]
\end{aligned} \tag{C.50}$$

The plasma momentum equation (C.17) can be further simplified by assuming a uniform plasma pressure, $p_e + p_i = p$, and by considering that neutrals are unaffected by the magnetic field, meaning their motion is purely random ($v_n \ll v$).

Therefore, the final system of *Semi-1D Fluid Equations for Plasma Flow* is established as

$$\boxed{\begin{array}{l} \dfrac{d}{ds_{\parallel}}\left[nv\right] = S_p(s_{\parallel}) \\ \dfrac{d}{ds_{\parallel}}[m_i nv^2 + 2k_B nT] = -m_i v\overline{\sigma v_{in}} n_n n \\ \dfrac{\partial q_{\parallel}}{\partial s_{\parallel}} + \dfrac{\partial q_{\perp}}{\partial s_{\perp}} = Q_R + Q_E \end{array}} \qquad \text{(C.51)}$$

The unknowns of the system (2.5) are the bulk velocity (v), the plasma density (n), and the temperature (T) in the parallel direction.